



Unikue **AICT**

System ADMINISTRATION *Made Easy*



Oguntola Yusuf

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INTRODUCTION

According to Merriam-Webster's dictionary, administration is the management of affairs. System administration therefore, is the management of the affairs of a system and the system in this reference is the computer system.

System administration is a subject matter that most authors has found writing books about superfluous because they believe computers should be managed by experts and whoever wishes to become an expert in system administration should thus attend technical trainings; but one thing they have failed to realize is the fact that when users mismanage their system, it might be a tough time troubleshooting problems on that system by an expert when the system finally develops one.

I have therefore decided to glean the few managerial skills I have learnt over the years into this small piece to assist the populace in managing the computer system they have gotten from their hard-earned money.

This book is organized using simple and easy to understand approach so that readers would not find it boring. I have also made it purely practical and the practice steps are explained to the atomic stage so that every reader can pick up their systems and start practicing.

Most of the topics discussed contain solutions to most of the problems users would have paid technicians to solve. The topic on installation covers almost everything you would need to know before, during and after operating system installation including the available types of installation.

With device and disk management you'd learn how to manage devices attached to your PC and your storage drives respectively. In security, you'd learn how to secure your computer system and your data against threat and in performance configuration you'd learn how to tune the performance of your system to your desired taste. Backup and Recovery explains the steps required to back up your data and how to recover from disasters.

I have decided to discuss solutions to some technical problems under technical fixes, while Customizing Windows explains how to customize your Windows to make it look exactly the way you want.

Comments, corrections and recommendations are all welcome from constructive and destructive criticizers as no one is an ocean of undisputable knowledge.

Sit back and enjoy this fact-filled piece!

Oguntola Yusuf
2014

BOOT PROCESS

This Chapter covers:

- + Boot process
- + BIOS
- + CMOS
- + BIOS Instructions
- + POST
- + BIOS update
- + When and how to update BIOS
- + Boot order

When you press the power button on your PC (Personal Computer) to start it or wake it or launch it or whichever term you decide to use, we technically say that the PC is booting. To a lame man, the screen gets lightened and you see Windows or any other OS (Operating System) get displayed; is that all? Definitely NO!

The following happens when your PC boot:

1. The system's motherboard receives power from the power pack and transfer the required amount of charge to the processor.
2. The processor then runs the BIOS's instructions.

What is BIOS?

BIOS is Basic Input/Output System. It is a program that allows us to gain access to various configuration settings of the computer hardware and gives the PC access to the motherboard and all that connects to it. The BIOS is stored on a ROM (Read Only Memory) which is seen as a chip on the motherboard.

ROM as the name implies suggests that information saved on it can't be modified, it can only be read. ROM also saves information permanently unlike RAM (Random Access Memory) which loses all information stored on it after power loss (this is because RAM requires some amount of charge to successfully store information). The BIOS has predefined settings which can be seen and modified by pressing the correct key on the keyboard as soon as the PC starts up. Don't get that? This is what I mean:

When your PC starts, check the left bottom corner or the top right corner of your screen for information telling you the correct key to press to enter BIOS setup. This key is the Escape key (esc) key on HP systems and some other systems and F2 key on dell

systems. It might be DEL (delete key) in your own case or some other function key because it differs from one PC to another.

Pressing the correct key displays the BIOS setup for your PC from where you can modify the configurations on the BIOS.

Wait a minute! Did I just say that from this screen you can modify the BIOS settings when I mentioned earlier that it is saved on ROM and ROM content can't be modified?

Yes, i actually meant that when I said it, but isn't that contradictory? This is what happens:

The BIOS has got something known as the CMOS (Complementary Metal Oxide Semiconductor) which is used for saving modified configurations/settings since the BIOS can't save or modify its ROM's content. The CMOS itself requires some amount of charge to successfully save information kept on it; it therefore has a battery attached to it (known as the CMOS battery) which produces the required amount of charge.

Now that we understand what the BIOS is; what instruction is executed from the BIOS by the processor? The following instructions are executed:

- The BIOS configures hardware components with the configuration settings in its ROM and modifications in the CMOS.
- The BIOS also check for other ROM BIOS. This is because some hardware on the motherboard has got their own BIOS which contain configuration settings peculiar to them and their drivers. E.g most video card has their own BIOS, because the PC BIOS can't take into account all video cards available in the world and load drivers for them, the video cards thus have their own BIOS which they can supply to the PC BIOS and the PC BIOS trust them to know best about themselves.
- The BIOS then executes the POST.

POST is "Power On Self Test". This is a test that checks all hardware components to see if they are in place and correctly configured. For a successful POST, a PC must have three things else the PC can't run POST; these are:

- Processor
- RAM
- Video card (to display error messages)

POST can notify the user of errors with beep sequences which should be monitored e.g a long beep and three short beeps indicate a memory failure on AMI BIOS systems. The sequence should be noted and the motherboard's manufacturer's website checked for correct interpretation or if your PC comes with motherboard manual, then check it when you notice beep sequences.

Let's start fixing anyway! Can we update the BIOS? Yes.

How?

Check your PC's manufacturer's website for something known as the flash BIOS. This is software you can download and run from within Windows to update your BIOS. But wait a minute; you must be sure you need an update before you do it. How?

- Check the release date of the BIOS update; it might be exactly what your PC is currently running.
- Only install an update if
 - You find out your PC is malfunctioning and might be errors from the BIOS.
 - There is a new hardware (e.g an expansion slot) which your present BIOS does not support.
 - There is a new technology which your present BIOS does not support e.g when the plug and play (PnP) technology was released with Windows 95, PC manufacturers had to release BIOS updates to support this technology.

One rule of thumb to always remember is: **“If it's not broken, don't fix it!”**

NB:

- Don't do anything during BIOS update i.e
 - Don't remove power from the PC
 - Don't turn off the PC
 - Don't touch anything

This is because your PC will get screwed! You could get corrupt BIOS and troubleshooting might be very tough, in fact, you might need to get help from the motherboard's manufacturer. So beware!

- Note your BIOS settings before proceeding with an update because every setting would be restored to their default after a successful update.

Remember: “If it's not broken, don't fix it!”

Hope you still remember where we came from? We are still discussing the boot process! Now step three.

3. The operating system is then loaded.

The BIOS settings (which you learnt how to check and modify earlier) have got a setting known as the “Boot order”; this is a list of the order of preference for devices to be

checked for a valid operating system. A PC's boot order might look similar to the one given below:

CD/DVD

USB hard disk/USB floppy

HDD

This implies that the BIOS would check the CD/DVD drive first for a valid operating system, if it hasn't got one, then it proceeds to check for a USB hard disk/floppy with an operating system, if there is no USB it would then check the HDD (Hard Disk Drive) e.t.c

NB: If you have problems seeing the BIOS settings on your PC even after pressing the correct key and your PC is running Windows 8, it's just because you were not fast enough; you might need to try and retry. This is because Windows 8 is configured to start up very fast and even start up without the keyboard which is why pressing the key late might not work. Press and hold down the key before you turn on the PC.

The disks (CD) has got a boot volume from where OS can be loaded while the hard disk has got an MBR (Master Boot Record), the BIOS search for these to know if they've got valid OS. More often, the hard disk is the warehouse for operating system.

Once a valid operating system is found, the PC loads it and everything looks fine. It is now that you see your Windows 7 or Windows 8 logo or the logo of whichever operating system you are running.

You can see a lot happens during start-up!

INSTALLATION

This Chapter covers:

-  What to know before installing Windows
-  Versions and editions of Windows
-  PC and OS architecture
-  Clean installation
-  Upgrading
-  Migrating
-  Downgrading
-  Hardware requirements
-  Hardware specifications
-  Multi-booting
-  Hyper-V

This section describes when and how to get a system loaded with a valid operating system. Let's get into business.

To start with, the following are the available types of installation:

- Clean Installation
- Upgrading
- Migrating
- Downgrading

Each of these would be discussed in detail but before then, we've got to know this:

Windows operating system has versions and editions. When I talk about version, I mean versions like: Windows 95, Windows 98, Windows NT, Windows 2000, Windows XP, Windows Vista, Windows 7, Windows 8 e.t.c. Now, each of these versions (though not all) has got editions. For example, Windows Vista and Windows 7 have:

- Starter edition
- Home Basic edition
- Home Premium edition
- Professional edition
- Ultimate edition

Windows 8 has three editions which are standard, professional and enterprise editions.

It should be noted that these editions are listed in order of advancement, therefore the most advanced Windows 7 is the ultimate edition and for Windows 8 is the professional edition for single users and enterprise edition for an enterprise (according to Microsoft).

One thing a user should understand and decide first is which version and edition he/she want, and then proceed with installation.

Another bone of contention is the architecture of the processor and the desired operating system. In this age, two architectures are in vogue, these are:

- 32 bit (x86) and
- 64 bit (x64)

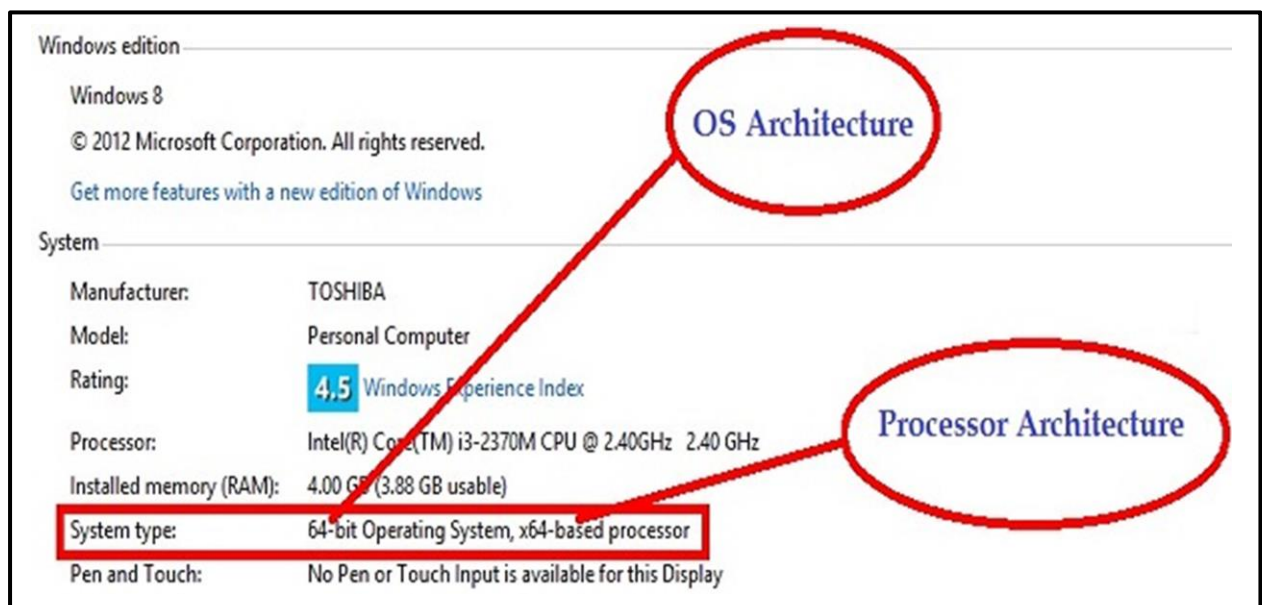
Note that a PC with a 32 bit based processor can only install a 32 bit OS (Operating System) but a 64 bit based processor can run either a 32 bit or a 64 bit OS.

➡ Digression:

Ques: How do I check my PC's architecture?

Ans: Go to "My Computer" (press a combination of Windows logo key and E), right click on any white space and choose properties.

From the properties page, check for "System type" as shown below:



How to choose installation type:

1. **Clean installation:** This is applicable in all cases provided that the PC's hardware meets the minimum hardware requirement for the OS of interest. It should be noted though that this installation type formats the drive upon which the installation is made but saves old

programs and system information in a “Windows.old” folder. Therefore, if you don’t care about losing files, go for clean installation.

2. **Upgrading:** This is used to move from one Windows OS to a later OS with the condition that the architecture of the OS must be maintained and the edition must also be maintained. i.e

A Windows 7 32-bit OS can only be upgraded to a Windows 8 32-bit and not 64-bit. Similarly, Windows vista Professional 64-bit can only be upgraded to Windows 7 Professional or Ultimate 64-bit.

Also:

- i. Windows XP can’t be upgraded to Windows 7 directly except by the upgrade path below:

Windows XP —————> Windows Vista —————> Windows 7

The exception to this rule is that Windows XP SP (Service Pack) III can be upgraded directly to Windows 7. (Check the service pack the same way we checked system’s architecture above)

- ii. Windows Vista can be upgraded to either Windows 7 or Windows 8 directly
iii. Windows 7 can be upgraded to Windows 8
iv. Windows XP can be upgraded directly to Windows 8

Upgrading does not affect user’s files and documents. It is therefore the recommended installation option if its conditions can be met.

3. **Migrating:** This is similar to clean installation as it requires no pre-condition except that the hardware must meet up with the minimum hardware requirements of the desired OS. This option also formats the drive but keeps the content of user’s library (Documents, Pictures, Music, and Videos), account settings, e-mail and other user selected files.

4. **Downgrading:** This is the opposite of upgrading i.e from one Windows OS to an older OS. This is done either by clean installation or migrating.

Having known the advantages and disadvantages of each installation type, you can then decide which type you would love to choose.

Installation requirements:

- The PC where installation would be made
- The installation media (Where the OS would be installed from). This could be any of:

- CD/DVD
- USB
- Network
- WAIK (Windows Automated Installation Kit)
- Image
- WDS (Windows Deployment Service) etc

This book would only discuss installations with CD/DVD and USB.

- To install from a disk, get the disk containing the desired OS
- To install from a USB flash, the following must be done to prepare the USB for installation:

- Make the USB bootable
- Copy the files of the OS onto the USB.

How do we make USB bootable?

Steps:

- i. Run **cmd** (command prompt) as an administrator (search for cmd in the start screen, right click on it and choose “Run as administrator” or press ctrl + shift + Enter)
- ii. In the command prompt window, type **diskpart** and press enter

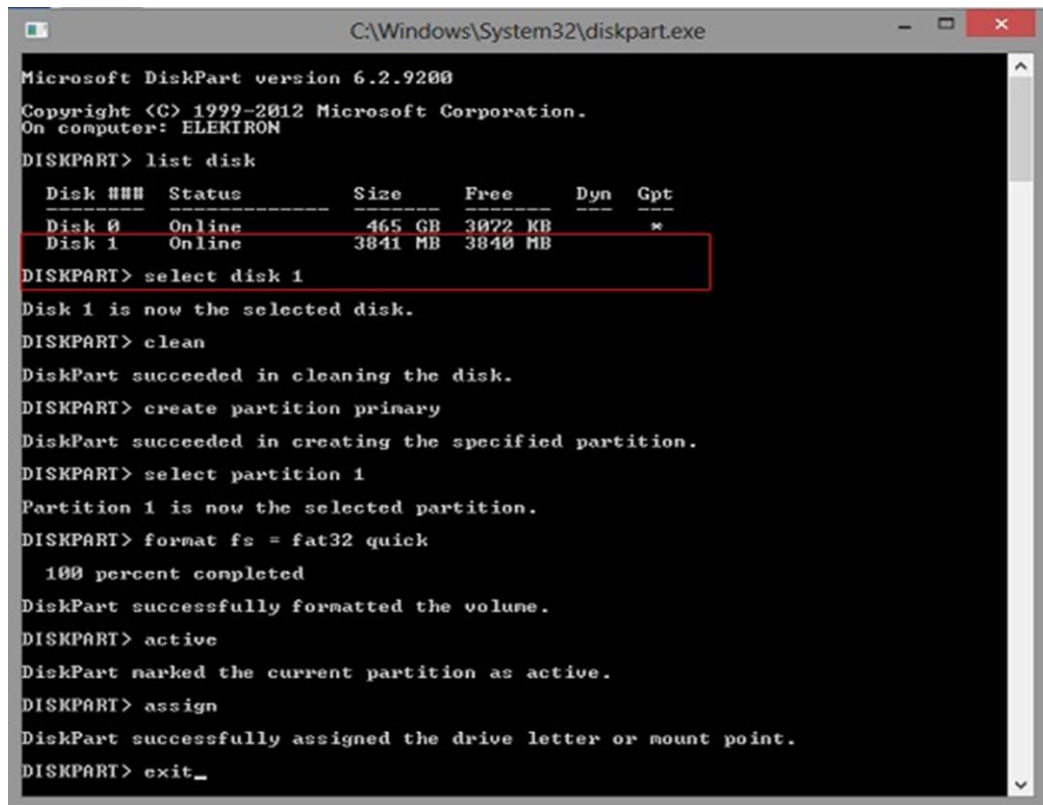
NB: In place of steps 1 and 2, you can simply search for **diskpart** and run it.

- iii. Type **list disk** and press enter

This would show a list of disk presently attached to your PC, the USB should thus be attached before now. Check to see the disk number of the attached USB (confirm with the disk size). This is very important as any error might damage the PC’s hard disk. Make sure the USB is 4GB or more. Assuming the USB is listed as disk 1 as in the example below

- iv. Type **select disk 1** [disk 1 is the USB’s disk number] and press enter
- v. Type **clean** and press enter
- vi. Type **create partition primary** and press enter
- vii. Type **select partition 1** and press enter

- viii. Type **format fs = fat32 quick** and press enter [The quick keyword is optional]
- ix. Type **active** and press enter
- x. Type **assign** and press enter
- xi. Type **exit** and press enter



The screenshot shows the Microsoft DiskPart utility window. The title bar reads 'C:\Windows\System32\diskpart.exe'. The main window displays the following text:

```
Microsoft DiskPart version 6.2.9200
Copyright (C) 1999-2012 Microsoft Corporation.
On computer: ELEKTRON

DISKPART> list disk

   Disk ###  Status         Size      Free      Dyn  Gpt
   -----  -
   Disk 0      Online         465 GB    3072 KB
   Disk 1      Online        3841 MB    3840 MB

DISKPART> select disk 1
Disk 1 is now the selected disk.

DISKPART> clean
DiskPart succeeded in cleaning the disk.

DISKPART> create partition primary
DiskPart succeeded in creating the specified partition.

DISKPART> select partition 1
Partition 1 is now the selected partition.

DISKPART> format fs = fat32 quick
    100 percent completed
DiskPart successfully formatted the volume.

DISKPART> active
DiskPart marked the current partition as active.

DISKPART> assign
DiskPart successfully assigned the drive letter or mount point.

DISKPART> exit_
```

Type exit again if you did this through cmd.

Your USB flash is now bootable! Copy the OS files either from a disk or a folder on your PC to the USB and get ready for installation.

But before we begin to install, there is one more thing:

Every edition of every version of Windows has got maximum hardware specifications and minimum hardware requirements.

Maximum hardware specifications are the maximum values or type of hardware configurations you can use in your PC once you install that type of OS while minimum hardware requirement refers to the minimum configuration of hardware you must have before such OS can be installed.

Focusing on Windows 7, the maximum hardware specifications is given below:

Windows 7 Starter

- 2GB RAM

- 1 physical CPU (Multiple cores on a single CPU supported)

Windows 7 Home Basic

- 4GB RAM (x86)
- 8GB RAM (x64)

Windows 7 Home Premium

- 2 physical CPU
- 4GB RAM (x86)
- 16GB RAM (x64)

Windows 7 Professional and Ultimate

- 2 physical CPU
- 4GB RAM (x86)
- 128GB RAM (x64)

This implies that a 32-bit Windows 7 ultimate PC can't run 8GB RAM. If your PC has 8GB RAM, you can't use it all. When you check the system properties as described above, it would read "8.00 GB (x.xx GB usable)" where x.xx would be a value less than 4. For example, if x.xx is 3.88 as in the example above, the PC would only make use of 3.88GB of the available RAM.

The minimum hardware requirements are:

Windows 7 Starter and Home Basic

- 1GHz CPU
- 512MB RAM
- 16GB disk drive (32-bit) with 15GB free space
- 20GB disk drive (64-bit) with 15GB free space
- DirectX9 GPU and 32MB graphics memory

Other editions (i.e Home Premium, Professional and Ultimate)

- 1GHz CPU
- 1GB RAM
- 40GB disk drive with 15GB free space
- DirectX9 GPU Windows Display Driver (WDDM) driver, Pixel shader 2.0 hardware, 32 bits per pixel, 128MB graphics memory.

This implies that a system with 512MHz CPU and 512MB RAM can't install Windows 7 Home Premium nor Professional nor Ultimate.

What about Windows 8?

Microsoft said any PC that run Windows Vista successfully can run Windows 8 and technically, Windows Vista has more hardware requirements than Windows 7. Therefore, if your PC can run Windows 7, then it can run Windows 8.

Now that everything is ready, let's start installation

CLEAN INSTALLATION

Remember, this would format your drive and move previous installation files into a Windows.old folder.

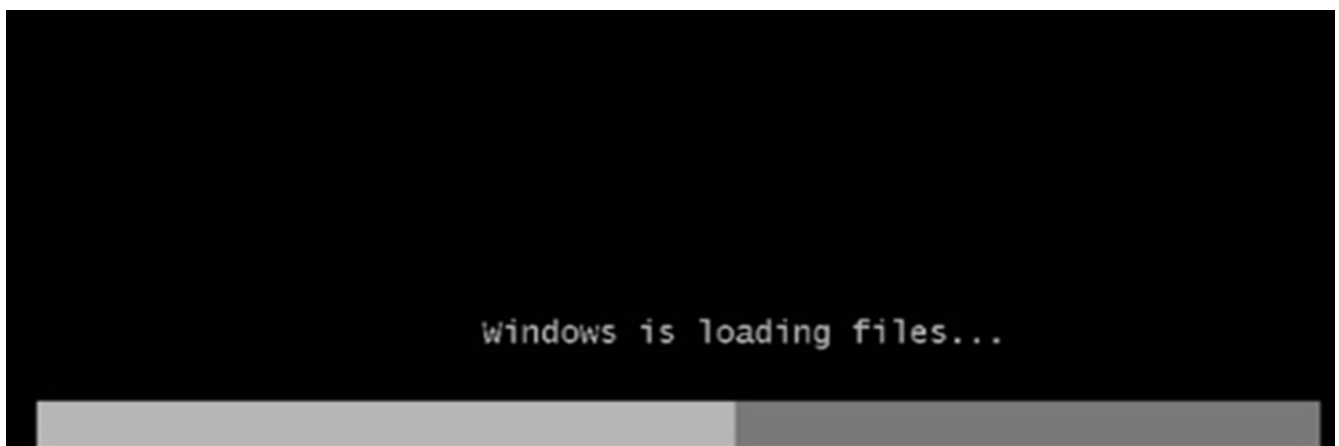
If you are ready to go, then follow the steps below (we would be installing Windows 7 in this example. Follow the same procedure to install other versions of Windows):

1. Insert the disk or USB into the PC
2. Restart the PC and get ready to launch BIOS setup interface [By pressing the correct key on the keyboard as described under boot process].
3. From the BIOS setup, change the boot order to make sure the CD/DVD drive option is above the hard disk if you want to use CD/DVD or make the USB option be above the hard disk to use a USB.

How do you change the position?

Press F5 to move down & F6 to move up or vice versa depending on your PC. It might be "U" for up and "D" for down, so make sure you check the information displayed at the bottom of the screen or on the right pane for some PC.

4. Save changes made to BIOS and exit by pressing F10 or the correct key as instructed by your PC's BIOS setup.
5. Once you exit setup, if everything is fine, you should see the screen below:



If there is a message prompting you to press any key to boot from CD/DVD, then press any key on the keyboard as fast as possible.

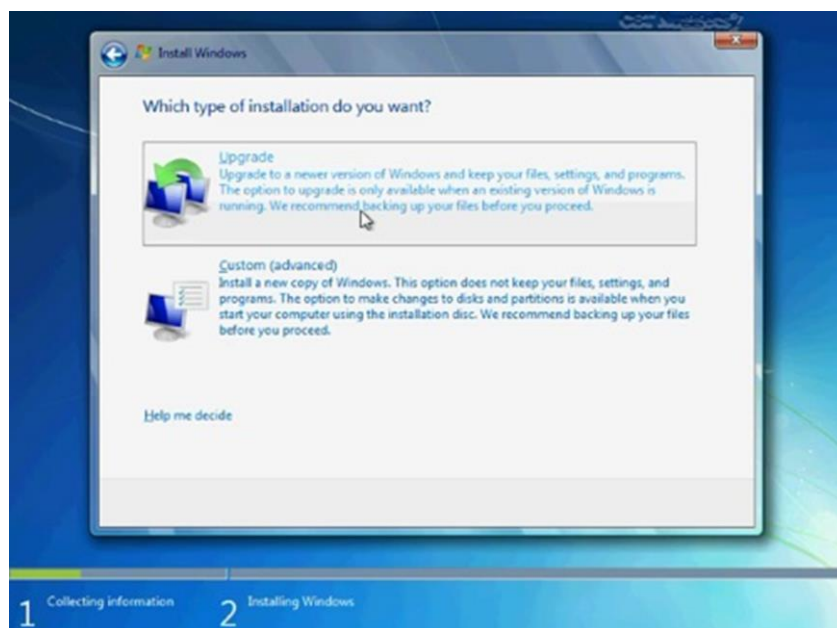
6. The next screen should be a screen asking you to choose your language, choose the desired language and click next.
7. The next screen should be the one shown below with the “Install now” button. Click on “Install now” to continue.



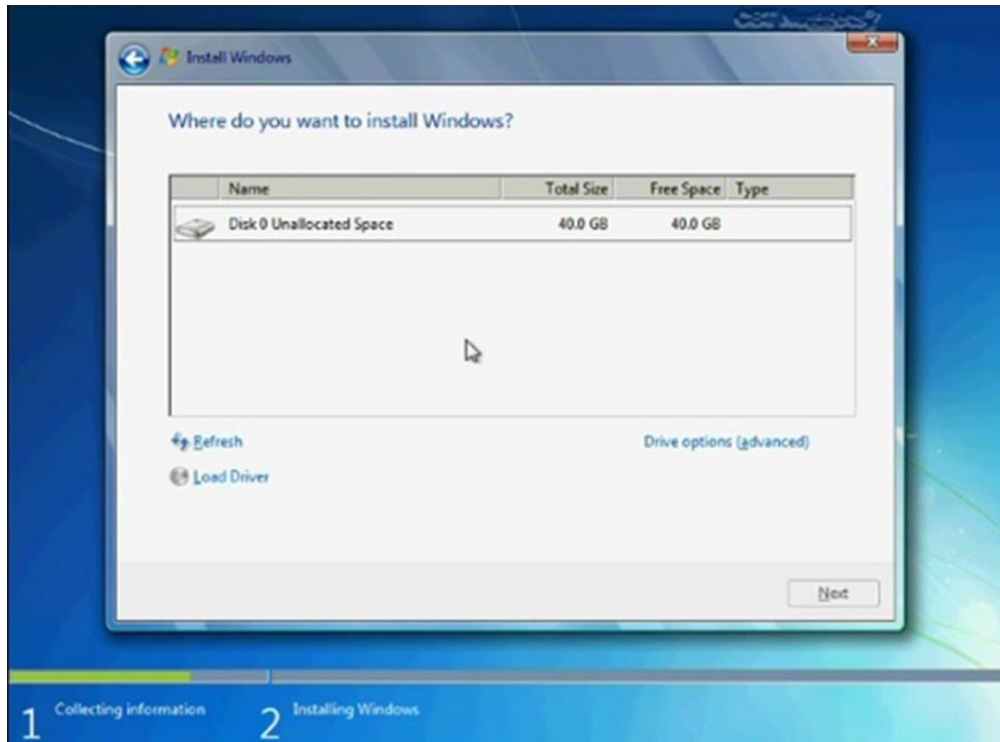
8. The next screen might differ depending on the content of your installation file. If you have a screen asking for the desired OS, choose the correct one and click next.

The next screen might prompt you to accept licence agreement, do so and click next.

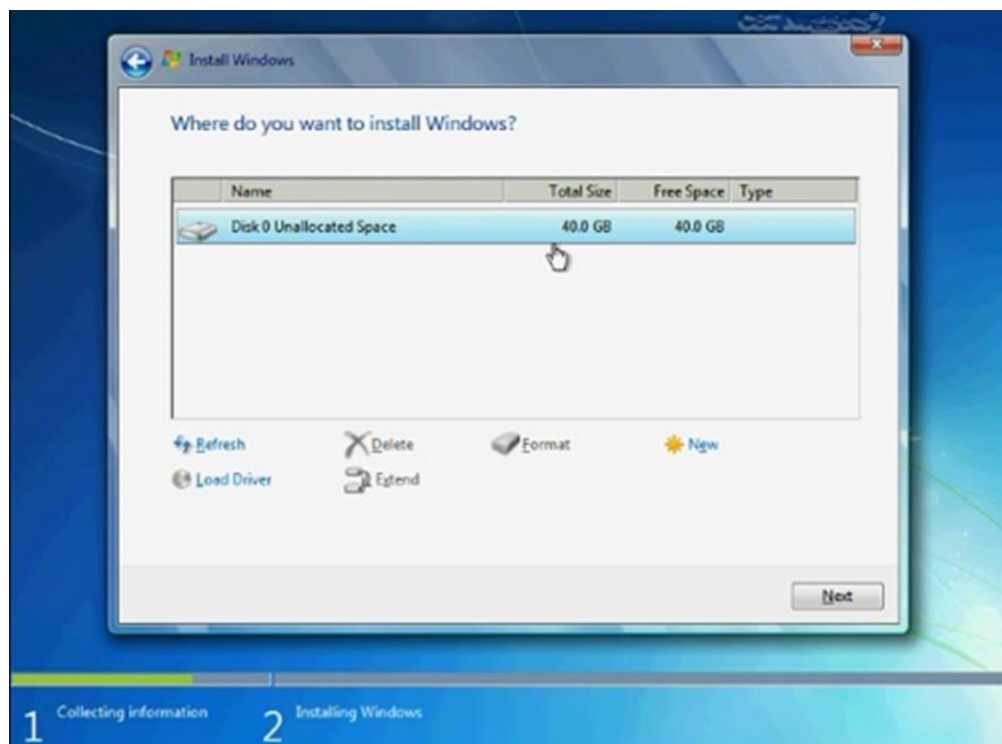
9. The screen below should be the next screen, choose “**Custom**” to continue



10. The next screen would ask you to choose the drive where you want Windows installed as shown below: choose the correct drive and click “Next”



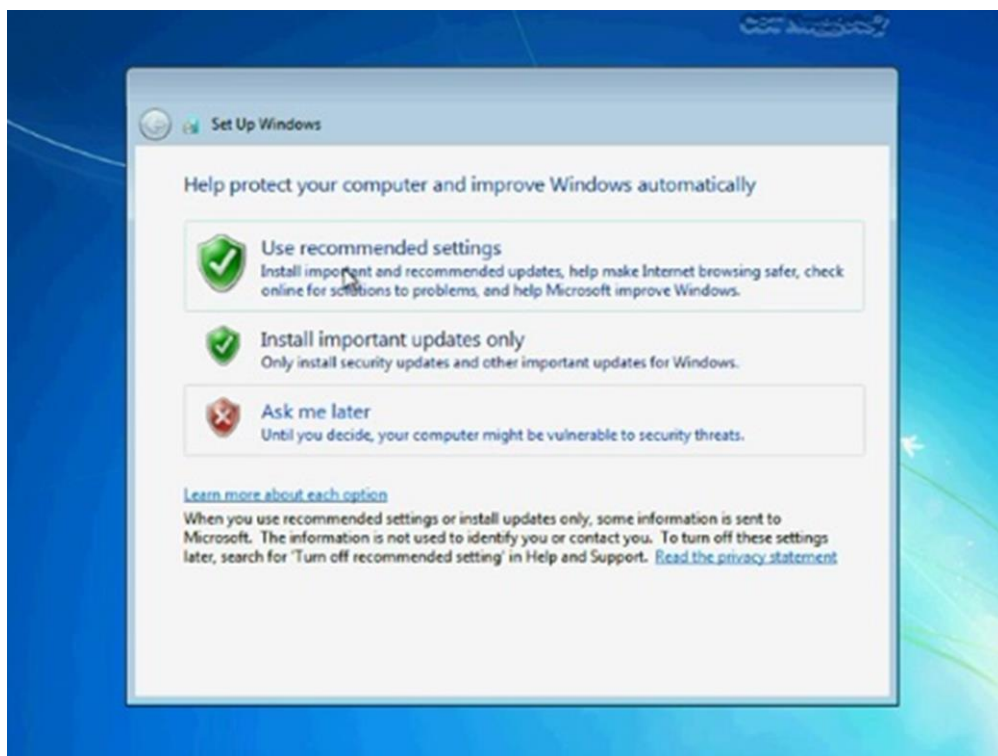
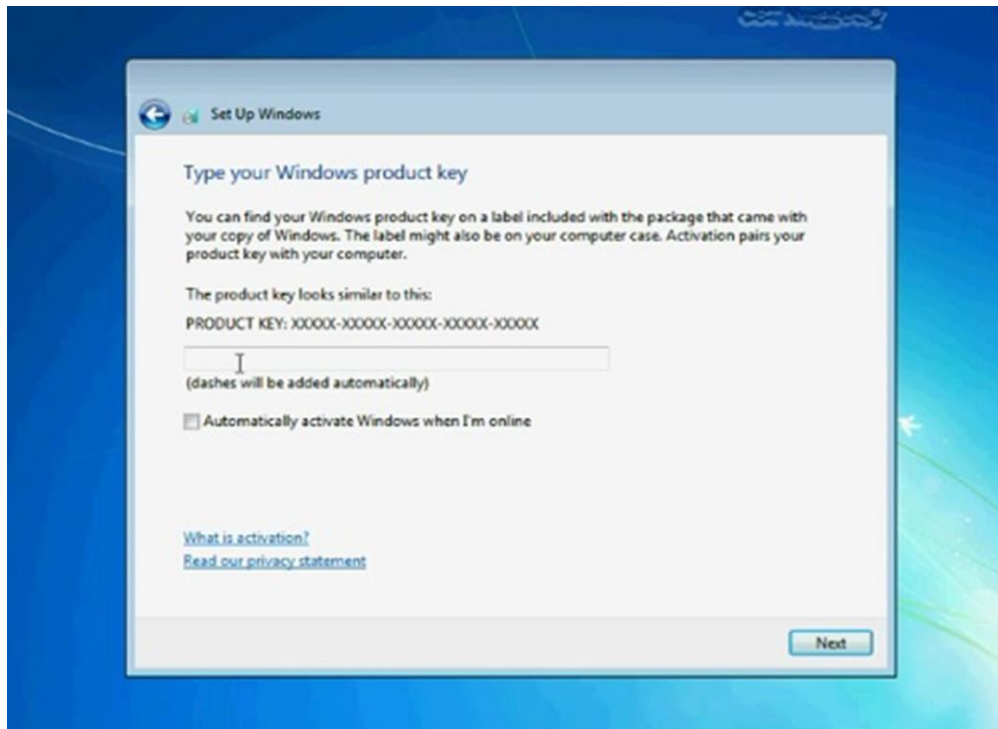
If you wish to format the drive before proceeding, choose “Drive options” from the screen above and you would get the options below, click format and click yes from the prompt that pops up. Click “Next” when all is ready.



Windows installation should restart after this step and your PC would restart several times during the process; you can remove the disk or USB flash once Windows restart. Do not

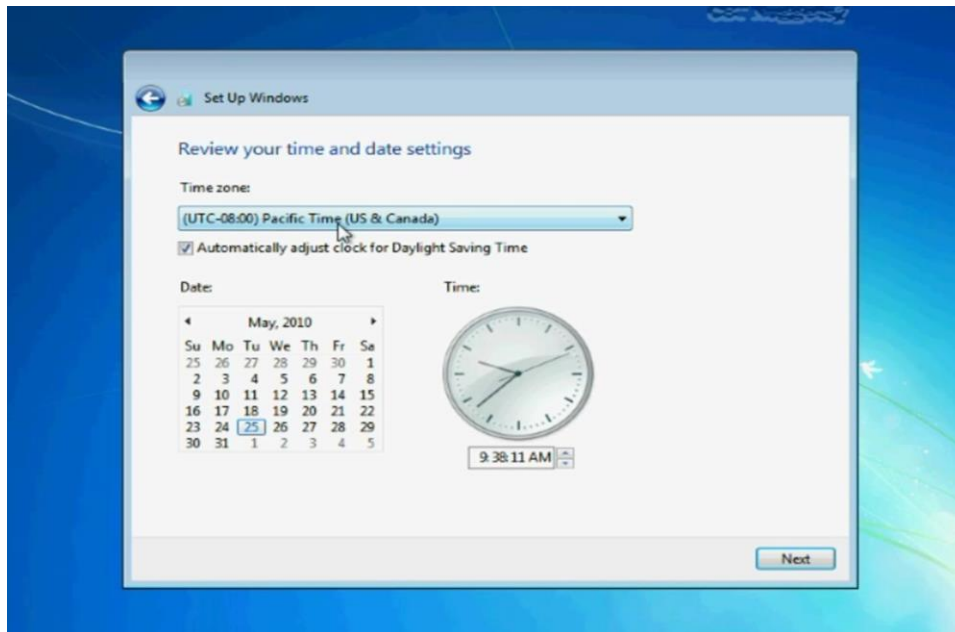
press any key again if you are prompted to press any key to boot from CD. Just sit back and watch the installation process.

11. Wait until installation is done and you get a screen prompting you to enter product key as shown below or the screen asking you to choose your desired setting. Choose “Recommended settings” and click next (you may not see the product key’s page if the installation media you are using has been pre-activated).



NB: If you do not have a product key now, you can choose to get it online once you connect to the internet or try Windows for 30 days (if it's Windows 7). You can't proceed without a product key in Windows XP and in Windows 8. The product key page would come earlier in the installation of Windows 8

12. Once you choose to use recommended settings, the next screen should be the one below asking for the time zone, choose "West Central Africa" from the Combo Box and click next.



Now Windows installation is ready!

There should be a screen asking for your password, supply it or just click next. Click Next for every other screen too until the Windows desktop is displayed.

That's all about Clean Installation! You never thought it was this simple? Yes it is, and that's all!

UPGRADING

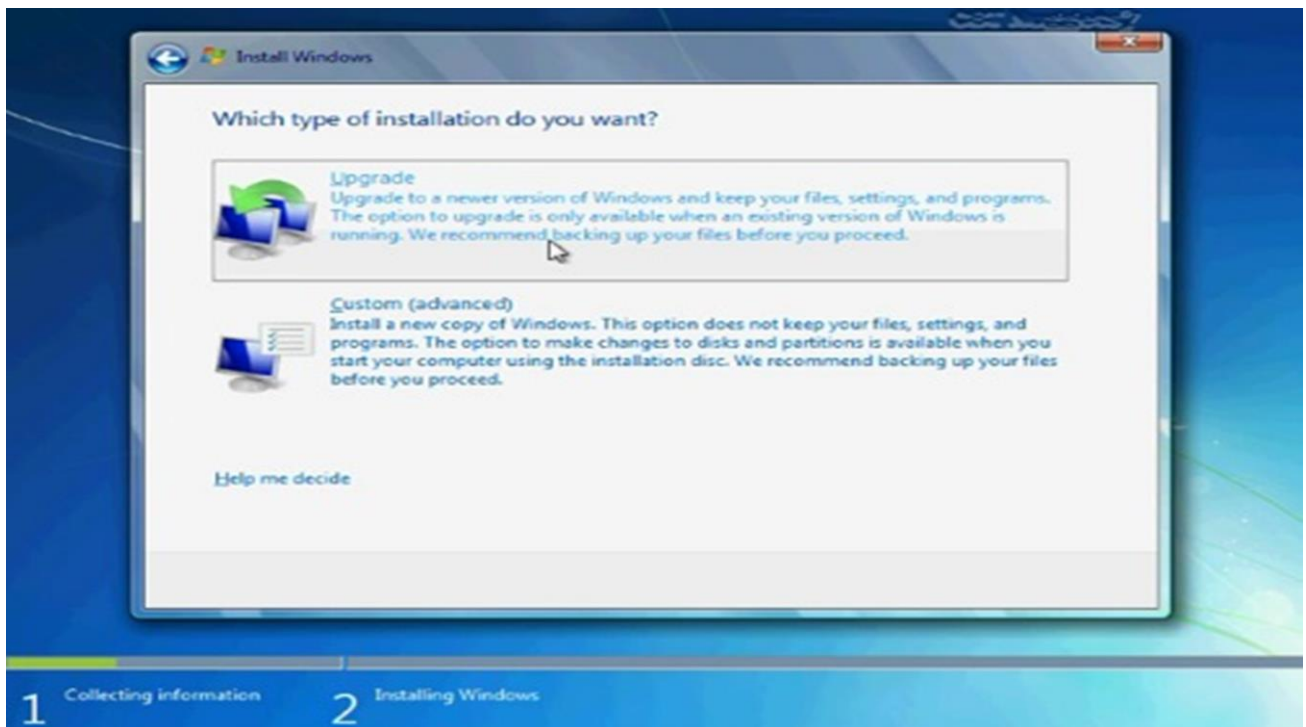
I hope you still remember upgrading rules, architecture and edition maintenance? OK let's go then!

If you can perform clean installation, then you can do upgrading. Mind you I'm talking about you and not your PC! I mean if you know how to perform clean installation, then you can upgrade a system successfully.

Unlike clean installation, upgrading does not require you to change any boot order, all you need is the installation media and you are good to go. In fact, you can even do upgrading with

Operating System files already copied to your system. Once the media is ready, follow the steps below:

Step 1: Insert the disk or USB into the PC, the auto run should launch automatically. If it does not or if you want to upgrade from a folder on your PC, then open the folder or disk or USB and double click setup (do not restart your PC). Walk through the installation like we did for clean installation. The only difference here is, once the screen below is displayed, you should choose upgrade and not custom.



Step 2: Click next and follow the prompts until installation is complete.

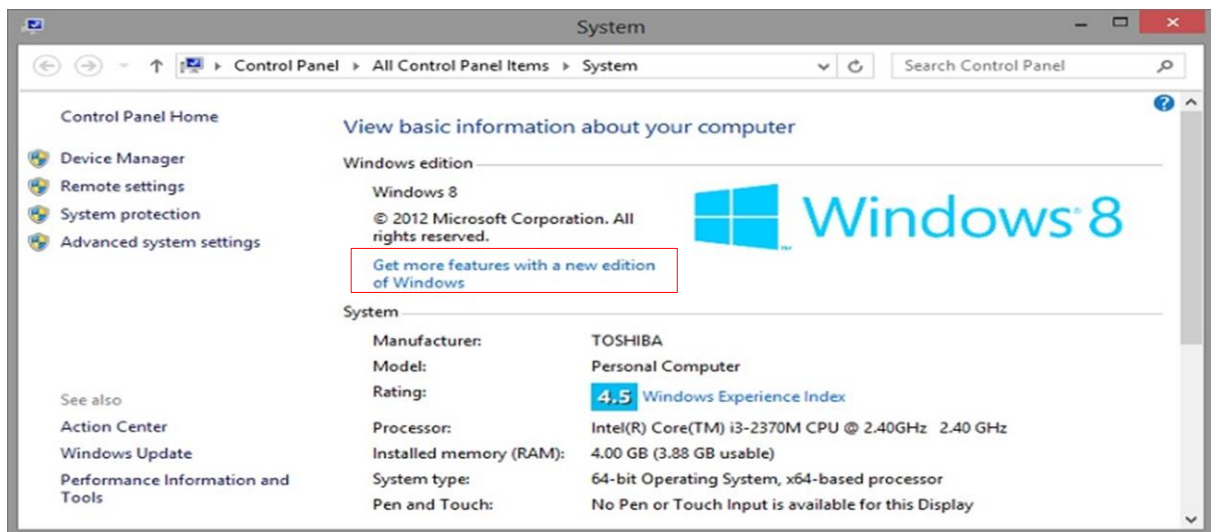
Now, you've got a new PC.

Important points to note:

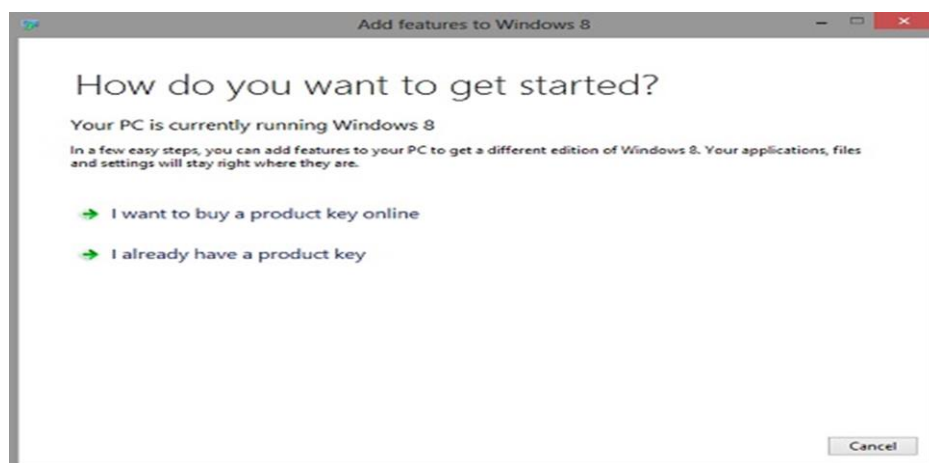
- You can upgrade from Windows XP directly to Windows 8 but only your personal files would be backed up, you would lose your settings and ALL software installation.
- You can upgrade from Windows Vista directly to Windows 8 but only your personal files and settings would be backed up, you would lose ALL software installation.
- Upgrading from Windows 7 to 8 is just pretty easy and you'd lose nothing except the Windows 7 interface (lol).

Edition Upgrade

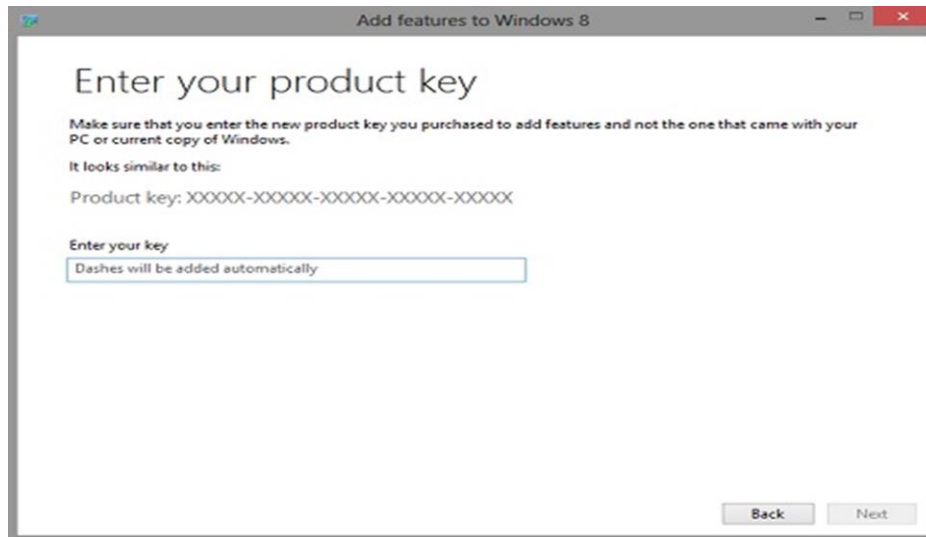
Another type of upgrading is to upgrade the same version of Windows from one edition to another e.g upgrading from Windows 7 Home premium to Windows 7 Ultimate. To do this upgrade, open system settings (open Windows explorer, right click on any white space and choose properties) and then click “Get more features with a new edition of Windows”. If your copy of Windows does not have this, then you are probably running the latest edition of that version of Windows else search for “Add features to Windows” and launch it.



Windows would now launch “Add features to Windows” setup, you would be given two different options on how to get the product key for the upgrade, one is to purchase the key online and the other is to enter a product key you have earlier purchased.



If you choose option one, you would be redirected online but when you choose option two Windows would give you an interface to supply your product key as shown below:



Supply the product key for the edition you wish to migrate to and click “Next” once Windows inform you that your key works; then agree to license terms and click “Finish” for the upgrade to begin. Upgrading might take up to hours so get your PC plugged into an AC power supply.

To check if the upgrade was successful, go back to system properties and see the type of Windows you are currently running, it should have changed from what it was to the edition that has the product key you supplied. You might lose your activation after a successful upgrade, so be prepared to re-activate Windows if you are prompted to do so.

MIGRATING

This is a little bit similar to clean installation like I mentioned earlier, in fact one of those things you would do during migration is clean installation. Note that two PCs are required for migration, one old PC and one new PC. Don’t get scared here, your PC is the old PC and is equally the new PC. How?

Before migration, you’ve got an old PC and after migration you’ve got a new PC. So when I talk about old PC, I’m referring to your PC that presently run the OS you wish to migrate away from and when I talk about a new PC, know that I’m referring to that same PC but with the new OS you migrated to. Migration has three major steps. I mean 3! And that’s all. The steps are as stated below:

Step 1: Prepare the old PC for migration

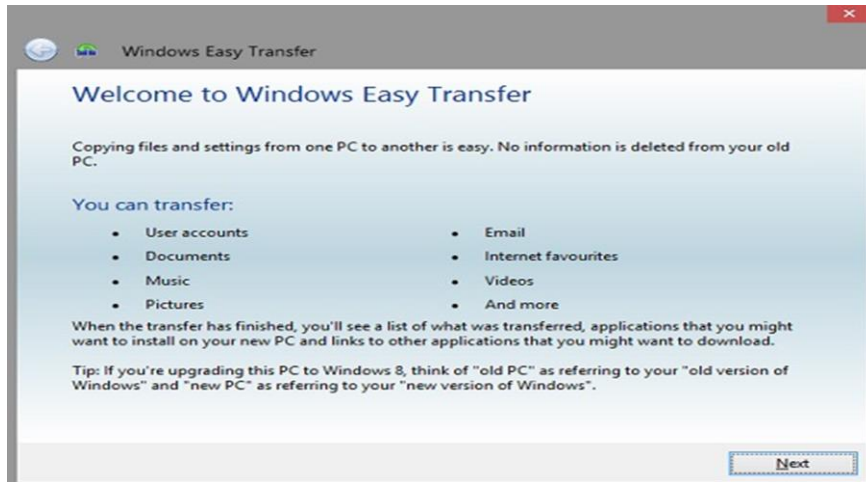
Step 2: Perform clean installation

Step 3: Carry out the necessary steps on the new PC.

That’s all! Now Step 1:

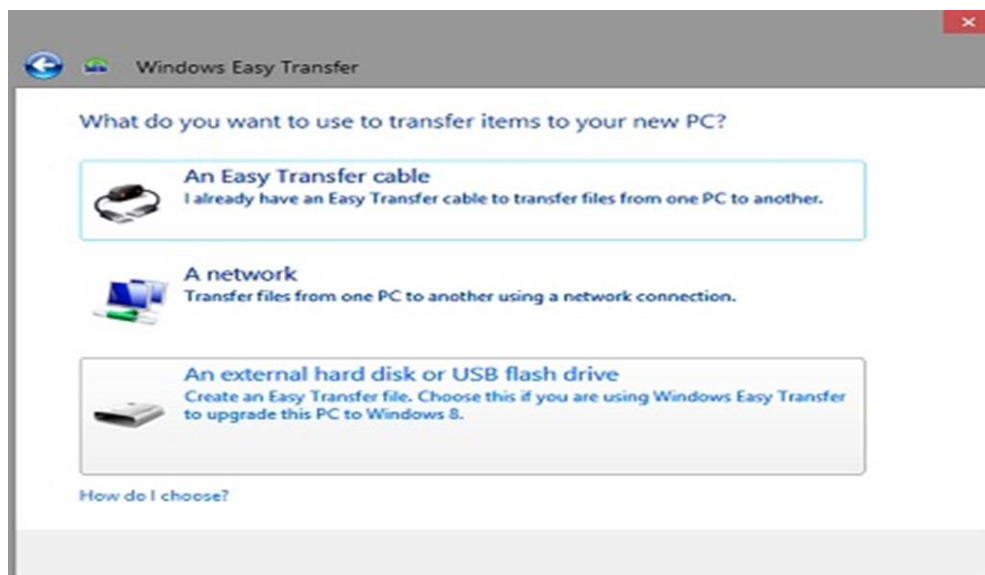
STEP 1:

If your old PC is running Windows Vista or newer versions of Windows, then press the Windows key and search for “Windows Easy Transfer” and launch it. You would have the screen below displayed:



Notice from this screen the types of files and settings Windows easy transfer would keep for you. The “And more” listed above includes the way you have designed your desktop, your wallpaper and some other things but not your personal files. That implies that all files outside Windows library (Documents, Music, Pictures, and Videos) would not be spared! They are all going (Actually you can add them up to the files to be spared; you would learn how to do that in a minute).

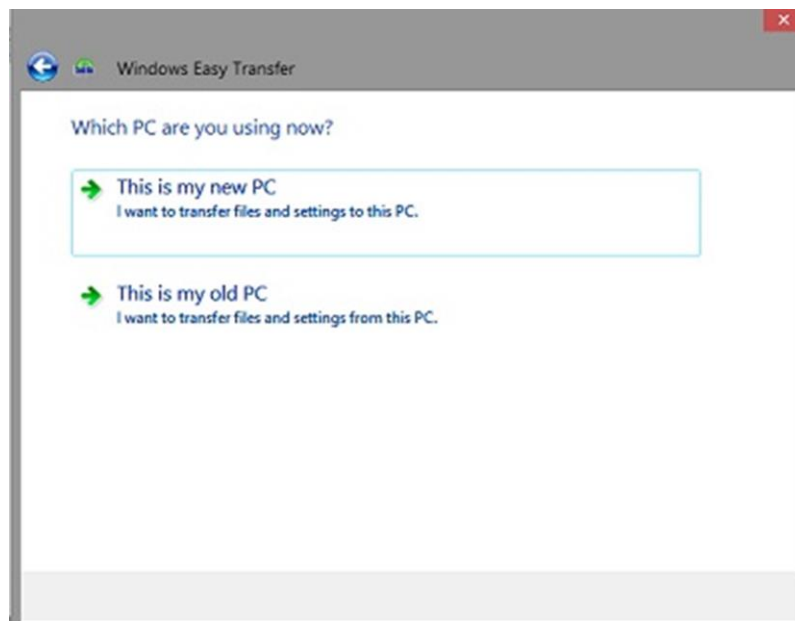
Click next to continue and you have this:



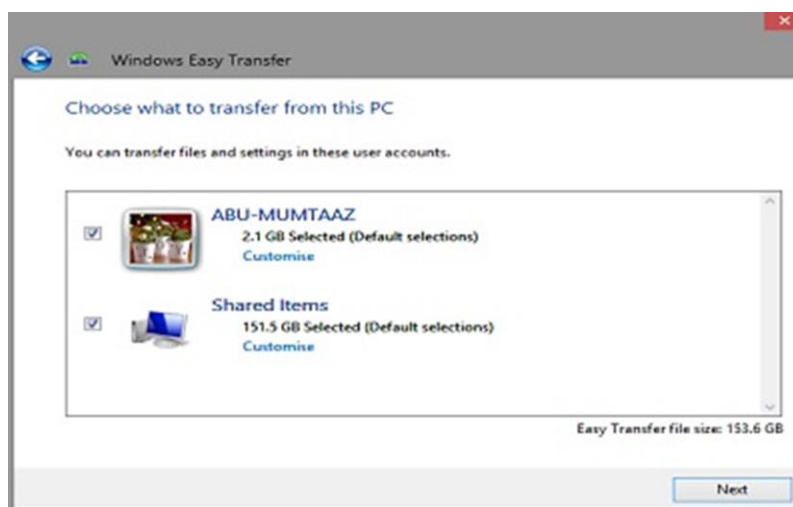
Choose “An external hard disk or USB flash drive”.

Let me point out something from here; migration requires that you have an external location or another partition on your PC where you are going to move some “Windows Easy Transfer” files to, and the method of transfer is what we’ve just chosen from the screen above.

Once you choose that option, the next screen would be the one displayed below:



Do you know the option to choose from here? Did you just say Old PC? Yes, you got it right. And I hope you still remember the reason for that? That is because your PC is currently running the OS you wish to migrate away from, so it's still old. Once you choose "This is my old PC", you would then see a screen similar to the one below:



This screen tells you all that would be transferred and the size of the files. If you wish to add some more folders, then choose "Customise" and select the folders containing the desired files from the screen that pops up. Once you click "next", the following screen prompts you to supply your system's password twice, supply that and click "Save", a screen then pops up asking you to choose the directory you wish to save the file to. Be sure to choose the external drive or the partition you wish to use (make sure you do not choose the same partition we are going to install Windows into) if you have another hard drive partition on your PC. Once this is done, proceed to step 2.

STEP 2:

Perform a clean installation on your PC with the OS you wish to migrate to, exactly as described under clean installation above (make sure you choose the correct drive when you are prompted during installation. Don't choose the same drive where you saved the Windows Easy Transfer file), and then proceed to step 3.

STEP 3:

Perform the required steps on your new PC! What are the required steps?

Launch "Windows Easy Transfer" again and follow the steps as shown above but remember that when the screen appears asking you which PC is this, choose "This is my new PC" and then click "next" to continue, choose the location of the Windows easy transfer file and watch Windows transform your PC and make it look the same way the old one did.

Wait a minute there! What if your old PC does not have Windows Easy Transfer? Maybe you are using Windows XP; what you would do is to get the files of the OS you wish to migrate to, right click on the disk or USB containing the files and choose open, then find the folder named "Support", open it and find the folder named "MIGWIZ" in the support folder, from the MIGWIZ folder find the application named "MIGWIZ" and double click it to run, now you've got the Windows easy transfer interface.

NB: Make sure the OS you've got have the same architecture with the one your PC is currently running, else the MIGWIZ setup won't run.

Now that you know how to perform clean installation, upgrade and migration, I've got a question for you (since you have not asked since!). Can you run more than one operating system on a single PC? Let me help you out; the answer is YES, it's called **Dual Booting** if you want to run two OS and **Multi Booting** if you want more than two. But how is that possible?

You actually have two options to multi-boot (I hope you know dual booting can also be referred to as multi-booting.... Just grammar things!); one is to use a software called "Virtual Machine" and the other is to install them separately on the same PC.

If you wish to use virtual machine, then download one from the internet and install it on your PC or use Hyper-V (if you are running Windows 8. Hyper-V would be discussed shortly, just relax), once you have it installed you can run multiple OS by launching your virtual machine and clicking "Create new Virtual Machine" or similar message to install a new OS **into** the virtual machine.

Let me give an example here to make the picture clearer: You are currently running Windows 7 but you equally wish to run Windows XP, Windows 8 and Ubuntu Linux, then install virtual machine on your Windows 7 PC, get the disk of the other OS or there ISO files, launch the virtual machine and choose to create a new machine, you would see a prompt asking you to choose the

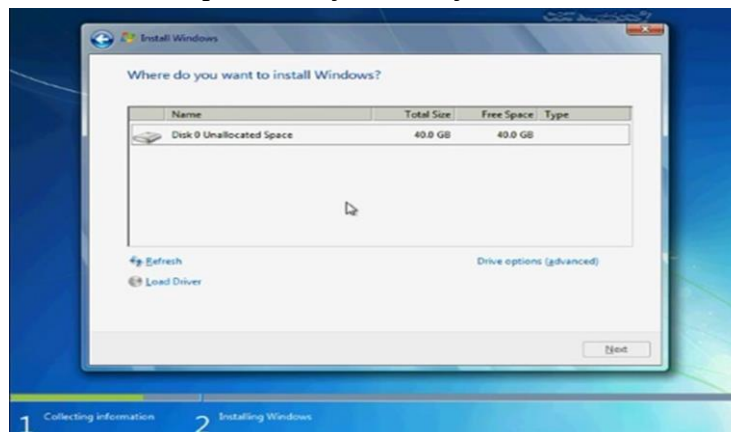
location of the OS, choose your disk drive or the location of the ISO file and continue with installation, you would get series of screens exactly the same or very similar to the ones described under clean installation. You would repeat the same process for all OS, now you are running 4 OS on a single PC but you can't launch any of the remaining three unless you launch Windows 7 first, then launch the virtual machine and choose "Start-up Virtual Machine" from within the virtual machine to start up (boot) the OS you wish to launch.

The only problem here is, the more OS you install inside your virtual machine the worse your system performance becomes as the virtual machine takes up part of your hard disk for each of the OS and part of your RAM too!

The second method to multi boot is the best option technically speaking and that is installing each OS separately. How do you go about that? Very simple, let's go:

Step 1: Partition your hard disk to have a number of partition equalling the number of OS you wish to run, (don't make the mistake of counting the recovery partition and other system partition as part of the partition you have) and make sure you do not convert your hard disk to a dynamic disk during partitioning because dynamic volumes can't install Windows. What are dynamic volumes? How do you make hard disk partition? You would learn all this and many more under DISK MANAGEMENT.

Step 2: Perform a clean installation with the OS of choice but make sure when you get to the screen where you are to choose the desired partition (the screen is shown below for remembrance purpose), choose one of the new partitions you have just created and click next.



Do the same for all OS you wish to install and every time, you get one more OS on your PC. After a successful installation of all the OS, once your PC start you would have a screen prompting you to choose the OS you wish to start among the varieties of OSs you have on your PC.

One advantage of this method is; whichever OS you start, you have the same hardware configuration as every other one and you would be using your PC with the best possible performance. Technically, the only disadvantage is; you have to restart your PC every time to change your OS.

One rule of thumb to master under this method is that: “the old OS **MUST** be installed first”, therefore, you have to install the OS in order of release e.g Windows XP before Windows 7 and Windows 7 before Windows 8. Se-fi-ni!!!

Hyper-V

Hyper-V is the virtual machine Microsoft shipped with Windows 8. Are you surprised? It's surprising, Windows 8 came with a whole virtual machine and you might never notice it (you'd know why in a minute).

Hyper-V is as good as any virtual machine software out there, in fact, it's better than most of them as you'd see in its configuration. This is an implication that you can actually run as many as possible Windows on your Windows 8 PC without any stress (provided you'd love to use virtual machine), Microsoft has saved you the extra effort you'd need to download a third-party virtual machine which might even come with a troublesome driver that can make your OS unstable.

Before you start jubilations though, Hyper-V has got requirements before you can even imagine running it on your PC. The requirements are as given below:

1. You must be running Windows 8 professional or enterprise edition (it's not available in standard edition of Windows 8).
2. You must be running 64-bit operating system (i.e 64-bit Windows 8 professional or enterprise).
3. Your PC must have at least 4GB of RAM
4. Your PC must support SLAT (Secondary Level Address Translation) hardware virtualization.

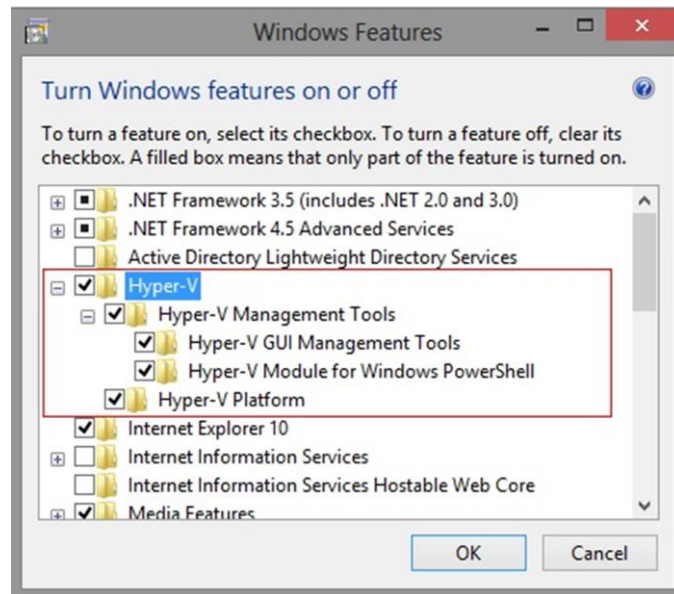
It is possible that your PC supports SLAT hardware virtualization but is not turned on by default. For you to use Hyper-V, you'd need to turn on this feature. To turn it on, restart your computer and edit the computer's BIOS settings. Look for options indicating virtualization, SLAT, Extended Page Tables (EPT), or Rapid Virtualization Indexing (RVI) and click to enable it.

Hyper-V is turned off in Windows by default, which was why I said most users would never notice it. To use it, you'd have to install it.

How would you install Hyper-V?

1. Open control panel.
2. Click on Programs.
3. Click on “Turn Windows features on or off” under “Program and Features”.
4. Check the box beside Hyper-V from the list of features and make sure it's marked. If the mark is in form of a square, then some of Hyper-V's feature has not been selected, expand

it by clicking the plus sign to see which feature is not selected and select all. If any of the features is disabled, then your PC probably does not support SLAT hardware virtualization.



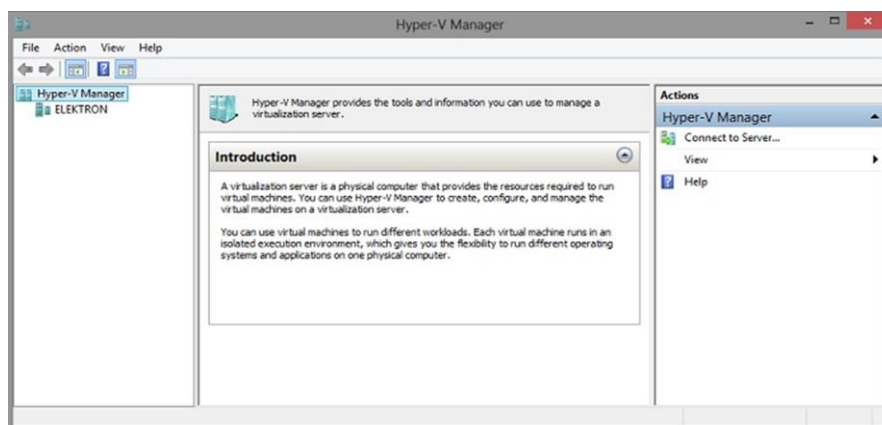
Click on Ok to proceed and watch Windows configure your PC. Windows would search for the required files on your PC and if it's not found (this happens only on rare occasions), Windows would attempt to download them from the internet.

Your PC should restart during the configuration process.

5. Hyper-V should now be installed successfully.

Now that Hyper-V is installed, you should create a desktop icon to make accessing it easier. To do this, go to your start screen (press the Windows logo key), scroll to the last end to see Hyper-V Manager, right click on Hyper-V Manager and choose open file location; at the file location, right click on Hyper-V Manager and choose "send to" and choose desktop.

Now double click on Hyper-V Manager on your desktop to launch it. You should have a screen similar to the one below:



If your computer name is not displayed in the left pane, then your PC does not meet one of the requirements of Hyper-V. Sorry you were not informed about that before now.

Choose Action from the menu bar and choose connect to server or double click your computer name from the left pane to start communication. You may be presented with a prompt asking you which virtualization server you want to connect to, choose local computer and then click OK.

Creating Switches

In networking terms, a switch is a networking device for connecting two or more computers together. Hyper-V's switch does this exact job without requiring you to purchase any hardware.

In Hyper-V, your PC (running Hyper-V) is called the host, while every other operating system you install in Hyper-V are the guest.

Without switches, guests would not be able to communicate with one another or with the internet or with their host. Therefore, you should create a switch before creating the first virtual machine (installing a new OS in Hyper-V is referred to as creating a new virtual machine).

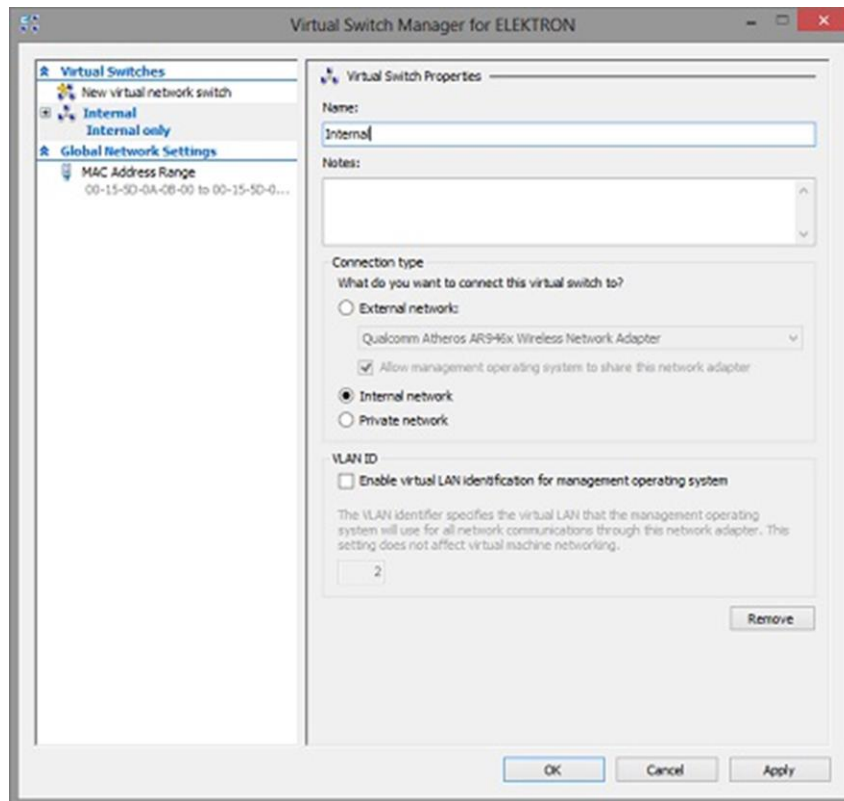
There are three types of virtual switches in Hyper-V:

- i. **Private:** With a private switch, guests can communicate with each other, but they can't communicate with the host or the internet. Private switches are ideal if you need to isolate the guest from the host and internet especially for the purpose of testing malicious softwares.
- ii. **Internal:** Guests can communicate with each other and the host but cannot communicate on the internet. With this type of switch, guests can infect the host with malwares.
- iii. **External:** Guests can communicate with each other as well as the host and the internet. This is important if your VM needs to download updates or drivers from the Internet.

You'd create one of these switches following the procedures below and then connect your VMs to it when creating VMs.

To create your first switch, follow the steps below:

1. From Hyper-V Manager, click on Action (in the menu bar) and click on virtual switch manager.
2. Choose the type of switch you would want to create from the list presented and then click Create virtual switch. You'd be presented with a screen to give the switch a name as shown below. Give it your desired name and click OK.

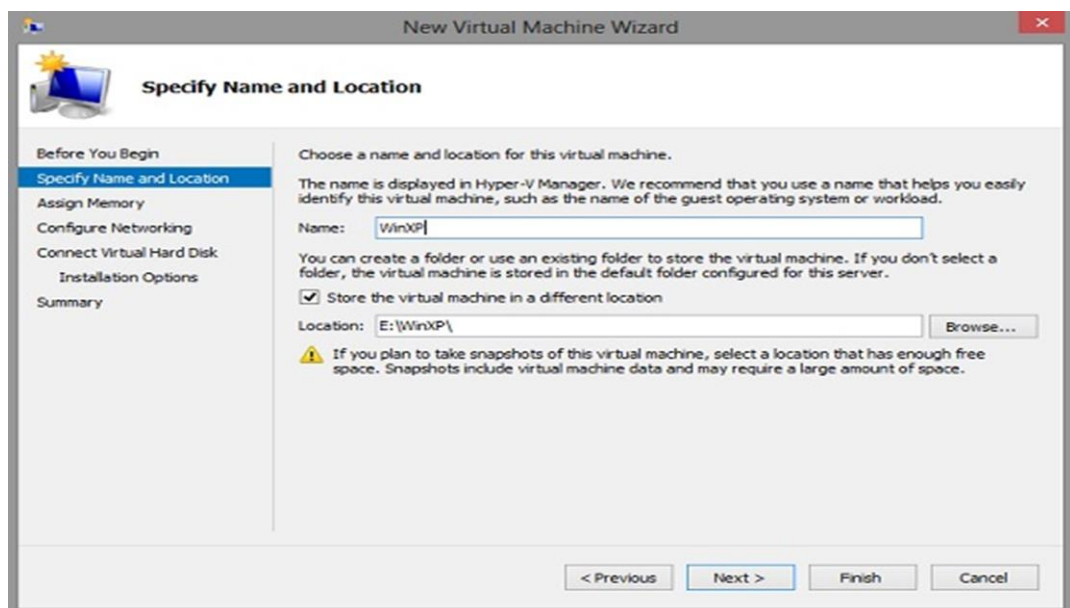


Now, let's create our first virtual machine.

For the purpose of this example, we would be installing Windows XP.

To do this, follow the steps below:

1. Launch Hyper-V Manager
2. Choose Action from the menu bar, choose New and choose Virtual Machine
3. The first page which is "Before you begin" would be displayed, just click on Next.
4. The next page should be similar to the one shown below where you'd be required to name your virtual machine and then specify the location where it would be installed.

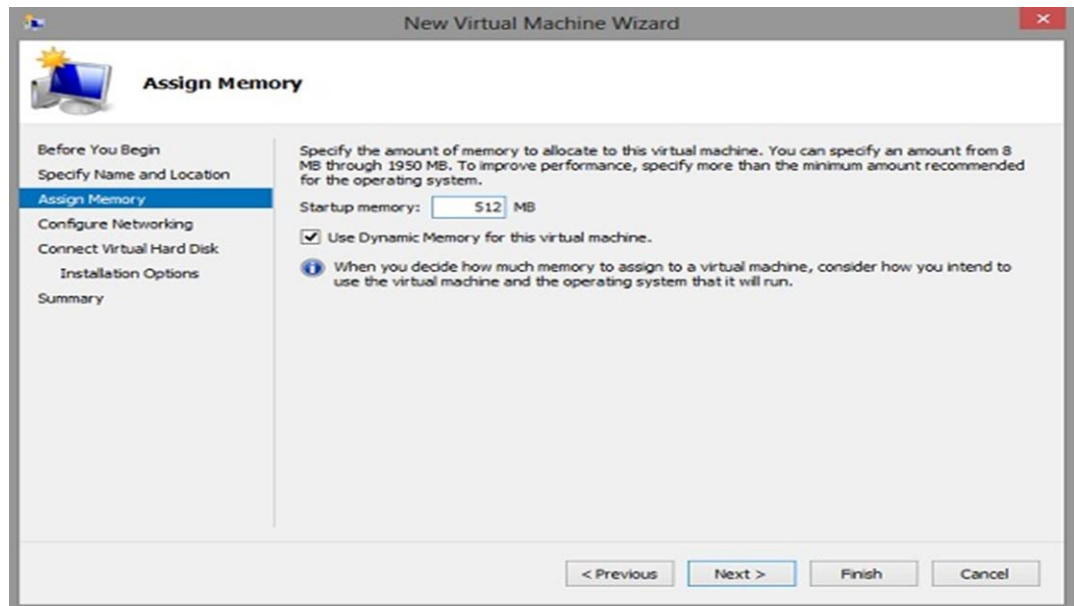


I have given the name WinXP to my VM and I've equally created a folder in drive E (which I also named WinXP) where the new VM would be stored. I would talk about the implication of this later.

If the box to type your location is not enabled, check the box "Store the virtual machine in a different location" and then click on Browse to choose the location of the folder.

Click Next to continue.

5. The next screen is the screen where you would be required to choose the amount of memory to be assigned to your new VM which is referred to as the start-up memory.



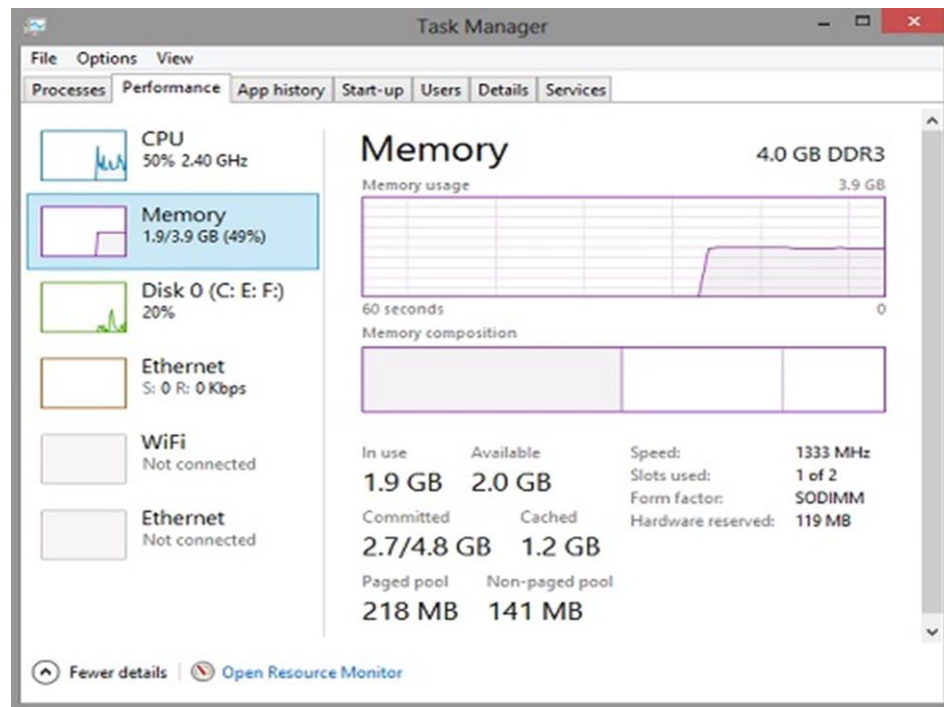
The default value specified in that box is 512MB and I've left it as 512MB. Why?

Because every OS has got a minimum amount of memory (RAM) it should be allocated for optimum performance. The table below gives an analysis of the size of memory required for the common types of OS:

Operating system	Minimum memory	Recommended memory
Windows 98	16 megabytes (MB)	24 MB
Windows XP	64 MB	128MB
Windows Vista	512 MB	1 GB
Windows 7	1 GB (32-bit) or 2 GB (64-bit)	1 GB (32-bit) or 2 GB (64-bit)
Windows 8	1 GB (32-bit) or 2 GB (64-bit)	1 GB (32-bit) or 2 GB (64-bit)
Ubuntu Linux	64 MB	512 MB (with a desktop environment)
Linux Mint	512 MB	2 GB
Arch Linux	192 MB	512 MB (with a desktop environment)
Fedora	512 MB	1 GB

Specifying more memory (than the recommended) would improve the VM's performance but reduce the performance of the host PC. That implies that you have to be very careful when making the decision to allocate memory to your VM, you have to understand the size of memory your PC is currently using and the amount that's free.

To see the amount of free memory in your PC, start task manager (click more options if you need to do so), click on the performance tab and click on memory as shown below:

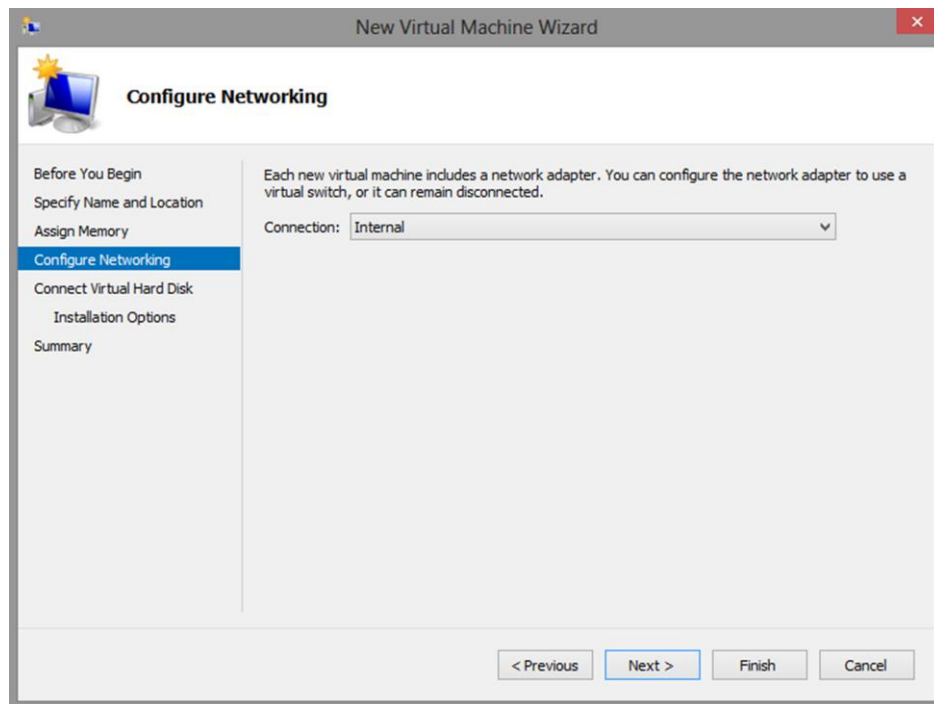


From the above example, the PC is presently making use of 1.9GB of memory and 2.0GB of memory is free and as I've noticed, this is always the value when my PC is using memory the most. That implies that I can allocate up to a maximum of 2.0GB to my VM without affecting my PC's performance. I am allocating lower anyway, since Windows XP does not require up to that.

Notice from the Hyper-V settings that I checked the box to use dynamic memory; this is because operating systems usually would not exhaust all its memory while performing its task, i.e my Windows XP after start-up might not require up to the recommended 128MB of RAM (and I even allocated 512MB), so making it dynamic would enable the host (my PC) take charge of the remaining RAM that XP isn't making use of. This is a way of being economical with your memory. This would enable you run more VM in less memory.

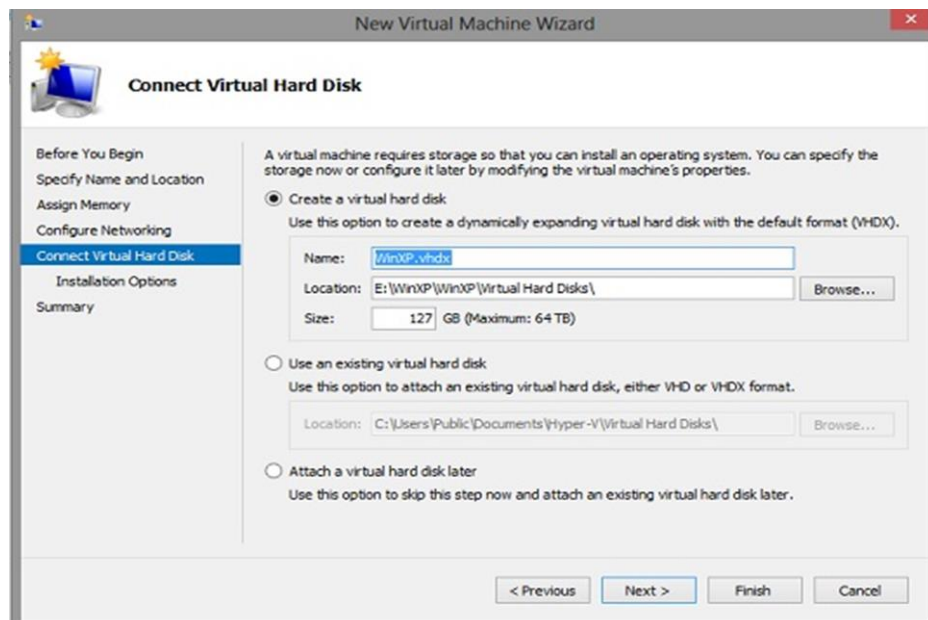
Dynamic memory can also increase the memory allocated to a VM if it needs more after start-up. Though after creating the VM, you can change the values of the minimum and maximum memory the VM can be allocated from its setting.

- The next screen is where you'd select the connection you wish to use (the virtual switches we discussed earlier).



I simply chose the internal switch I created earlier and clicked Next.

- The next screen is where you'd choose the virtual hard disk to be allocated to your VM. There are three options you can choose from (as shown below). You can allow Hyper-V create a new virtual hard disk in the path you specified earlier or choose a virtual hard disk you had created or choose to attach a virtual hard disk later.



Now, the question is; what is virtual hard disk?

Virtual hard disk is the hard disk your VM would use (just like your PC also has a hard disk). It is saved as a file on your PC's hard disk with a .vhd or .vhdx extension. Once the

VM is created, it would continue to save its files in that virtual hard disk which you would see as a file.

What's the difference between .vhd and .vhdx?

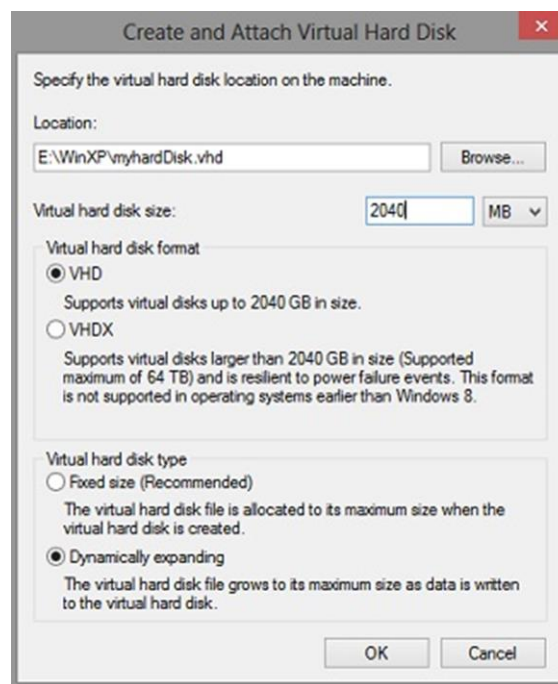
.vhd can save files up to a maximum of 2TB while .vhdx can save file up to a maximum of 64TB, .vhdx can also protect the guest OS against corruption when there is power loss from the host. Now you see the difference!

There are also two other options you have for creating the virtual hard disk. Your virtual hard disk can either have a fixed or dynamic size. If it's fixed, it would eat up the size of memory you specify from the partition you place it on your hard drive but if it's dynamic, it would continue to expand as files grow until the maximum value is reached (2TB for .vhd or 64TB for .vhdx) or the partition is filled up. That is why you should keep your virtual hard disk in a partition with substantial amount of memory.

Now, back to the three options:

If you choose option 1; for Hyper-V to create the virtual hard disk itself, it would create the virtual hard disk with the same name you gave your virtual machine with the appropriate extension in a folder having the same name as your VM inside the folder you specified earlier. i.e. **E:\WinXP\WinXP\WinXP.vhdx** in the case of this example.

If you'd love to create the virtual hard disk yourself and then attach it to your VM, then open disk management console (search for **diskmgmt.msc** from the start screen), click on Action from the menu bar and choose "create vhd". Select the appropriate location (by clicking on Browse), size and type as shown below:



I have decided to make it dynamic and I named it “myhardDisk”.

You can then select “use an existing virtual hard disk” in Hyper-V and then browse to the location you saved the hard disk to choose it.

Note that if you choose to attach the virtual hard disk yourself, you’d not be given the option to install an operating system now but you can do that later.

For convenience purpose, I have decided to ask Hyper-V to create the virtual hard disk so that I can install the OS straight away.

8. The next step is to install the desired Windows XP. I have chosen to install from an ISO image because I have one. If you don’t, you can install from a disk.

Choose the correct location of the Operating System and then click Next.

9. Click finish from the next screen to create the virtual machine.
10. Once the virtual machine is created, right click on it from Hyper-V manager and click connect to install it.
11. Follow the steps 1-10 above to install more operating system (create more virtual machine) into Hyper-V.

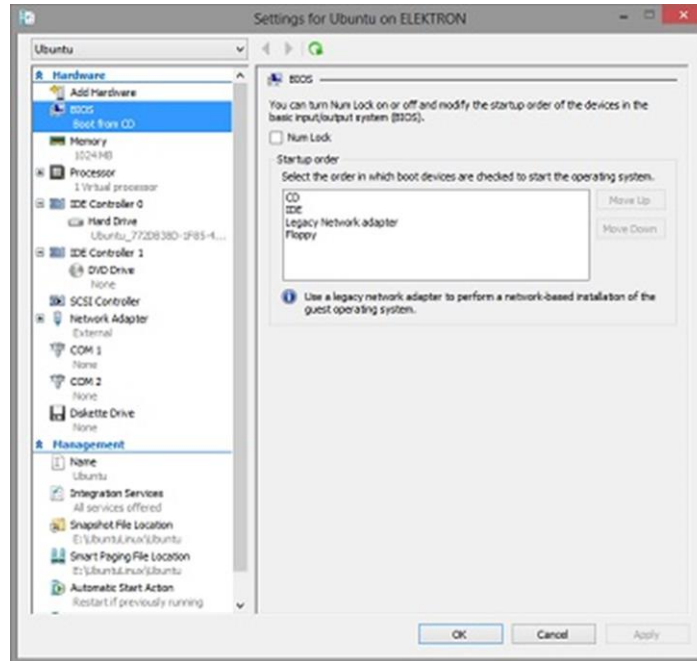
After creating VMs, you can always start them from Hyper-V Manager or Hyper- V Virtual Machine in the start screen. You can also create a desktop icon for Hyper-V Virtual Machine.

Hyper-V settings

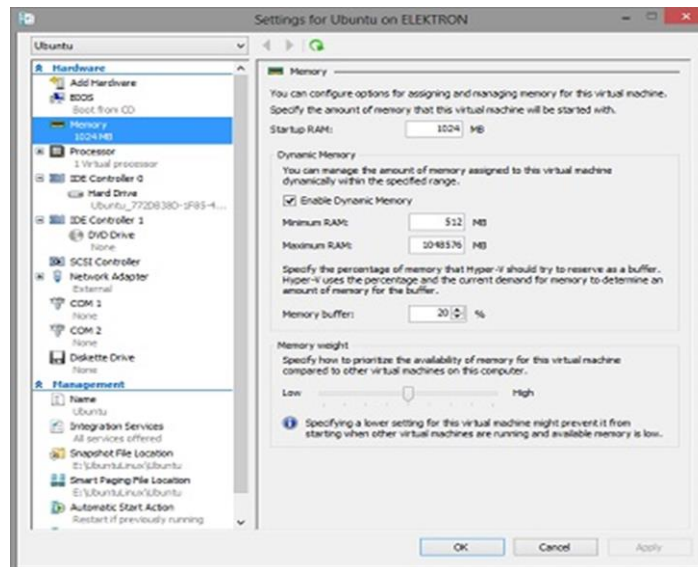
You can change virtual machine’s settings from within Hyper-V Manager or the virtual machine itself. To view the settings page, open Hyper-V manager, choose action from the menu bar and click Settings or right click on the virtual machine and choose settings.

From this interface you can then configure a lot of options as you’d do for a normal PC. Note that when the virtual machine is running, you’d not be able to modify most of the settings.

One of the interesting settings is the boot order of the virtual machine; just as your PC (host) also have a boot order. From the left pane in the setting screen, click on BIOS to view and modify the boot order (don’t alter the boot order if you do not really need to do so).



You can also alter the amount of memory allocated to your virtual machine by clicking on memory.



All other settings too are straight forward and once you choose them, Windows would give you an idea of what they do and how to modify them. Just make sure you know what you are modifying before you start modifying nonsense! That was just a joke anyway.

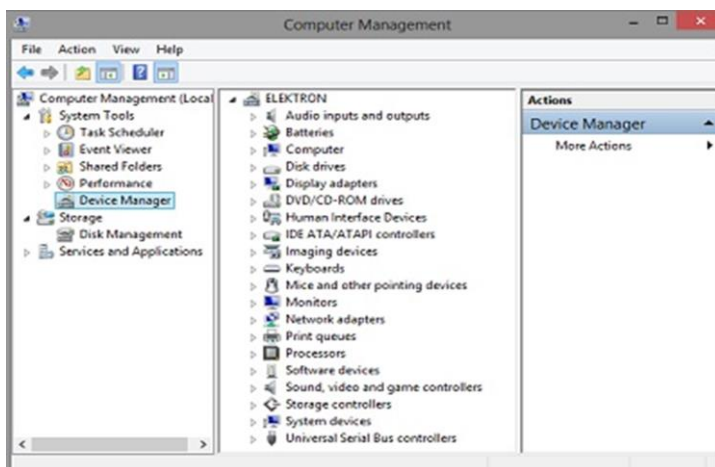
DEVICE MANAGEMENT

This Chapter covers:

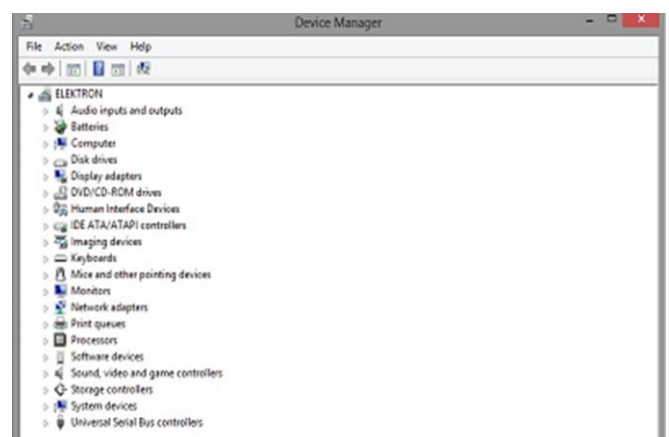
- ✚ Driver management
- ✚ Viewing and resolving driver problems
- ✚ Driver's signature
- ✚ Driver error codes and solution

All hardware attached to your PC are referred to as devices, this includes your keyboard, mouse, monitor, processor, CD/DVD drive, removable drives e.g USB etc. Each of these hardware require a software to “attach” them correctly to the Operating System (OS) so that they can perform there functions effectively. Such software which serves as a link between the devices and the OS is known as a driver, without a driver, hardwares cannot function as expected and as such every hardware has its own driver. Removable drives today are often plug and play (PnP) because they come with drivers which automatically get detected by Windows and get installed automatically.

Windows has a special application for device management which comes with every Windows OS. This app is known as the device manager and can be accessed either through its own interface i.e. as “Device Manager” interface or through “Computer Management” interface. To start device manager, search for “**devmgmt.msc**” or go to System properties and click “Device Manager” and to start computer management, search for “**compmgmt.msc**” or go to my computer, right click on the computer icon in the left pane and choose manage. Once computer management is up, click “Device Manager” from the left pane. Both interfaces are shown below:



Computer Management



Device Manager

The device manager has four different views which are:

- Devices by type
- Devices by connection
- Resources by type
- Resources by connection

You can switch between these views by clicking on view from the menu bar and selecting the appropriate view. The default view is “Devices by type”. Windows also hide some drivers by default (these include drivers that make Windows function properly), these drivers can be seen by selecting “show hidden devices” from the view menu.

Uses of device manager:

- Viewing devices
- Viewing drivers
- Managing drivers
- Resolving resource conflict among drivers

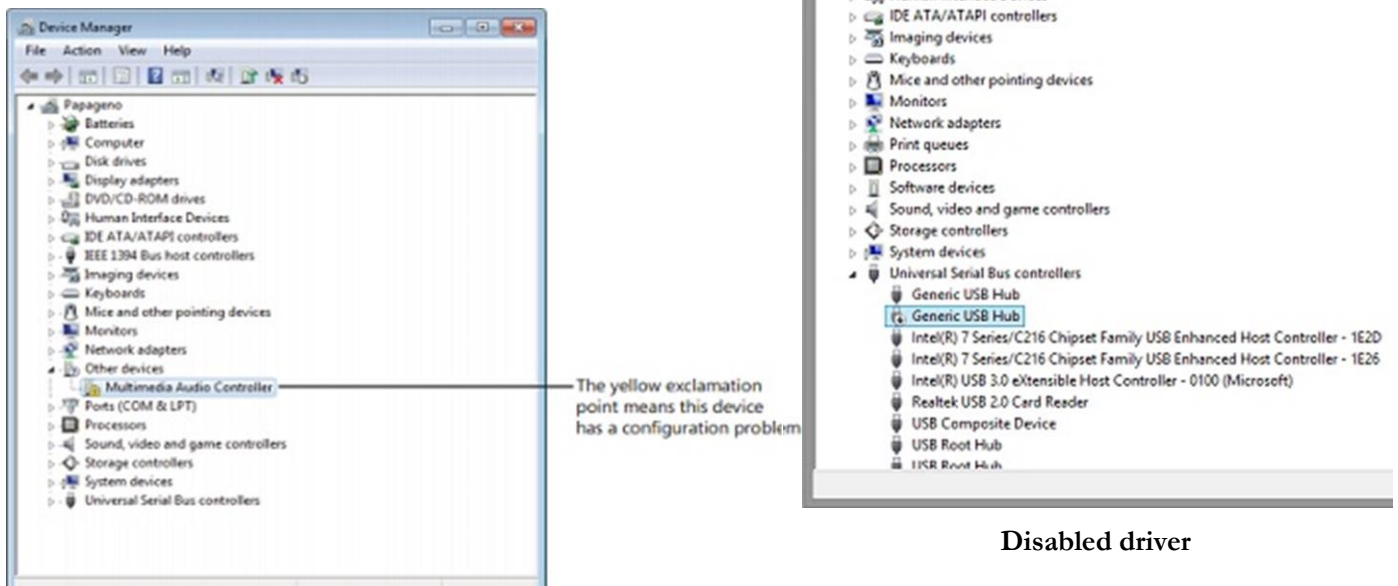
When you choose to view “Devices by type”, Windows gives you a list of devices presently attached to your PC, each of these devices has a small triangle beside it which you can click to view the drivers attached it.

One major thing we need to learn here is how to manage drivers (Note that device management in Windows is an extensive topic to discuss, therefore the most important and easiest things to do with device manager are what this book would discuss).

Why do I need to manage drivers? This question has got answers that can result into a whole book but the major reasons are that your drivers might be disabled, you might need to install a driver, you might need to check for resource conflict among drivers, you might need to check the certificate status of a driver, you might need to roll back a driver, you might need to add legacy devices and many more. In fact, it might interest you to know that, the fact that your web cam is not working, your Windows 7 does not have a sleep option enabled, you can't connect to a network, you can't change the volume level on your PC, you can't change the brightness status of your PC and many more might be as a result of a faulty driver; now you see why you should learn this? Let's get into business (I'd try to organize and explain this as simple as possible, God helping me..... I'm ready if you are ready!).

Once you open device manger interface, Windows gives you a view that solves almost every problem you might have, how do I mean? One thing you should look out for is a driver that has an arrow down or a yellow exclamation mark beside its icon, all such drivers would

be shown by default and you do not need to start checking each device driver to see this. What do those icons mean? Here is the picture of what I'm talking about:

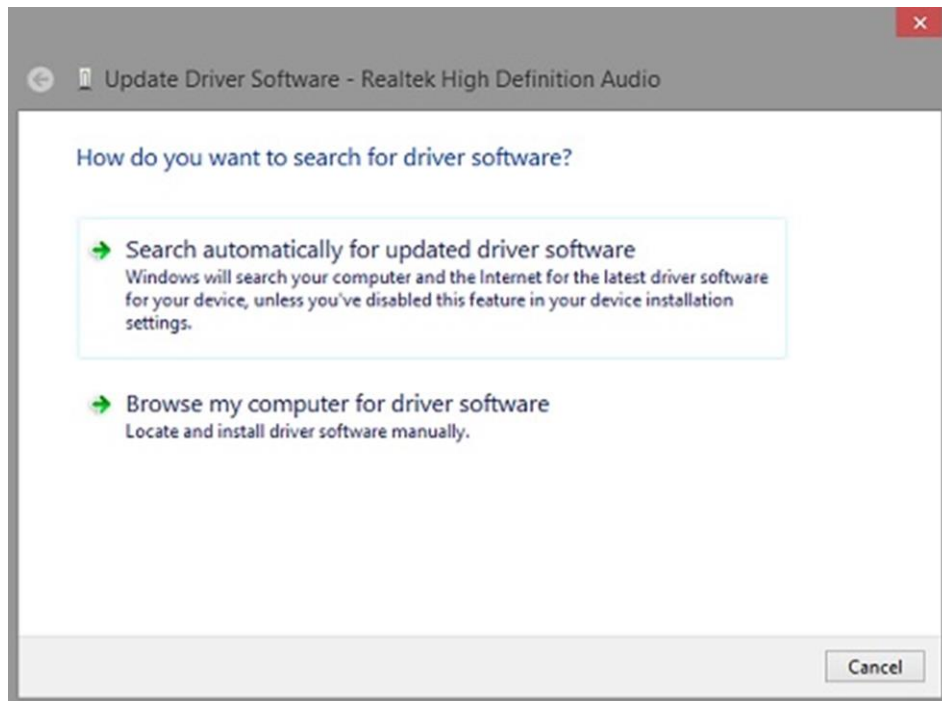


Get this straight! An arrow down indicate a disabled driver, and a yellow exclamation indicates a driver that require an update (update in driver management might mean to find the latest update to the driver and might also mean to install the driver. Don't worry yourself about that though!), because it has a configuration problem, often times, drivers with configuration problem are grouped together as "Other devices". What should you do to these drivers? When you right click on most drivers, you have five options:

- Update driver software
- Disable
- Uninstall
- Scan for hardware changes
- Properties

If the driver has got configuration problems, the option to go for is "update driver software", once you choose this, you get a screen asking you if you either want to "search automatically for updated driver software" or "browse my computer for driver software" as shown below. To use the first option, you have to be connected to the internet as Windows search your PC first for the driver and then to the internet if there is no updated driver on your PC, this option is the option of choice if you don't even know which driver to install because Windows would probably find one online, download it and install it for you. If you are very sure about what you want on the other hand, you might go to the website of the device

manufacturer to download the driver yourself (e.g. maybe you are having problem with your web cam manufactured by HP, you can go to HP's website to download the correct driver), then choose option two and choose the location where you saved the driver on your PC for Windows to install it for you.



Mind you, when Windows download drivers for you, it downloads Microsoft's version of that driver and not the manufacturer's version except in cases when Microsoft does not have the driver but when you download drivers yourself, you might download the manufacturer's version. The problem with this is stability because manufacturer's versions are often times unstable; you want to ask me why? This is because Microsoft made Windows, and therefore understands what the OS require to communicate with a particular hardware. Secondly, there is what is called signatures in driver technology, Microsoft has a lab known as WHQL (Window Hardware Quality Lab) where drivers are tested for stability with Windows OS and then signed, those drivers signed by Microsoft are very stable but the drivers you would download from the manufacturer's website MIGHT be one signed by a third party which might not even be recognized by Windows, and that's where the problem lies.

Before I discuss driver's signature, let me give you this hint: You can easily purchase software known as "driver pack" from software vendors which you can just run on your PC and would help you scan your PC for missing or out-dated drivers and then update the driver for you without any stress, and that's all!

Now to driver's signature, trust me, I'd only discuss the important things i.e. once I discuss it, it's important.

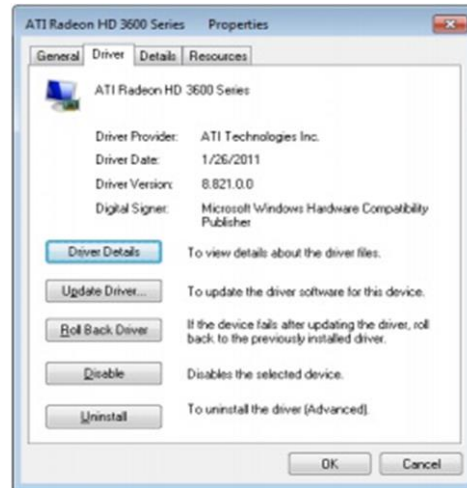
With respect to signature, there are two types of driver: Signed and Unsigned drivers. Why should I bother you with this info? Because Microsoft said: "90% of BSOD are caused by unsigned drivers". What is BSOD? BSOD means Blue Screen Of Death, it's the blue screen that often notifies you that "Your PC ran into trouble and need to be restarted". Now you see what I mean? If you install unsigned drivers, you might endanger yourself. But I don't install drivers now! Is that what comes to your mind? Actually, you do very often without noticing. Once you attach a hardware to your PC for the first time, be sure to check the bottom right corner of your screen, you would see a message like "installing device driver software", and the driver get installed. But don't get scared, Windows always check the signature, and if it's broken or from an untrusted company, you would be informed and prompted to decide if you want to proceed with such installation. But still, somehow, your PC might still be running unsigned drivers if you are running a 32-bit OS, that is because Microsoft said you can't install unsigned drivers on 64 bit OS (I must confess that there is actually a technical way to do that if you really need to do it), but remember that it is detrimental to run unsigned drivers on your PC.

If you've got some troubles with your PC like BSOD and you wish to check if you've got unsigned drivers, then follow these steps to create a .csv file containing info about all drivers on your PC which you can view with Microsoft Excel. Assuming you have a drive E on your PC (we would be saving the .csv file there, so change E to which ever drive letter you've got if you don't have a drive E), then let's go:

- Run **cmd** (command prompt) as an administrator
- Type: **driverquery /si /fo csv > E:\drvsign.csv**
- Press enter and then exit cmd

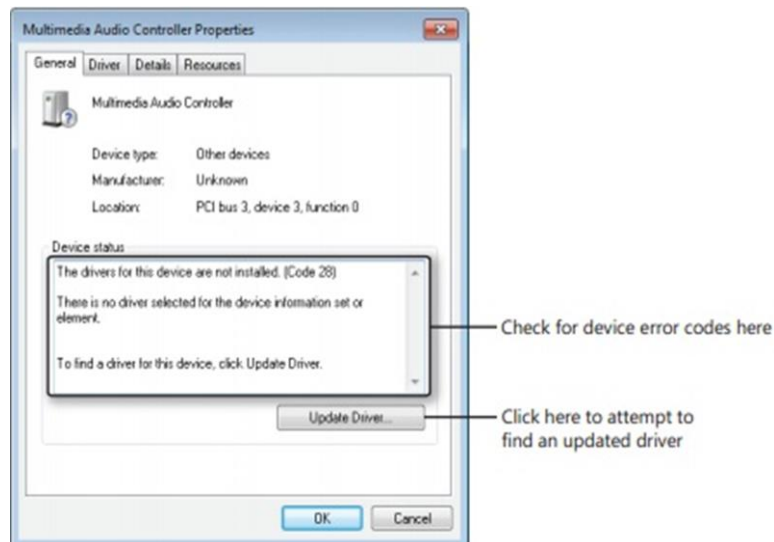
Now you have a file named "drvsign.csv" in your drive E, open this file with Excel and check the column named "Is signed" to see the drivers whose value for that column is "FALSE", any such driver is unsigned.

Once you get hold of the unsigned driver, you need to be sure it's the cause of trouble. How? Go to device manager, locate that driver, right click on it and choose disable. Now restart your PC and use it for a while to see if the problems still persist, if it does not, then that driver is the black sheep of your driver family. What should you do next? Go to device manager, right click on the driver and choose properties. Now you would have a multi-tabbed properties panel, from the tabs choose the "Driver" tab, you should have a screen similar to the one shown below:



From this screen, choose “Roll back driver” to undo installation if you just installed it or “Uninstall” if the roll back option is not enabled.

Furthermore, to check problems with a particular driver, right click on it and choose properties, from the general tab, read the content of the status area as shown below:



Notice that an error code is attached to the driver’s problem, the various error codes and possible solutions are given in the table below:

Error Code	Error Message	What to do about it
1	This device is not configured correctly. (Code 1)	After downloading a compatible driver for the device, click the update Driver button and follow the wizard’s prompts to install the new driver.
3	The driver for this device might be corrupted, or your system might be running low on memory or other resources. (Code 3)	Check available memory and, if necessary, close some programs to free up RAM. If you have sufficient memory, try uninstalling and reinstalling the driver.

10	This device cannot start. (Code 10)	Try restarting your PC, if that doesn't work, click the update driver button to install updated drivers if any are available. The troubleshoot button might provide useful information as well.
12	This device cannot find enough free resources that it can use. If you want to use this device, you need to disable one of the other devices on this system. (Code 12)	The device has been assigned one or more I/O ports, IRQs or DMA channels used by another device. This error message can also appear if the BIOS is configured incorrectly (e.g if a USB controller doesn't get an IRQ from the BIOS). Check BIOS settings. Use the resources tab in the driver's properties to identify the conflicting device.
14	This device cannot work properly until you restart your computer. (Code 14)	The driver has probably been installed correctly but cannot start until you reboot your PC.
16	Windows cannot identify all the resources this device uses. (Code 16)	A legacy device is improperly configured. Use the resources tab to fill in the missing details.
18	Reinstall the drivers for this device. (Code 18)	Click the update driver button to start the update hardware wizard and reinstall driver
19	Your registry might be corrupted. (Code 19)	Incorrect or conflicting information is entered in the registry settings for this device. Try uninstalling and then reinstalling the driver. Try using system restore to roll back the configuration to a point when the device worked properly.
21	Windows is removing this device. (Code 21)	Wait for a few seconds, and then refresh the device manager view. If the device continues to display the same message, restart your PC.
22	This device is disabled. (Code 22)	The device has been disabled using device manager. Right click on the driver and choose enable.
24	This device is not present, is not working properly, or does not have all its drivers installed. (Code 24)	This is a catch-all error that can be caused by bad hardware or corrupt or incompatible drivers. This message also appears after you use the remove device option.
28	The drivers for this device are not installed. (Code 28)	The driver installation for this device might have been restricted by group policy, contact your administrator on how to resolve that (you'd get a hint in disk management). If that's not the problem, then download another driver, after downloading a compatible driver for the device, click update driver and follow the wizard's prompts to install the new driver.
29	This device is disabled because the firmware of the device did not give it the required resources. (Code 29)	This is most commonly seen with SCSI adapters, third-party disk controllers, and other devices that supply their own BIOS. Check the documentation for the device to learn how to re-enable it.
31	This device is not working properly because Windows cannot load the drivers required for this device. (Code 31)	Windows was unable to load the drivers probably because it is not compatible with your version of Windows. After downloading a compatible driver for the device, click to update the driver and follow the wizard's prompt to install the new driver.
32	A driver service for this device was not required	The driver has been disabled. The start type for this service is set to disabled in the registry. If the driver

	and has been disabled. (Code 32)	really is required, change the start type in the BIOS by using the BIOS setup. Or change the start-up to “enabled” from msconfig or task manager. If the device previously worked properly, use “system restore” to return to a working configuration.
33	Windows cannot determine which resources are required for this device. (Code 33)	This error typically indicates a misconfigured legacy device or a hardware failure. See the documentation for the device for more information
34	Windows cannot determine the settings for this device. Consult the documentation that came with this device, and use the resource tab to set the configuration. (Code 34)	This legacy device requires a forced configuration. Change the hardware settings (using jumpers or a software utility), and then use Device Manager’s resources tab to set the forced configuration.
35	Your computer’s system BIOS does not include enough information to properly configure and use this device. To use this device, contact your computer manufacturer to obtain a firmware or BIOS update. (Code 35)	This error is specific to multiprocessor systems. Check with the system manufacturer for a BIOS upgrade.
36	This device is requesting a PCI interrupt but is configured for an ISA interrupt (or vice versa). Please use the computer’s system setup program to reconfigure the interrupt for this device. (Code 36)	IRQ translation failed. This error usually occurs on Advanced Power Management (APM) machines. Check BIOS settings to see if certain IRQs have been reserved incorrectly. Upgrade to an ACPI BIOS if possible.
37	Windows cannot initialize the device driver for this hardware. (Code 37)	After downloading a compatible driver for the device, use the update driver option in device manager to install it.
38	Windows cannot load the device driver for this hardware because a previous instance of the device driver is still in memory. (Code 38)	Restart the PC
39	Windows cannot load the device driver for this hardware. The driver might be corrupted. (Code 39)	The driver is missing or corrupted, or it’s in conflict with another driver. Look for an updated driver or reinstall the current driver. If the device worked previously, use System Restore to roll back to a working configuration.
40	Window cannot access this hardware because its service key information in the registry is missing or recorded incorrectly. (Code 40)	Information in the registry’s service key for the driver is invalid. Reinstall the driver.
41	Windows successfully	This error occurs with legacy devices because plug

	loaded the device driver for this hardware but cannot find the hardware device. (Code 41)	and play cannot detect them. Use device manager to uninstall the driver and then use the add hardware wizard to re-install it
42	Windows cannot load the device driver for this hardware because there is a duplicate device already running in the system. (Code 42)	Restart the computer
43	Windows has stopped this device because it has reported problems. (Code 43)	A driver has reported a device failure. Uninstall and reinstall the device. If that doesn't work, contact the device manufacturer.
44	An application or service has shut down this hardware device. (Code 44)	The device has been halted by an application or a service. Restart the PC
47	Windows cannot use this hardware device because it has been prepared for "safe removal", but it has not been removed from the computer. (Code 47)	The device has been prepared for ejection from PCMCIA slot, USB port, or docking station. Unplug the device and plug it in again or restart your PC
48	The software for this device has been blocked from starting because it is known to have problems with Windows. Contact the hardware vendor for a new driver. (Code 48)	Contact the hardware vendor for a compatible driver.

DISK MANAGEMENT

This Chapter covers:

-  Partitioning hard disk
-  Hiding & un-hiding disk
-  Checking disk for errors
-  Cleaning disk
-  Defragmenting disk
-  Disk compression
-  Drive options
-  Storage spaces

Storage devices are the most precious device on a computer system to many users, a failed processor might not be a tragedy because the very first question an expert would be asked after a PC failed to start is: can I still recover my files? If the answer is yes, then the person feels relaxed. It is therefore pertinent to understand how disks (storage devices) can be managed. Disks in this regards include hard disks, flash drives and other removable media etc.

One of the most important maintenance rules to be mastered is “keep everything organized” i.e. Music, Pictures, Videos, e-Books and System files are not supposed to exist in the same folder or on the same drive. Just imagine packing “all” in drive C and your PC get infected with a virus; a rootkit to be specific (Rootkit is a special type of virus that can only be best removed with clean installation after wiping the drive with cipher or other tools), you stand the chance of losing everything you’ve got because you haven’t been organized. So the question is how do I organize my PC?

Let me mention this first, when you open Windows explorer (by pressing window key + E or opening my computer from the desktop), the first drive labelled C which has the Windows icon next to it is the drive running Windows, this drive consist of all system files and your installed program files. Make sure this drive does not contain any other data except these two i.e. System files and installed program files (located in Program Files (x86) and/or Program Files depending on your OS architecture). The folders you should expect in C are:

- PerfLogs
- Program Files
- Program Files (x86) [if you are running a 64 bit OS (Operating System)]
- Users
- Windows

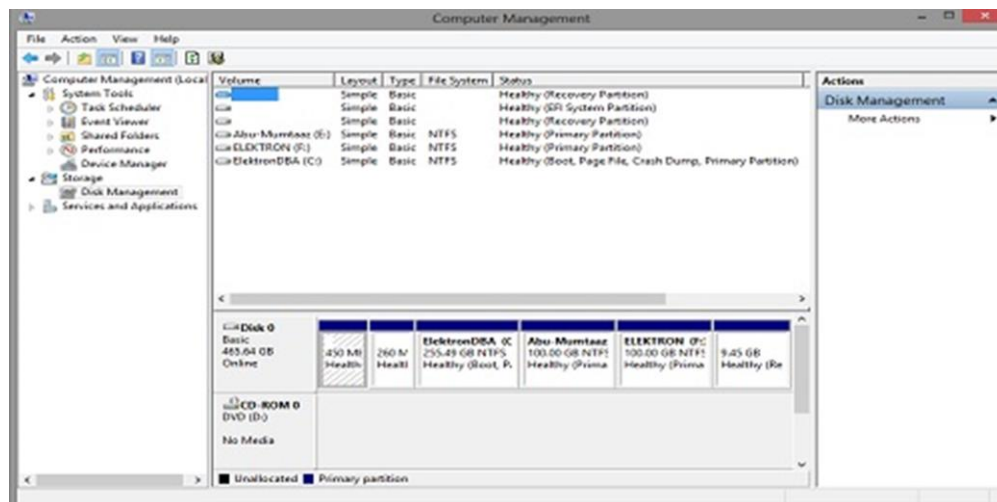
Some PC manufacturers also create a folder in C by default once you install Windows e.g. you might have a folder named “Toshiba” if your PC is Toshiba. Some software also create folders in C directly e.g. Oracle creates a folder called Oracle, wamp server creates a folder called wamp, octave creates a folder called software etc.

Apart from these types of folder, no other file or folder should be in your drive C (for organizational purpose), notice that the above list does not include a file, they are all folders, so files does not even come into the picture at all, they are not expected there.

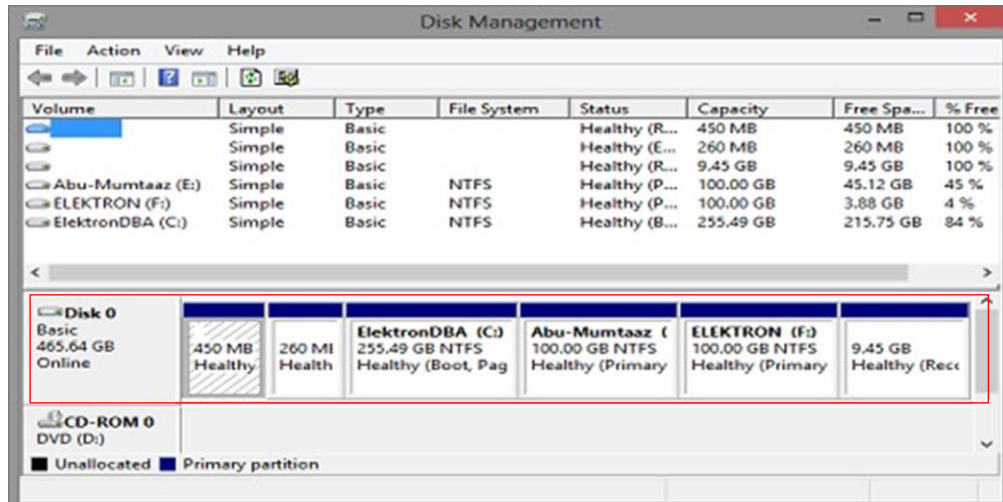
If they should not be in drive C where then should they be?

You have to consider creating more partition on your hard drive, so that other partitions apart from C would be dedicated to saving other files and folders. How do we make disk partitions then? Let’s start from there. We are going to be doing this using the disk management console; to access this console:

1. Open computer management and from the left pane choose disk management. To open computer management, go to “My Computer”, right click on computer from the left pane and choose “manage” OR simply search for “**compmgmt.msc**” from the start screen or from command prompt. You should get an interface similar to the one below:



2. Search for “**diskmgmt.msc**” from the start screen or from command prompt. You should get an interface similar to the one below:



Notice from the interface the section highlighted in red that show the disk divided into sections with each drive (section) having a colour on it, a blue colour indicates a primary partition, a yellow or orange colour indicates a volume on a dynamic disk, a black colour indicate an unallocated disk, other colours would be indicated at the bottom of the screen.

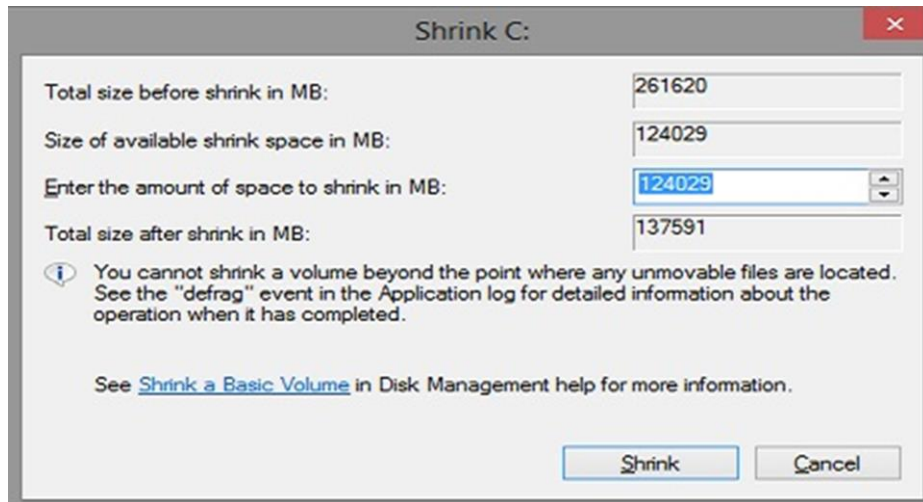
This screen would tell you how many partition(s) you presently have on your hard drive even though you would not see up to that from Windows explorer, this is because some partitions are hidden by default for management purposes. Do not tamper with the following drives if you've got one of such: "Reserved partition", "Recovery partition", "System Reserved" and "EFI system partition". The logic is if you've never created a partition on your PC then the only partition you can tamper with is your drive C.

Follow the steps below to create a partition:

- Right click on the drive you wish to partition e.g. C

NB: To create a partition, you are going to "shrink" out some of the free spaces on an existing partition. Wait a minute, did I just say shrink? Yap, that's the word, you are going to shrink it. So if you have a drive of 300GB (as an example) probably as your C, then you can shrink out 100GB and call that F or some other letter you wish, you would now have two drives: C which would now be 200GB and F (or something else) which would be 100GB.

- Choose "shrink volume" and wait for Windows to query your volume for available shrink space.
- Once this is done you should have an interface similar to the one below:



From this interface you can decide the amount of space you want to shrink out in MB. Notice that a maximum value has been set by Windows and you can't exceed that value.

- Once you have decided the amount of space to shrink, click on "Shrink" and the partition is created.
- Once the partition is created, you would see it listed among the partitions you have coloured in black; this implies it is still unallocated. Right click on this unallocated partition and choose "New simple volume".
- Follow the wizard and just click "next" until "finish" and the new partition is created. One of the interfaces you would see in the wizard is an option for you to choose the drive letter, a default letter is assigned which you can change to whichever letter you wish to use.

What if I decide I don't want this partition anymore?

Follow the steps below to return the partition back to where it was "shrunk" from (if there is any word like that).

- Right click on the new partition and click delete volume. Note that this would cause you to lose the content of the drive, so backup anything you have on it to another drive. Once the volume is deleted you would see it listed as unallocated with a black colour.
- Right click on the drive where you "shrunk" the new volume from and click "Extend Volume". Follow the extend volume wizard until the volume is extended.

Now that you have other partitions on your drive, save your files and folders in other partition apart from C to make your data more organized.

Let me ask you this: have you ever noticed that without partitioning, some PCs have got more partitions in Windows explorer than others? Yes, this should happen. This is because some PC hides other partitions apart from C while others don't. Those drives should actually be hidden.

If you wish to hide a drive, follow the steps below:

- Open disk management console.
- Right click on the drive you wish to hide and choose "Change drive letters and path".
- From the interface you get, click on "Remove" and remove the drive letter. Once the drive has no letter, it would not show up in Windows explorer.

To show the drive again, go to disk management and assign it a drive letter.

DISK MAINTENANCE

It is important to make sure that your hard disk has got no errors so that it's not going to fail anytime soon. The following are tips on how disks can be maintained.

1. Check your disk for errors.

When you notice strange behaviours in your drive then you should check if it has got some errors. How?

Method 1:

- Run cmd (command prompt) as an administrator
- Type: **chkdsk C: /f**

Where C is the drive letter you wish to check for errors.

- This should inform you that drive C's format is NTFS and can only be checked on next reboot, so press "y" to check on next reboot or "n" not to check it. Then press "y" without quotes to check your disk on next reboot.

This would check the drive for errors and also fix errors found.

If you change your mind about this and wish to cancel the scheduled check, then launch command prompt again with admin privilege and type: **chkntfs /x C:** to cancel check on drive C

Method 2:

- Open Windows explorer
- Right click on the drive you wish to check for errors and choose properties.
- From the properties tab, choose tools as shown below:

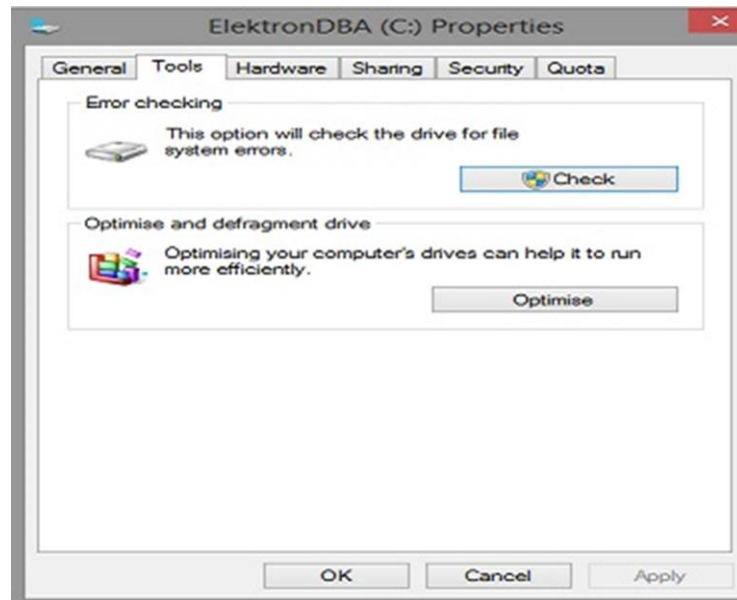
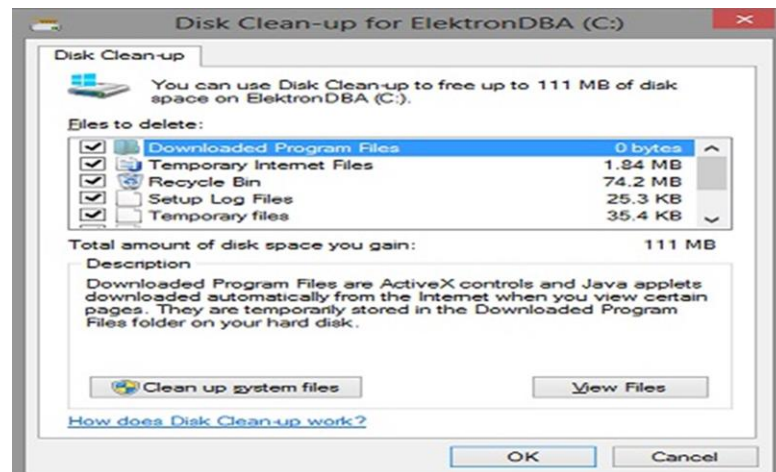


Fig. 1

- From the interface, click on Check and let the scan run.
 - If errors were found during the scan you would be informed and Windows would ask if you wish to fix the errors. Check the box and click to fix it.
2. One other thing you should always do often, (probably once in a week depending on how often you use your PC, update your PC and browse the internet) is to free up your disk space. This is because using some apps and browsing the internet leave some temp files in your hard drive. Files you delete also stay in your recycle bin if you do not choose to delete it permanently (always use Shift + Del key to permanently delete a file), Windows update files and some other ones take up appreciable part of your hard disk from few megabytes to gigabytes.

To free up disk space:

- Open control panel
- Click on system and security. If you do not have a category view in your control panel, click on “view by” from the top right hand corner of the screen and choose “category”.
- Under “Administrative tools” choose “Free up disk space” and select the drive you want to free, always bother yourself with drive C. You should have a screen similar to the one below when Windows is done calculating dump files on your hard disk.



- From the interface, choose “Clean up system files”. This would enable the access to clean up files that require administrative access. Check all boxes and click ok.

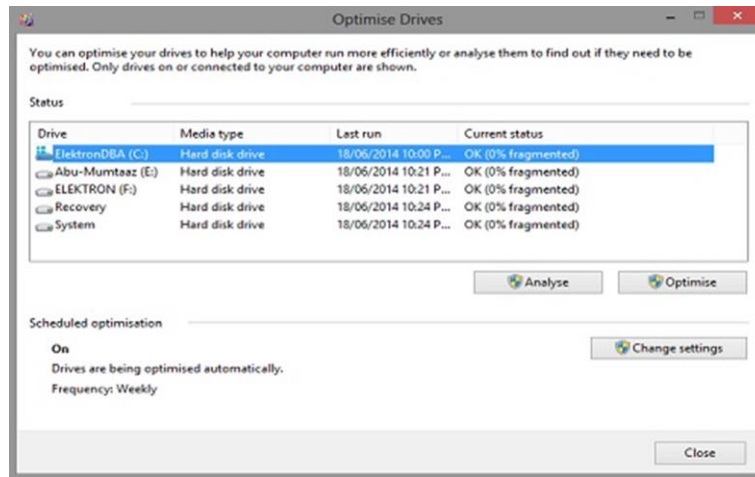
To be honest with you, Windows would still refuse to clean up some temporary files on your PC which might still be eating up a lot of disk space but you can actually do that yourself but make sure you delete them permanently (don’t send them to the recycle bin).

To do this, open Windows explorer and in the address bar type: **%temp%** and press Enter.

This would display the temp folder; delete everything in this folder permanently.

The presence of these files on your PC is the reasons why you ask yourself, I don’t have many files, why is my disk space running down fast? Now you can answer that question.

3. When you have performance issues with your PC, i.e. write and read operations get slower (in simple terms, it takes more time to copy and paste files), then your hard drive has got fragments (errors and scattered files) and you should remove them by defragmenting your hard drive. This should actually be done on weekly basis too depending on how often you use your PC. To defragment your PC, choose optimise from Fig. 1 above to get an interface similar to the one below:

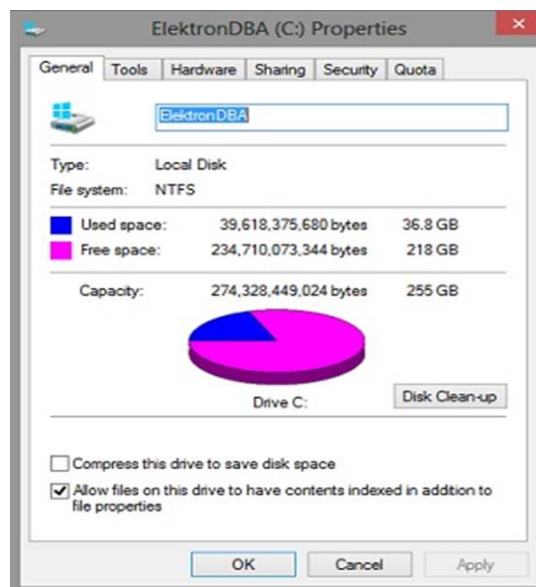


From this screen you'd notice two options: Analyse and Optimise. Microsoft recommend that you analyse your PC before you defragment (optimise) it, Microsoft also said that you should proceed to optimise the drive if it has more than 10% fragments from the result of the analysis.

If you are defragmenting your PC on a weekly basis, you might not need to analyse before proceeding with optimising, that's going to give your PC better performance.

Compressing Your Hard Drive

To save space on your hard drive, you can choose to compress data on it. When you compress your hard drive, it would reduce the disk space occupied by files on that drive giving you more space to save data. To compress a drive, open Windows explorer, right click on the drive you wish to compress and choose properties. You should have a screen similar to the one shown below:



Check the box "Compress this drive to save disk space" and then click OK.

DRIVE OPTIONS

Technically, there are two types of disk, these are:

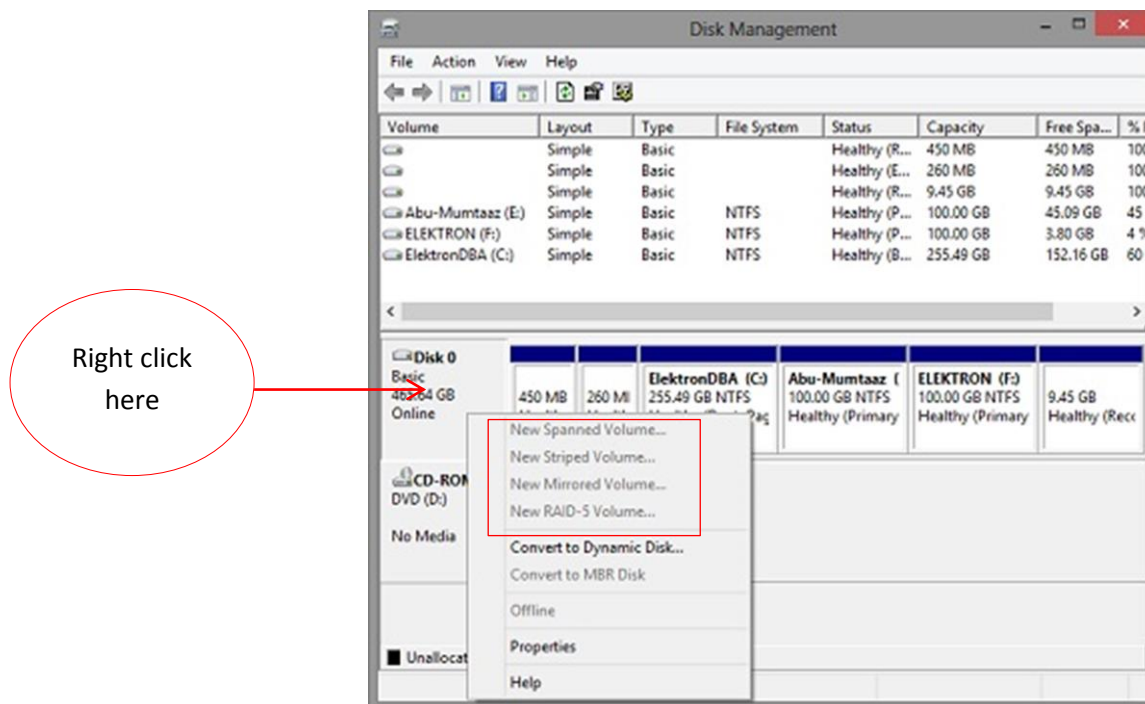
- **Basic disks:** These disks can have a maximum of 4 primary partitions, to create more than that, you can have a maximum of 3 primary partitions and 1 extended partition. The extended partition can then be partitioned to get up to 2million logical partitions. This is the only type of disk that can run Windows.
- **Dynamic disks:** These disks can't run Windows; they are created for administrative purpose. They can create a type of disk known as **RAID** (Redundant Array Of Independent Disks) disk.

Multiple drives can be mapped together to work as a single drive using RAID technology. To do these, connect the drives to the PC and open the disk management console, right click on one of the drives and choose one of the available RAID options which include:

- **RAID-0:** This is also known as **striped** volume. It has no redundancy (i.e. losing one drive amounts to losing all drive and their content), fast read/write. Striped volume uses amount of free space on a disk equivalent to the least space in one of its members, this is because all drives added up to a RAID-0 array must have the same size. For example, if you add three drives, one of 4GB, another of 6GB and the other of 2GB all into RAID-0, you'd end up having 6GB striped volume since the least member is 2GB; it would only make use of 2GB on the other drives too and give you the opportunity to use the remaining space on the other drives for other things you wish.
- **RAID-1:** This is also known as **mirrored** volume. It uses two disks (copies the content of one disk to another as a backup) so that when one disk fail, the other works and you still have your data intact. It has a 50% overhead (i.e. the amount of disk space you can save data to is half of what you have as the other half would be used for backup). It has a fast read performance.
- **RAID-5:** This uses a minimum of 3 disks. It saves data on 2 disks and saves parity information about the data saved on those 2 disks on the third disk; this third disk is called the rotating member. RAID-5 allows for fast read operation but CPU burdens during write operation.

Once you choose any of these options, Windows prompt you to select the drives you want to add to the array.

One other option Windows give to map multiple drives as one is **SPAN**, this only allow you to add all the drives up to make a **span drive** which you can then access as a single drive from Windows explorer. It has no redundancy and no performance benefit; it's only useful for temporary expansion.



Remember:

- Redundancy in disk technology refers to losing drive(s) without losing data.
- Only dynamic disks can use RAID, that's one of the reasons why the options to use it is not enabled in the image above.

STORAGE SPACES

In Windows 8, Microsoft created an equivalent of RAID to map multiple drives as one, this is called Storage Spaces. Storage Spaces takes whatever random drives you connect to it and combine them to make useful, and optionally redundant, storage. What it does is to create a pool of storage which consists of drives; you save your data into the pool and not into a particular drive, the pool then divide the data among drives in the pool as it wishes. You can add more drive(s) to the pool after creating the storage pool.

Storage Spaces is not a direct substitute for RAID though, because:

1. If you add a drive to a RAID array, the RAID array will integrate the new drive into the array by rewriting all of the existing files across every drive in the array but if you add a drive to a pool, Storage Spaces does not rewrite any of the existing files, it will use the new drive for future data.

2. You can use RAID to protect your system drive, but you can't use your system drive as part of a Storage Spaces pool.

Let's take some time to evaluate Storage Spaces before we start configuring it!

Before I even start evaluating, let me mention one thing I read from a boss (Mike Halsey), he mentioned in one of his books that Storage Spaces is not a well proven technology yet, therefore users should take caution in using it. Take caution in this regards implies that you should backup your data and don't trust them to Storage Spaces; you won't sue Microsoft if the pool fails and all data in it get corrupt somehow or something worst happen to the Storage Pool. I don't know how long it would take to prove such technology though (though he mentioned that technologies which we can trust our data with must have been used for years to test its efficiency), but remember it was launched with Windows 8, and as at the time of this writing, Windows 8 is barely 2years old.

Now, straight to evaluation.....

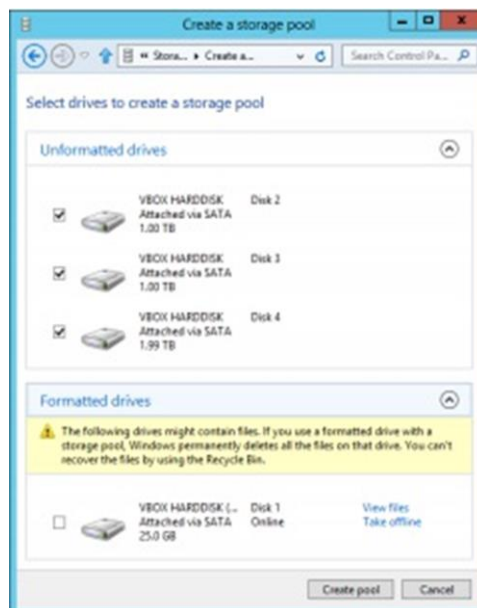
1. Unlike RAID, Storage Spaces uses a format that is not supported by earlier versions of Windows, therefore you can't move the drives that constitute a pool to an earlier version of Windows and access your data; it would just be empty. However, you can move all of those drives to another PC running Windows 8 (all of them), it would detect them automatically and configure them the way you did the configuration from the old PC so you can access your data.
2. Storage Spaces distributes your data across drives in its pool, therefore you can't take the drives to another PC separately and access the data on them, if you would have to move, then they must go together.
3. Your PC's read performance might be improved if it can communicate with the separate drives faster than with a single drive. For most PCs, however, you won't notice a substantial difference.
4. If you use resiliency (we would discuss this during configuration), write performance might be decreased.
5. If one drive fails and you have not configured resiliency, all your data is lost. You'd learn how to configure all that in a minute don't get scared.
6. Drives that are added to a pool must be formatted, which erases all the data from the drive. Therefore, you cannot integrate an existing drive with data into a storage pool. Instead, you can copy the data to your storage pool (assuming you have enough free space in the pool) and then add the drive to the pool.

Configuring Storage Spaces

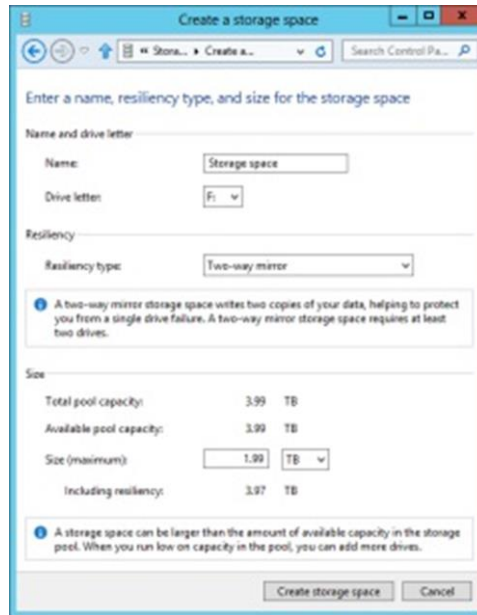
To access Storage Spaces configuration interface on your Windows 8 PC (it's not available on older versions of Windows), open control panel (Press Window logo key + X and then press P), view your control panel by Category and click on "System and Security", from System and Security, click on Storage Spaces to get the interface below:



When you click the link to create a new pool, Windows would prompt you to select the drives to add to the pool. As shown below, Windows automatically selects all unformatted drives. If you select a formatted drive, all the data from that drive will be lost forever, so be sure there's nothing you love on it.



Once you click “Create Pool”, Windows would prompt you to choose your resiliency option as shown below



Resiliency is the redundancy methods in Storage Spaces. I hope you still remember what redundancy means? With redundancy, you can lose drive(s) without losing data.

Since Storage Spaces would be making use of one or more drives, it is important that we configure redundancy on it. However, Windows changed the name here, they called it resiliency.

There are actually four resiliency options from the interface above, and each of them is explained below:

- ❖ **Simple (no resiliency):** Storage Spaces writes one copy of your data, so if a drive fails, you lose all the data in the pool. This requires only one drive and you can access 100% of your physical capacity. This is similar to RAID-0 with the only difference that you can use all the spaces on each of the drives unlike RAID-0 that would configure every drive to use the space on the least member.
- ❖ **Two-way mirror:** Storage Spaces writes two copies of your data, allowing you to lose one drive without data loss. This requires at least two drives, and you can access only 50% of your physical capacity. This is similar to RAID-1.
- ❖ **Three-way mirror:** Storage Spaces writes three copies of your data, allowing you to lose two drives simultaneously without data loss. This requires at least three drives, and you can access only 33% of your physical capacity (i.e. 1/3). This is different from RAID-5 because RAID-5 only saves parity information about data. This option on the other hand writes two copies of your data (i.e. 2 backups) which makes up three copies with your actual data so that when one fails you still have 2 other. If you use a 1TB drive you can only save data to a maximum of 337GB (33%).

- ❖ **Parity:** Storage Spaces uses mathematical algorithms to protect your data more efficiently than by simply creating additional copies of it. Parity requires at least three drives. The portion of the physical capacity that you can access is related to the number of drives you add. To calculate the maximum capacity you can access, take the reciprocal of the number of physical drives, and then subtract that from one. Basically, divide one by the number of drives you have to calculate the capacity used by parity. If you have three drives, you can access up to $1 - \frac{1}{3}$ (66%) of your physical capacity. If you have four drives, you can access up to $1 - \frac{1}{4}$ (75%) of your physical capacity.

Notice also from the interface above that the Size (Maximum) option is set by default to the maximum available capacity on the physical drives you have allocated. You can set it to a larger value, but I have no idea why you'd want to. For example, if you connect 4 TB of physical drives, you can set a 10 TB capacity. Naturally, Windows can't actually save 10 TB of files in that pool, but when you start to run out of physical space (for example, if you have used 3.9 TB of your physical drives) Windows will prompt you to add more physical drives to the pool.

NB: If you are using only one drive, you can't use resiliency.

Choose your desired resiliency and click "Create Storage Space" to create the storage space.

SECURITY

This Chapter covers:

- + Types and definitions of threat
- + Prevention against threats
- + Reliable anti-threat softwares
- + Removing threats from an infected PC
- + BitLocker drive encryption
- + User accounts and family safety
- + How to create a secure password
- + Recovering from a lost password scenario
- + Windows firewall

Things that serve as a threat to a computer system are said to be malicious, they are sometimes referred to as malwares (**malicious softwares**) as a general name and sometimes as threats. The most common security threats are discussed below:

- **Virus:** A virus is a program that can copy itself, usually by attaching itself to another object. Viruses can infect program files, documents (in the form of macro viruses), or low-level disk and file-system structures such as the boot sector and partition table. Viruses can run when an infected program file runs; they can also reside in memory and infect files as the user opens, saves, or creates the files.
- **Rootkit:** Rootkits are among the most worrying forms of malware, specifically designed to burrow into the most sensitive areas of the operating system, where they can evade detection and removal. The rootkit's job is to cloak the actions of another program—such as a backdoor or key logger—and prevent it from being detected even by an administrator.
- **Worm:** A worm is a program that replicates by copying itself from one computer to another on a network. Many modern worms also contain virus code, backdoors, and downloaders.
- **Spyware and adware:** Spyware and adware are terms that have been applied to a variety of unwanted programs, including advertiser-sponsored software that tracks a user's web surfing habits; programs that display pop-up ads or install unexpected toolbars, favourites, and links in your web browser; programs that redirect Internet Explorer to a search engine or home page that's different from the one you specify; and more. Note that spyware doesn't necessarily "spy" on you. A common characteristic of anything dubbed "spyware" is that it does its deeds—malicious or otherwise—without your informed consent. For the

purposes of this chapter, our definition of spyware is “any program that is installed without the user’s full and informed consent, often through deceptive means, and that displays advertising, records personal information, or changes a computer’s configuration without the user’s explicit permission.”

Some or all of this can be on a PC at a time and I think you should still remember the saying that “prevention is better than cure”, that is the main reason why you should always prevent your PC against all these threats. Developers have tried a great deal; they have developed anti everything you call threat, that’s why we have anti-viruses, anti-malwares, anti-spywares, anti-adwares and many more. The prevention mechanism is to install a trusted anti all these threats to be free from them, don’t let anybody fool you; no antivirus can prevent you against malwares (even though some prevent some malware) nor spywares nor adwares.

Which anti-threat software should you use?

To be honest, I am not a security expert but I’d give you info about the technical reviews I’ve read from expert on their websites and in some of their books. Experts said:

- AVG antivirus free available from www.free.avg.com

AVG has long been a favourite firm in the free antivirus category for its overall effectiveness in blocking malware and warning about dangerous websites. The current version of AVG is still as effective, but it now comes with some annoying advertisements for the full paid-for version. If you are prepared to tolerate pop-up advertising, AVG Anti-Virus FREE is still an excellent choice to protect your computer.

- Mike Halsey (Microsoft MVP for Windows)

- Norton Internet Security & Norton 360, available from www.symantec.com

The scanning engine in Norton 360 has won as many awards as Microsoft’s own Forefront, a corporate anti-virus scanning engine. This is a more comprehensive suite, with more notifications and warnings, so you’ll notice it more when using your PC, which some people might find annoying. It contains many useful features and additions, however, and comes with a license that allows you to install it on several PCs.

- Mike Halsey (Microsoft MVP for Windows)

- Kaspersky Internet Security, available from www.kaspersky.com

For IT professionals and enthusiasts who want more control over their computer’s security, Kaspersky is the package to have. It includes an excellent scanning engine with significant volumes of controls for technically-minded people.

- Mike Halsey

- Malwarebytes anti-malware available from www.malwarebytes.org

I recommend several anti-malware products, including SpyBot Search & Destroy and Spy-ware Terminator. Only Malwarebytes, however, has consistently dealt with hard-to-remove malware from PCs I've encountered. There are also free and for-purchase versions of the product.

- Mike Halsey

- SuperAntiSpyware

This removes hard to discover and hard to remove spywares and adwares from your PC.

- Recommended by Microsoft

One common question people ask often is if they can use more than one anti-threat software on their PC. According to experts, you should not use more than one antivirus for two major reasons:

1. **Performance issues:** Antivirus programs scan every file during write and read operations for viruses. This may slow down your PC's performance if you use more than one.
2. **Security issues:** There is going to be a "clash" between antivirus programs when you use more than one. I translate their statements to mean that they are going to fight over files for threats – lol.

As for other ones, antispyware, antimalware and others, it is recommended to use more than one if you have enough money to purchase them because many of them are not free.

But let me add at this point that many antimalware are actually malware, especially those that often come with softwares you didn't ask for, they scan your PC and tell you that you've got thousands of malwares and to remove them you have to purchase the full version from their website. Make sure you don't fall prey to such information, they are malicious!

Removing threats from your PC

How do you know if you've got viruses on your PC?

If you have often notice file corruption on your PC, if file icons change from their normal icon to a question mark or something similar or if files that work pretty well some weeks ago just aren't working anymore, then you are probably infected with viruses (seriously, I don't know if it's you that's infected or your PC).

If your PC has got viruses or malwares or other threats, what should you do?

If it's a malware, then get your PC scanned by a responsible antimalware such as malwarebytes. If it's a virus, then:

1. Get a good antivirus to scan your PC. I'd recommend "AVG internet security", it's not actually free but you can still use their free antivirus. Make sure you update the antivirus before you make use of it though. Your PC should then be safe after a successful scan is completed.
2. If the virus still doesn't get removed (probably it's a rootkit), then consider formatting your hard drive. But to do this, you should have other partitions on your PC, and then move your personal files and folders to the partition (provided you are very sure they've got no virus too. The secret is, move your personal files to other partition(s) and use an antivirus software to scan that partition, antiviruses do a great work in that respect) and then format the system drive i.e. drive C. If it's a deadly threat like rootkit, you might consider formatting your PC with "cipher" (I'm not going to discuss how to use that though; contact experts).

NB: Installing a good antivirus and other anti-threat software on your PC does not guarantee your safety against threats, one thing you must do always is to make sure you update your anti-threat engine very often, this is because threats (viruses, worms etc.) are released almost every day, anti-threat manufacturers therefore update their databases often as soon as new threat is released to make sure you are fully protected, that implies that there is no way you would be protected against recent threats until you update your anti-threat engine.

BITLOCKER

This is a security measure Microsoft has built into Windows to protect your PC and encrypt your drives from **theft** so that whoever steals your PC or your drive(s) has just stolen metals, provided your PC is not left logged on.

How does BitLocker work?

BitLocker prevents your PC from starting up until you have supplied your credentials. Do you just say you can hack system passwords? Let me tell you this:

Some PCs have got something known as TPM (Trusted Platform Module), this is a chip that is built into the PC's motherboard and is used for storing pins and passwords. If a PC has TPM, then BitLocker saves the PIN inside it (BitLocker is actually designed for this) and in such case, the PC would not run BIOS instructions until you have supplied your PIN, you can't open the BIOS settings or do anything on your PC until the PIN is supplied, TPM is not BIOS so you can't hack it, it's actually designed in such a way that if an attacker attempt to access

credentials on it, it would break apart in such a way that the credentials can't be retrieved again by anybody, not even the administrator of the PC, so it's really secure.

BitLocker only prevent booting on your PC, so you are not protected from theft once you leave your PC logged on as BitLocker decrypt everything after start-up. If you protect your drive(s) with BitLocker, it would only function if your drive is stolen and attached to another PC.

BitLocker has got some advancement in Windows 8, this include:

- It supports offloading drive encryption & decryption to specialized hard drives that have those features to avoid performance degradation.
- It supports encrypting only the part of drive with data on it. This is known as "Used Drive Only Encryption".

BitLocker is really secured but has got some few drawbacks. Let's review the drawbacks:

1. BitLocker slows down drive performance, this is because every data read from the drive(s) must be decrypted and data written onto the drive(s) must be encrypted.
2. If the lock key is corrupt or TPM is damaged, the PC would become unbootable.
3. Data recovery would become difficult once an encrypted drive corrupts.

Drive types that can be encrypted with BitLocker

- **System Drive:** You can use BitLocker to protect your system drive i.e. the drive running Windows. This is the primary reason why BitLocker was created.
- **Data Drive:** This benefits one if someone attempts to access the data on the drive(s) offline by starting the PC from a different PC or stealing the PC and connecting it to a different PC.
- **Removable drive:** It must be decrypted with password or key on every PC it is attached to. This protects data on your removable drive in case of theft.

When you configure BitLocker on your PC, you have to configure your choice of start-up mode. The options and usage is given below:

- **Enter a PIN:** This option makes use of a PIN. A PIN is made up of numeric characters (4 digits in Windows 8) unlike a password that is alphanumeric. This option is available only when the PC has TPM.
- **Enter a Password:** This option makes use of alphanumeric password. Available only when the PC has no TPM.
- **Insert a USB flash drive:** You'd be required to start your PC with your flash drive just like you'd start your car with a key. Your PC become inaccessible once the flash drive is lost.

- **Let BitLocker automatically unlock my drive:** This option only verify the system integrity and automatically start-up the PC. This option doesn't offer any protection from someone who steals your entire PC, but it does protect your data in the event someone steals just the drive from inside your PC. It also helps protect you from rootkits and other malware that might modify your system files. This option is only available when the PC has TPM.

Configuring BitLocker

To start BitLocker on your PC, follow the steps below:

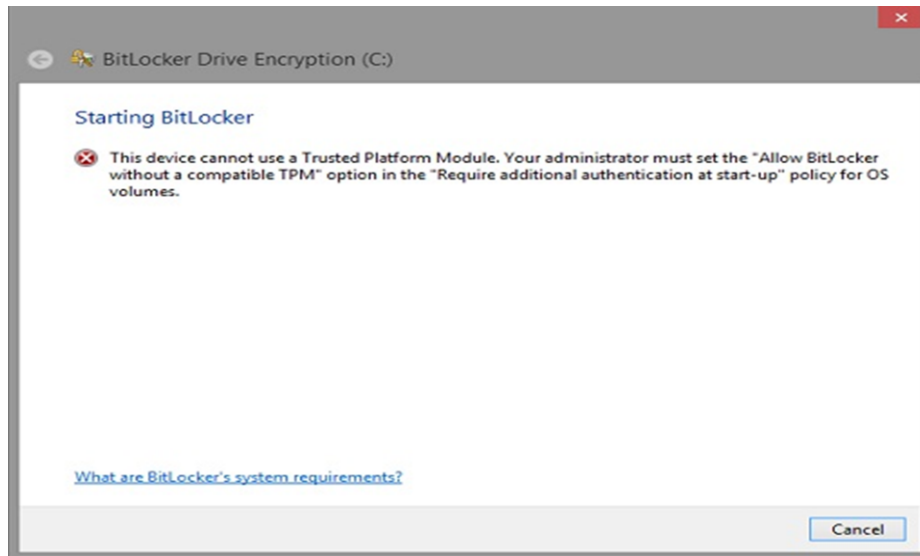
1. Open control panel and click on "System and Security"
2. Click on "BitLocker Drive Encryption" to get an interface similar to the one below.

BitLocker is not available in Windows 8 standard edition.



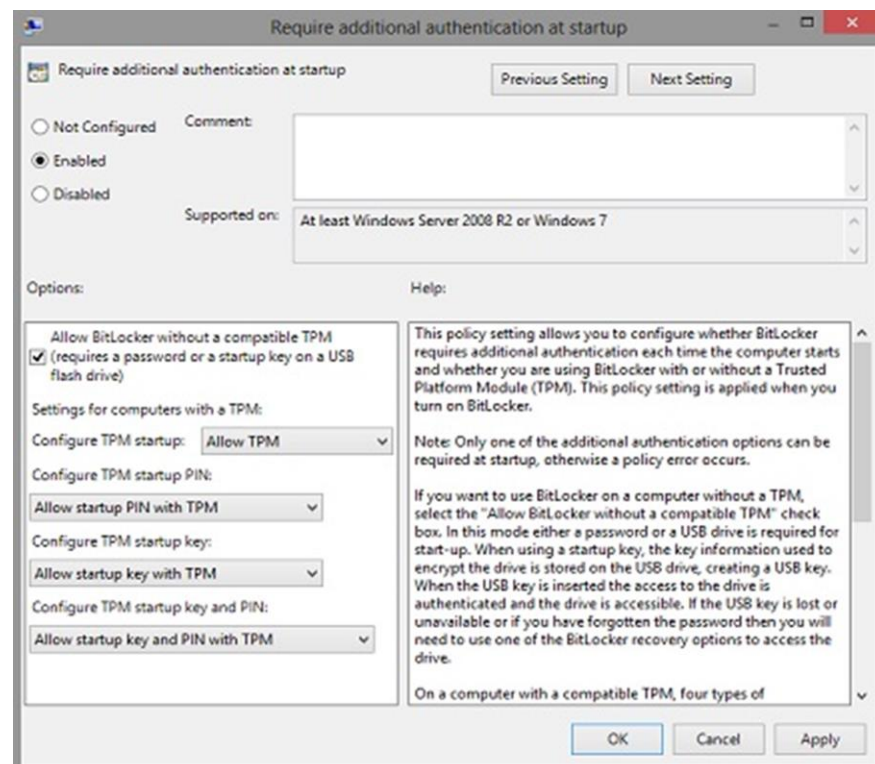
Windows lists all drives attached to your PC in that interface with an option to turn on BitLocker on any of them, when you use BitLocker on removable drives; it's called "BitLocker To Go".

When you click on "Turn BitLocker on" as shown in this interface, you'd get an interface similar to the one below except your PC has got TPM.



To use BitLocker without TPM, you'd have to configure that from group policy editor. To do that, follow the steps below:

1. Run **gpedit.msc**
2. From the left pane, expand "Computer Configuration", expand "Administrative Templates", expand "Windows Components", expand "BitLocker Drive Encryption" and click on "Operating System Drives".
3. Double-click "Require Additional Authentication At Startup", select Enabled, and select "Allow BitLocker Without A Compatible TPM (Requires A Password Or A Startup Key On A USB Flash Drive)", as shown below.

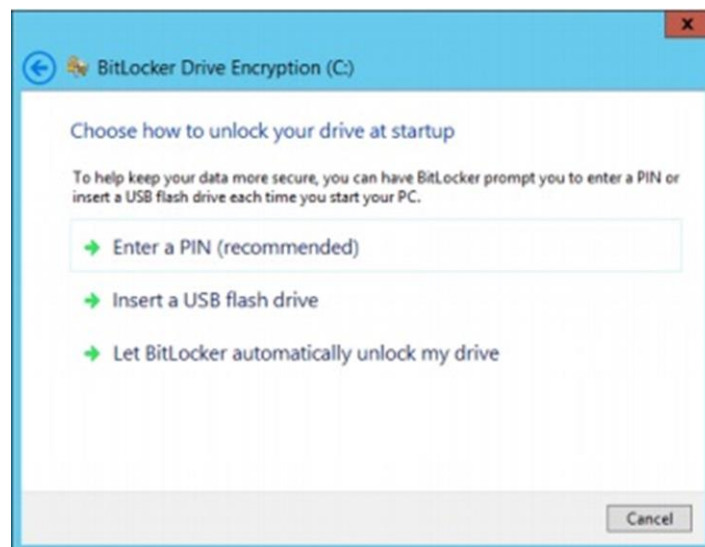


4. Click OK.

Once this is done, go back to the BitLocker interface and click “Turn BitLocker on”, you should now have something similar to the one below:



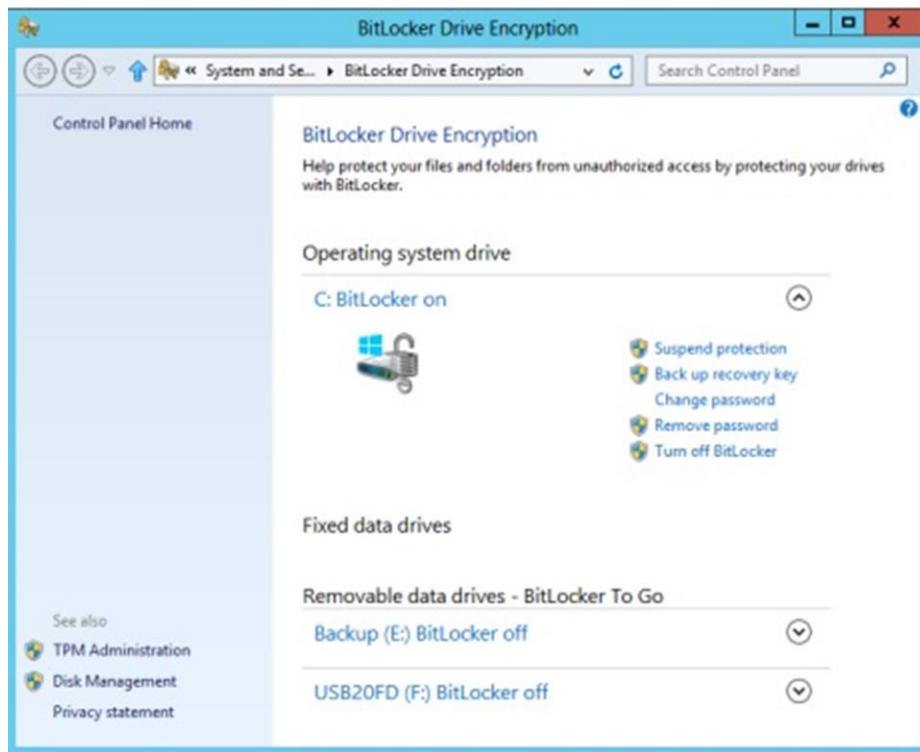
Assuming your PC has TPM, you would have gotten the interface below:



When you choose your desired unlock option, Windows would prompt you to back up your key for recovery purpose and you'd have to choose from four options, choose one of the options and then click next. The encryption would start once you restart your PC.

After restarting, launch the BitLocker interface, you should have an interface similar to the one below.

To back up your key, click on “Backup recovery key”, to change your password or PIN, click on change password or change PIN respectively.



Suspending BitLocker

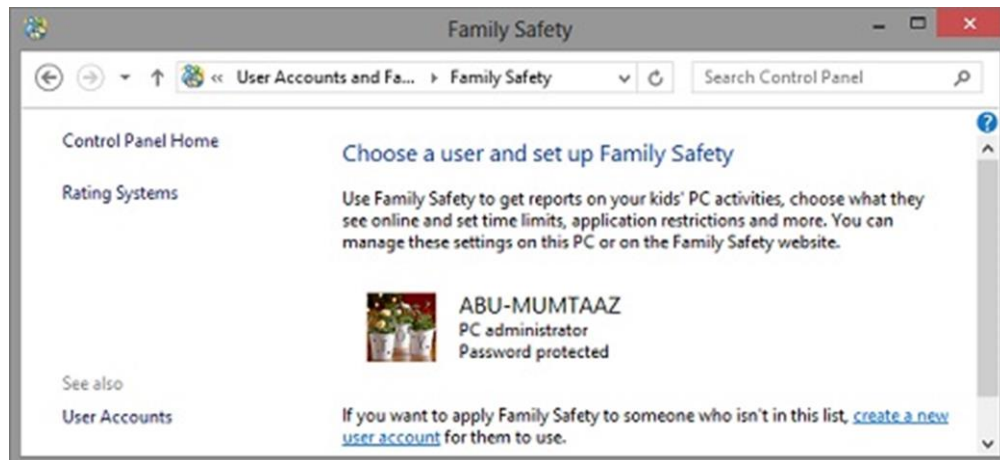
To suspend BitLocker, you have two options as shown in the image above:

- **Turn off:** Turning off BitLocker decrypts your drive. Of course, you lose the benefits of BitLocker, but you also lose the drawbacks. You'll be able to access data directly from other operating systems, and you won't suffer any performance penalty for encryption and decryption. If you turn BitLocker back on, it will need to re-encrypt all of your files.
- **Suspend:** Suspending BitLocker keeps your drive encrypted but stores the decryption key in plain text. Suspending BitLocker allows you to start Windows without a password, PIN, flash drive, or recovery key. If you're using a TPM, Windows will start even if system files have been modified. Suspend BitLocker when you don't need to access the files from other operating systems and you plan to re-enable it in the near future.

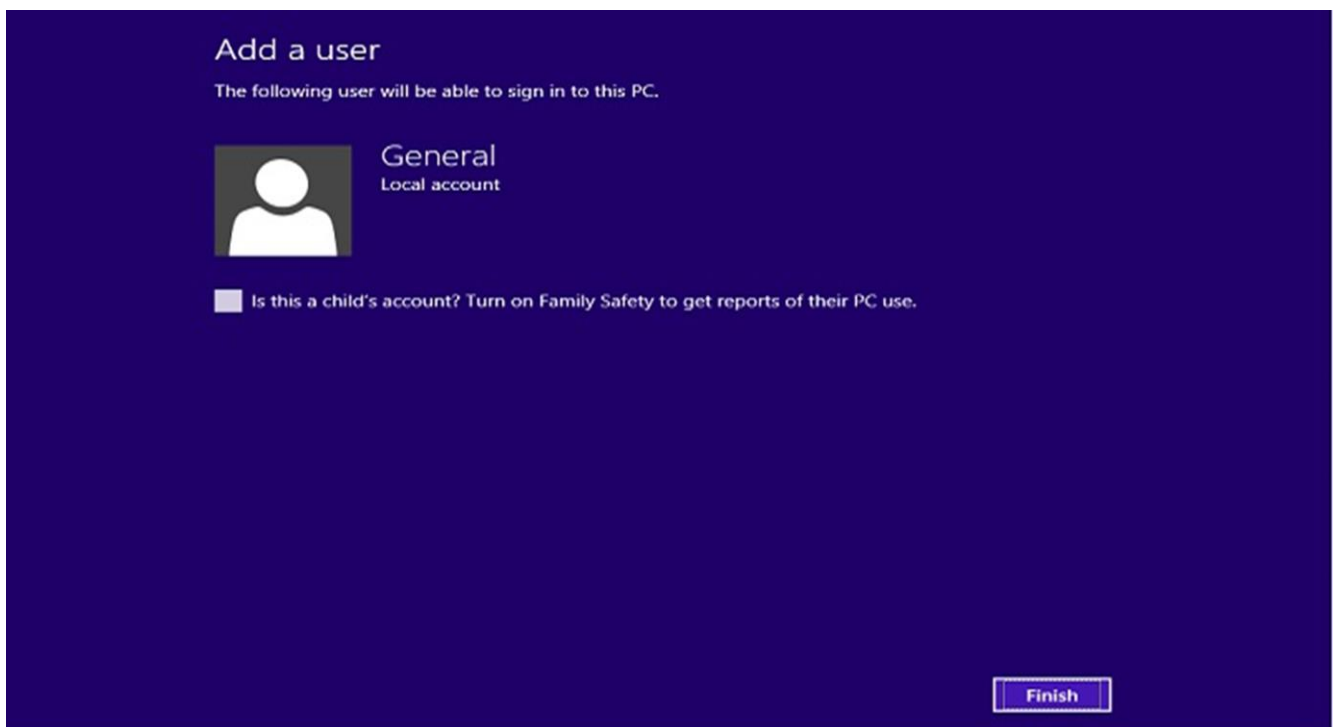
USER ACCOUNTS AND FAMILY SAFETY

Microsoft created a new way to manage user accounts in Windows 8 with something known as family safety (available as Parental controls in Windows 7). With family safety, you can configure what users have access to on your PC.

To start family safety, open control panel, click on “User Accounts and Family Safety”, click on “create a new user account”.



Choosing this option would open PC settings to create a new user account, supply the credentials and click on next and you should have the screen below:



Check the box to create it as a child account and add it to family safety and then click finish. You should then have the account created and listed among the list of user accounts you have from the page you left off. Double click on the new account and you should have an interface similar to the one below:

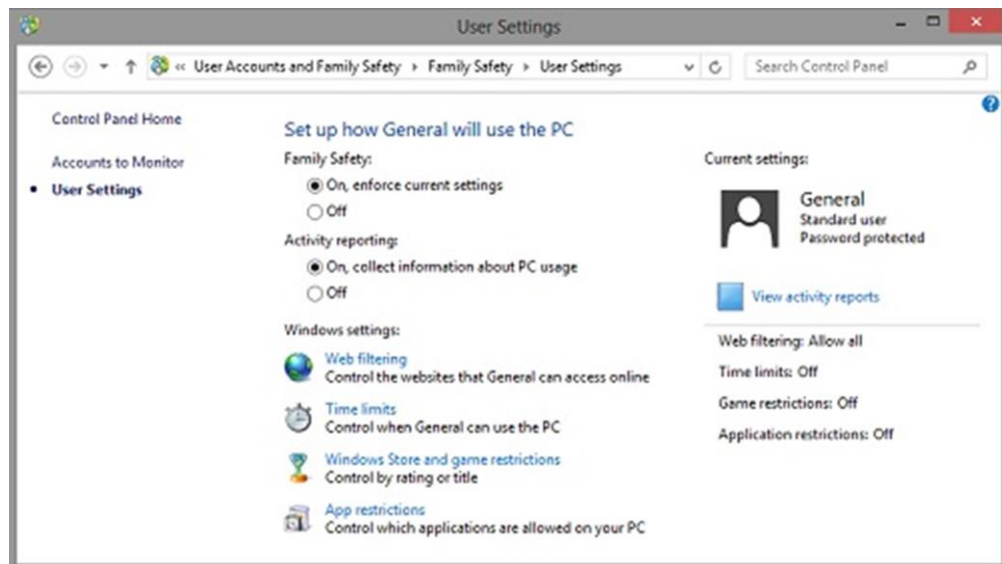


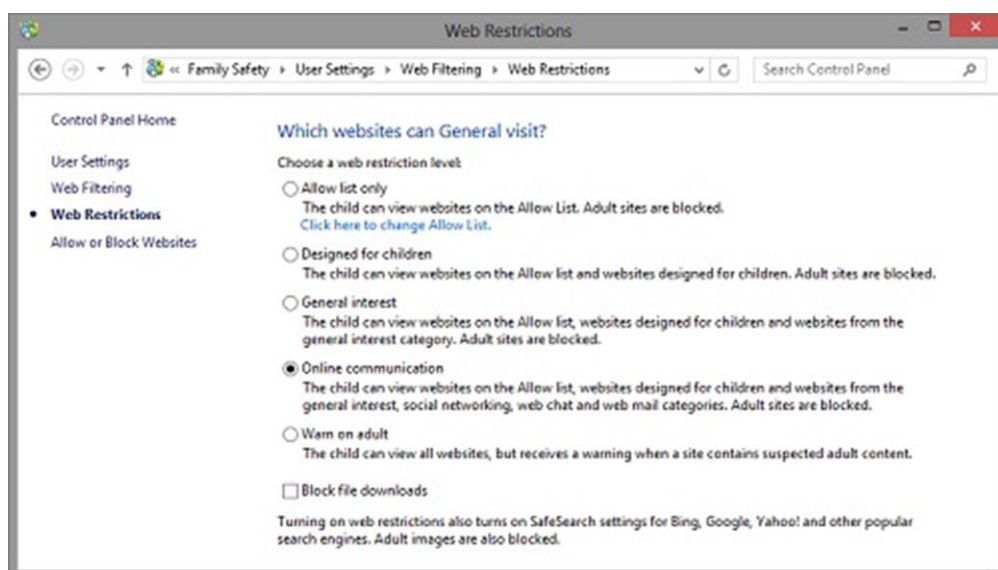
Fig. 2

To specify which websites the user can access click on “Web filtering”. From the interface that pops up, you have two options:

- **Open access:** user can access all website
- **Restricted access:** user can only access specified websites

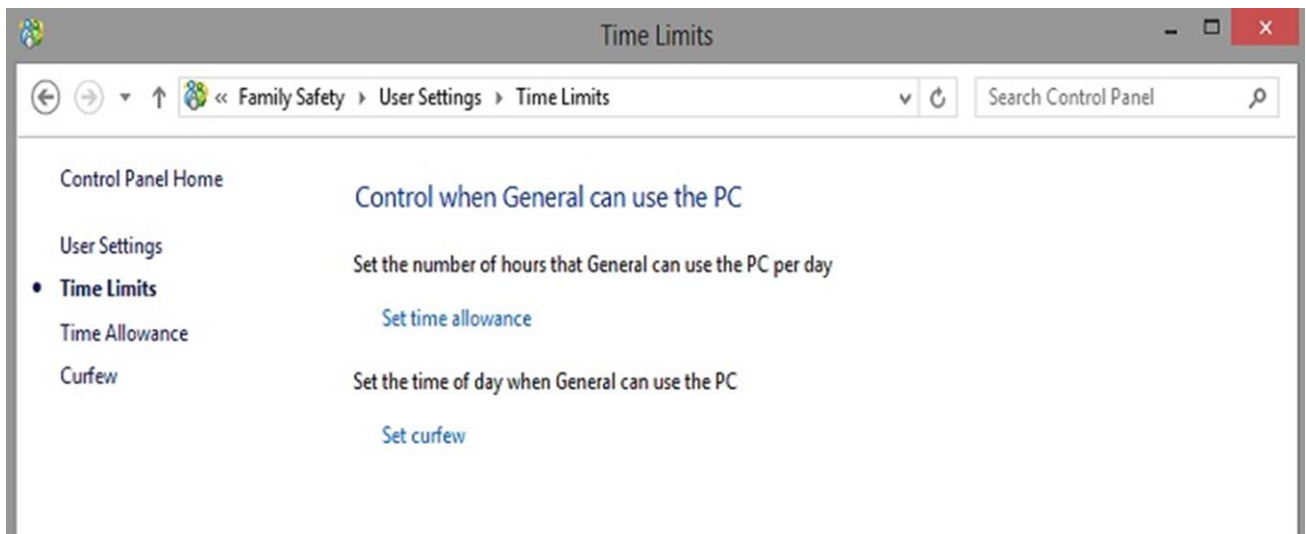
The default selection is for the user to have access to all website, to change this option; click on the option below the default to make the user access only specified websites.

When you click to specify the websites, you can either choose to restrict access by using rating and content types (click “set web filtering level” to do this) or by specifying the websites in a list (click “Allow or block specific websites” to do this). If you choose the first option, you have the screen below:



From the options above, you can make several decisions, each of the options have a description underneath. From the first option (Allow list only), you can create a list of website to allow the user through and a list of website to block, to do this, click on “Click here to change Allow list”, this would show you a page where you can supply website names and mark it as “allow” or “block”. You can check the box “Block file downloads” to block the user from downloading internet contents.

In family safety, you can also specify the time limit a user can spend on your PC, to do this, click on “Time limits” from Fig. 2 above and you should have the screen below:



From here, you can set the number of hours the user can use your PC and the time of the day the user can use your PC. When you choose “Set time allowance”, you are presented with two options to permit the user all day or specify a time limit; click to specify the time limit and then specify the number of hours during weekdays and hours on weekends the user can access your PC.

You can equally use the “set curfew” option to set the time range when the user can access your PC. To use this option, you have to drag your mouse to highlight a region from the graph shown (click on “Set curfew” to see the graph) which correspond to the range of time the user **can't** access your PC (highlighted area indicate restriction).

From fig. 2 above you can also set game restrictions and app restrictions. App restriction would enable you decide which apps the user can access on your PC, click on this option and edit it using the methods described for web limit above. You can also view a report about all events relating to the user’s activities on your PC by clicking “view activity report” from fig. 2 above.

NB: To use the controls that family safety offers in Windows 7, click on “Parental control” from “User Accounts and Family Safety” in Control Panel.

HOW TO CREATE A SECURE PASSWORD

A password is of little value if it's easily guessed by an intruder. Obviously, you shouldn't use your name or something equally transparent. However, even a random word provides little security against a determined intruder; this is because some hackers use tools that try every word in the dictionary. By observing the following guidelines, you can create a password that's difficult to crack in a reasonable amount of time:

- Use at least eight characters. Longer is better, which is why some security experts suggest using a pass phrase. A password or phrase can (and should) include spaces and punctuation; the maximum length is 127 characters.
- Use a mixture of uppercase letters, lowercase letters, numbers, and punctuation.
- Avoid writing words the way they exist in the dictionary.
- Avoid including your name or username in the password.
- Use random sequences instead of words, or intersperse numbers and punctuation within words: Sy5@3M &Dm!n m&d3 e&5y (System admin made easy) for example.

With a little thought, it's pretty easy to come up with a password that is memorable and secure. For example, start with a phrase about yourself or your hobbies—one that you can easily remember, such as “I'm addicted to Solitaire”. Make a few letter substitutions, misspell a word or two, and you come up with “I'm +Icted 2 \$ol!ta!re”. It's long, uses all four types of characters, contains no dictionary words, and is easy to remember—so you won't be tempted to write it on a sticky note attached to your monitor.

RECOVERING FROM A LOST PASSWORD

To recover from a lost password scenario in Windows, you have to create something known as a password reset disk BEFORE the password amnesia sets in. to create a password reset disk, you would need a removable media such as cd or USB flash which would serve as your key to unlock the PC. Once the media is available, attach it to your PC and open “User accounts and family safety” in control panel, click on the account you wish to create the reset disk for, from the left pane choose “Create a password reset disk” and follow the prompt.

Once you lose your password, from the logon screen supply a wrong password, Windows would inform you that your password is wrong and give you an option to reset it, you can then insert the removable media and click on reset password. Windows would then give you the option to change your password. You can use the password reset disk as many times as possible but once you create another one with another media, the old one becomes useless.

If you lose your password without creating a reset disk, you would need to contact an expert to bail you out.

WINDOWS FIREWALL

A firewall provides a barrier between your computer and the network to which it's connected by preventing the entry of unwanted traffic while allowing transparent passage to authorized connections. Antivirus softwares provides security against viruses on your PC i.e. prevents you from malicious softwares and files while firewalls monitors everything that goes on in your network, firewalls make sure you don't receive malicious packets and sometimes, you don't send malicious packets too (For the purpose of this book, everything that moves through the network is a packet though it has a more comprehensive meaning to networking experts).

Windows comes with a built-in firewall to protect you on the internet since most basic antivirus softwares do not have one installed, but once your antivirus software has a firewall, Windows firewall would automatically be disabled (except in some instances, probably when the software lacks some functionality especially IPsec protection) and network protection would be done by your antivirus. This is the case of most internet securities such as Norton Internet Security, AVG Internet Security and others, they are made up of both antivirus and firewall engines.

Windows firewall start-up immediately you boot your PC and takes the following actions by default:

- The firewall blocks all inbound traffic (anything coming in through the network), with the exception of traffic sent in response to a request sent by your computer and unsolicited traffic that has been explicitly allowed by creating a rule.
- All outgoing traffic is allowed unless it matches a configured rule.

Windows allow every outgoing packet by default because it believes you can only send malicious packets if your PC is infected and if your PC is infected it's the duty of your antivirus engine to take care of that but the firewall makes sure no malicious packet comes in to infect your PC by blocking incoming traffics reasonably.

NB: If you have a great difficulty accessing the internet or a particular website or having some internet related issues with your PC, it might be because your firewall is blocking that traffic (mind you, I said "MIGHT"). You might therefore need to amend some rules in the firewall settings which we would learn from the next section, one thing you can try is to disable your firewall and check if the problem still persist (remember you are open to all kinds of network related threat once firewall is disabled).

Managing Windows Firewall

This section would only explain how to block or allow an app or a packet through firewall and other simple management techniques.

To access Windows firewall, simply search for firewall from the start screen or open control panel, click on “System and security”, and then click on “Windows Firewall”, you should then have a screen similar to the one below:

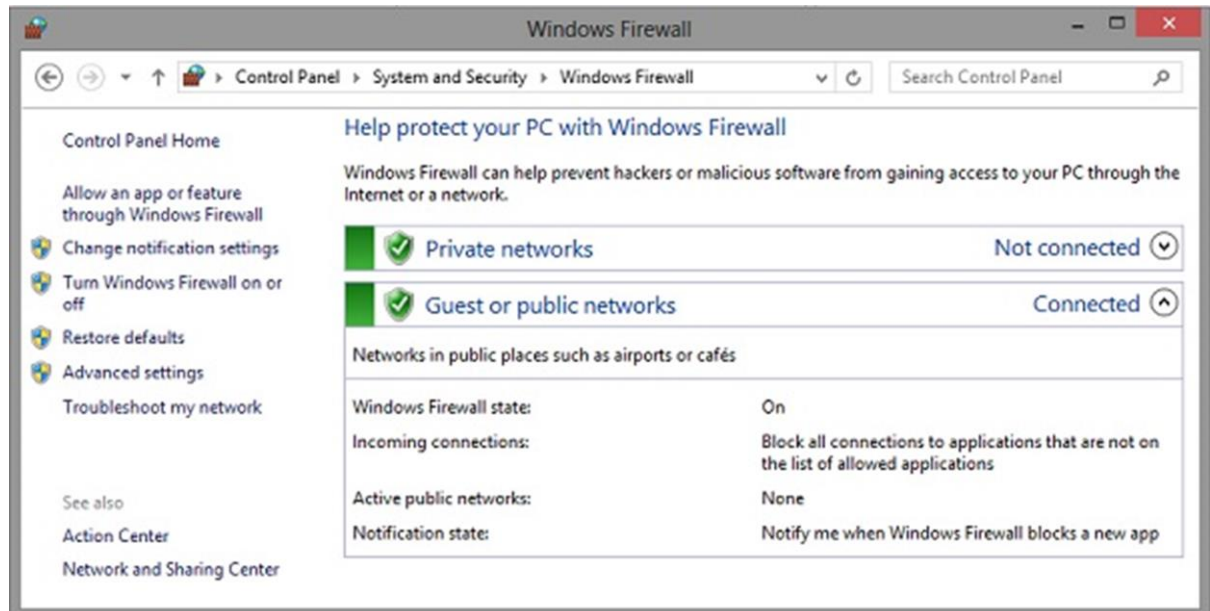
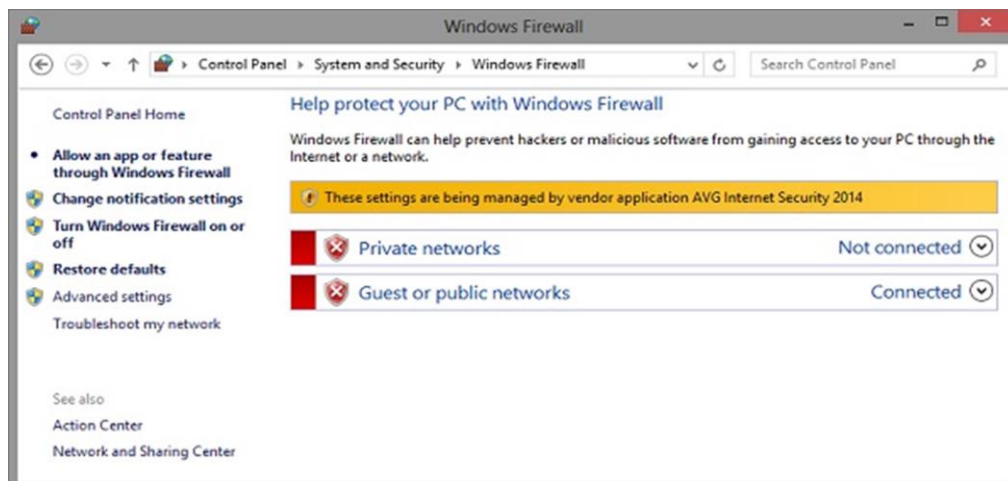
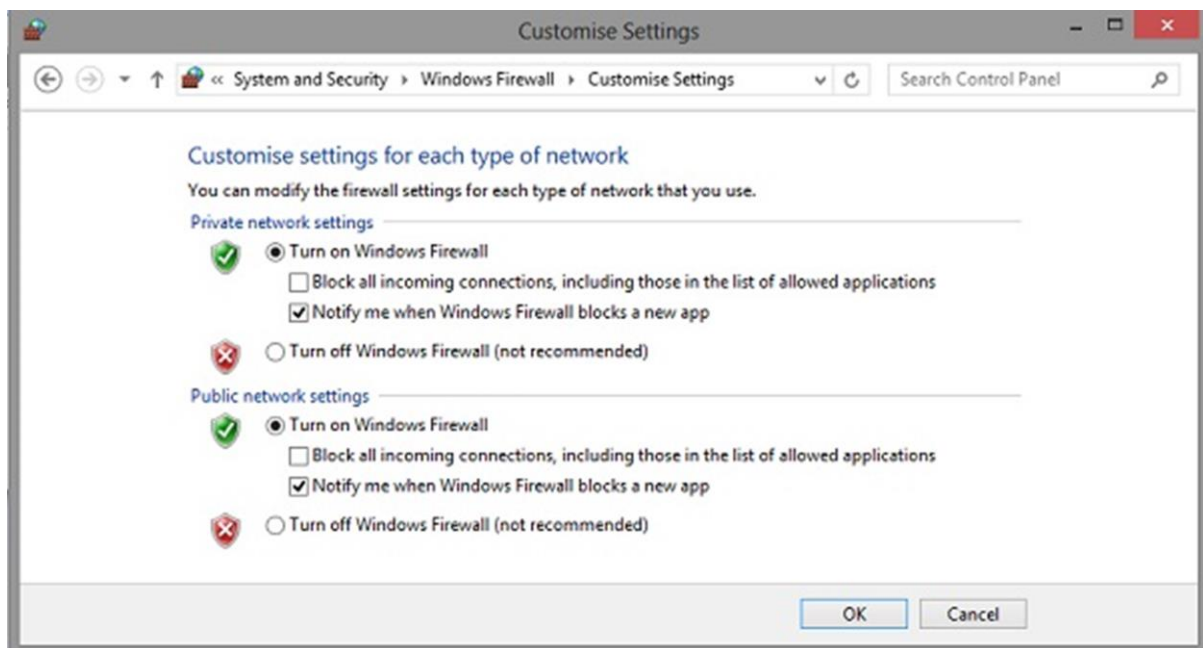


Fig. 3

If you have a third-party software managing firewall on your PC in which case you can't manage firewall with the explanations in this section but have to manage firewall from your antivirus interface, you would see a screen similar to the one below:

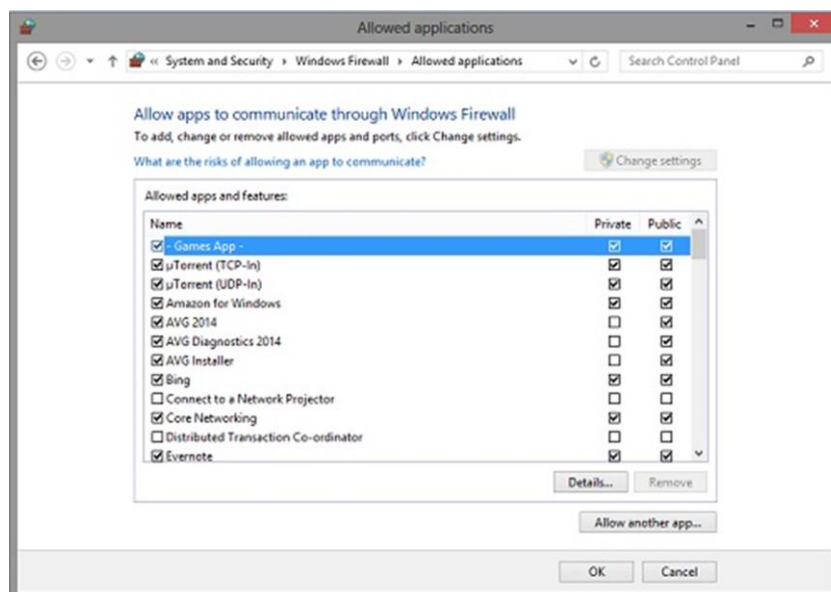


To enable/disable firewall, click on “Change notification settings” or “Turn Windows Firewall on or off” from the left pane of Fig. 3, you should have the interface below:



From this interface, click on “Turn off Windows Firewall” for the network you wish to turn off for (often, it’s both private and public network) and then click OK.

If there is an app on your PC that should have access to the internet e.g. a browser, but can’t browse, it may be because firewall has not allowed that app through the incoming traffic and can’t therefore receive packets. To allow an app through firewall, click on “Allow an app or feature through firewall” from Fig. 3, you should then get in interface below:



From this interface, click on “Allow another app” if the app/feature is not listed and you would be presented with an interface from where you can choose the app and then click OK to add it to the list of apps (You may need to click on Change settings to enable the “Allow another app” button). Once the app/feature is in the list, check the box for private and/or public in front of the app/feature to allow it through a private and/or a public network and then click OK.

You can also edit some advanced features by choosing “Advanced settings” from Fig. 3 and then choose to edit the inbound rules or outbound rules (You hardly have to configure any outbound rule(s) in windows firewall since Windows allow outgoing traffic by default).

If you have played around with your firewall settings, you can return back to the default settings which would change all rules you have configured to the default. To do this, click on “Restore defaults” from Fig. 3 above.

PERFORMANCE CONFIGURATION

This chapter covers:

-  Causes of poor performance in PC
-  How to analyse PC's performance
-  Finding and controlling troublesome app
-  Setting priority and affinity when starting an app
-  Speeding up start-up
-  Performance monitor
-  Virtual memory
-  Visual settings
-  Maintaining your battery

This chapter would explain how to enhance the performance of your PC, but before we get into the core of performance tuning, it is important that your PC has some basic configurations before we can think of making it work more efficiently.

The common causes of poor performance in a PC are highlighted below:

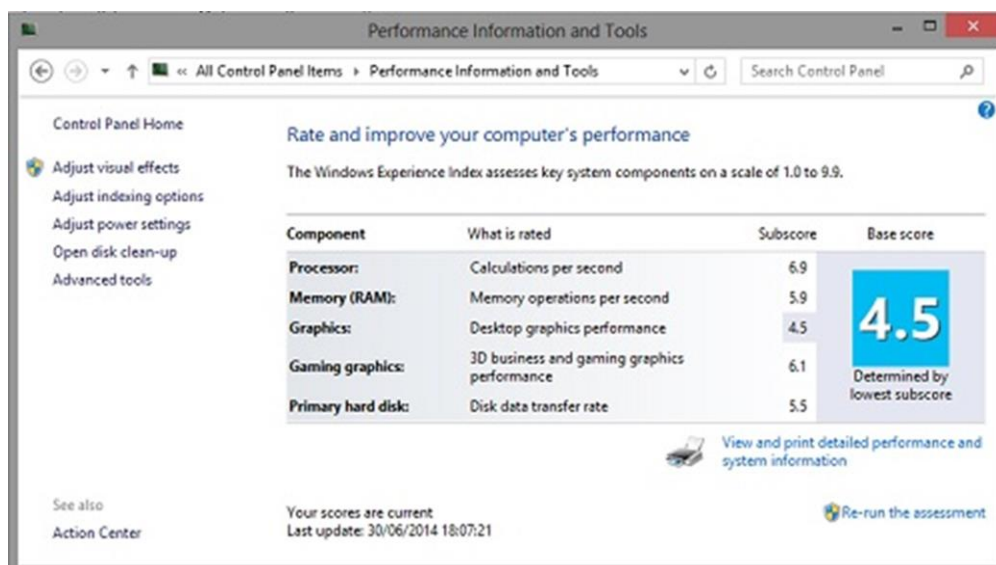
- **Defective hardware:** Just as memory & disk errors are the primary cause of system crashes, hardware-related problems also cause performance drag in a PC. You should run a check on your hard drive and run a memory test on your PC to confirm that both are functioning properly. Memory tests vary from PC to PC, you can check with your PC manufacturer on how best to evaluate your PC. Windows also comes with a tool to diagnose problems with your memory, to use this, search the start screen for **memory** and run “Windows memory diagnostic”; click “restart now and check for problems”.
- **Out-dated or flawed device drivers:** An out-dated driver might be the cause of performance degradation.
- **Inadequate hardware resources:** Sometimes, the hardware resources your system has can't complement for the works you expect the system to do (e.g. you may not have enough graphics memory to run game software).
- **Active background tasks:** Some system default apps, Windows maintenance app, antivirus softwares and some other apps run in the background and could lead to performance degradation, which is one of the reasons you should not use more than one antivirus software. When your PC is running slow, you might need to reduce the number of background tasks and the rate at which they run.

- **Out-of-control processes or services:** Sometimes, a program or background task that normally runs just fine will spin out of control, consuming up to 100% of CPU time or grabbing increasing amounts of memory or other system resources. In the process, of course, performance of all other tasks slows down or grinds to a halt. Knowing how to identify and kill this sort of process or service and prevent it from recurring is a valuable troubleshooting skill.
- **Malware:** Viruses, Trojan horse programs, spyware, and other forms of unwanted software can wreak havoc on system performance.

How do I analyse my PC?

To analyse your PC for performance rate/level do the following things:

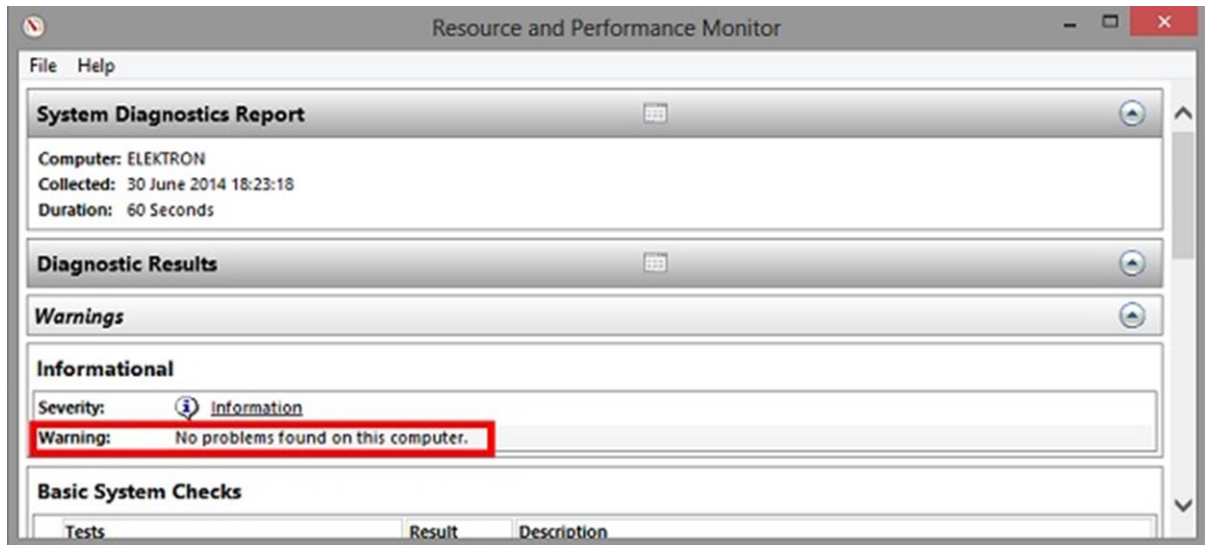
- **Use Windows Experience Index (WEI):** You can use WEI to analyse how efficient your PC is performing. WEI analyse your PC and rate it on a scale of 1.0 to 7.9 on Windows 7 and 1.0 to 9.9 on Windows 8 systems. To view this, go to system properties and click on “Windows Experience Index” beside ratings or go to control panel, view by large icons or small icons and choose “Performance Information & Tools”, you should have a screen similar to the one shown below:



The rating you would see as the “Base score” is the lowest sub-score in the rating. To run this test using the command line tool, open an elevated command prompt and type **winsat query** or **winsat formal** (while running on AC power), these two would give a more comprehensive analysis of system performance.

- **Generate a system health report:** This is a report that would tell you how efficient your PC is performing and what you should do to enhance it. To do this, open control panel (viewing by large icons or small icons) and then click on “Performance Information and

Tools” and from the left pane (as shown in the image above) click on “Advanced tools”. From the screen displayed, click on “Generate system health report” and wait for 60s for the report to be generated.



Check the region highlighted in the interface above to see if your PC have any error. If any error is detected, scroll down to “Basic system checks” section and click on the plus signs to see the error(s). Some other sections might also contain error descriptions.

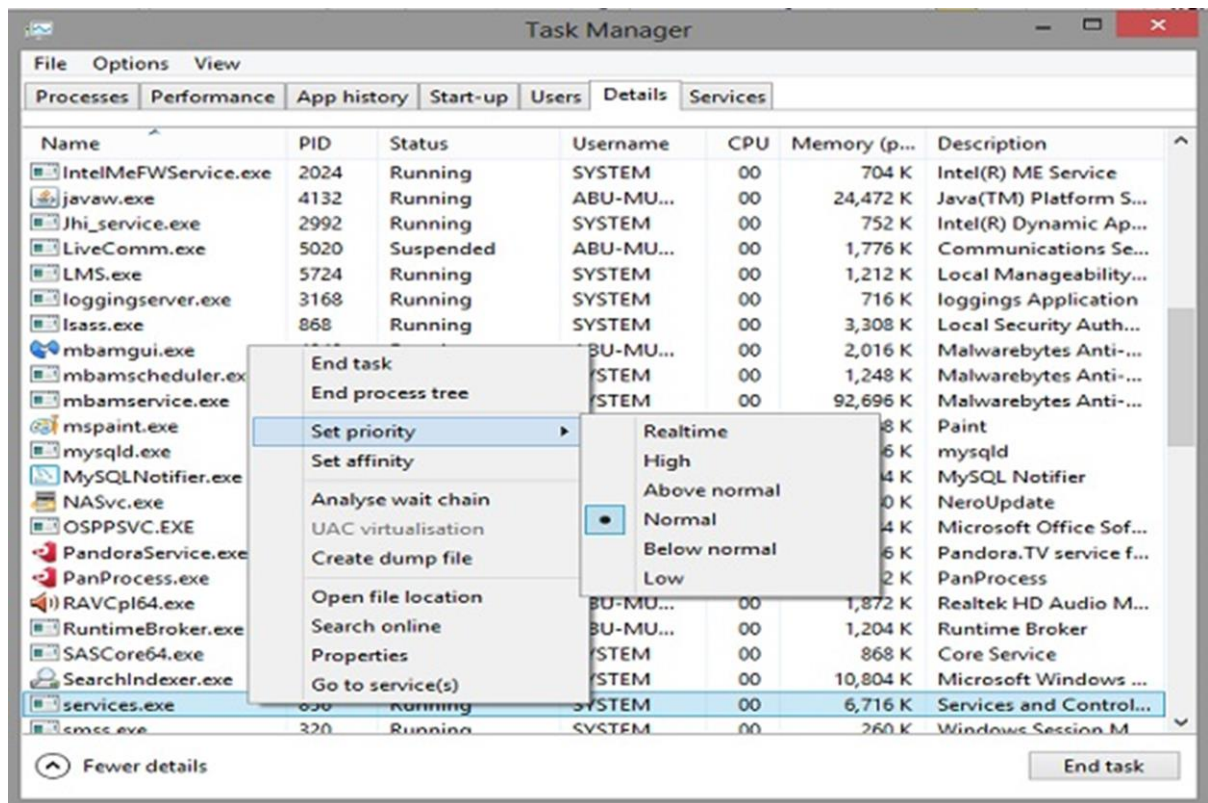
Finding and Controlling Troublesome App

Troublesome apps can be controlled by the Windows Task Manager (some of the explained features are not contained in Windows 7). To start the task manager, press a combination of **ctrl + shift + esc** keys or press **ctrl + alt + del** and then choose task manager or right click on the task bar and choose “Task Manager” or on Windows 8, press a combination of Windows logo key and X and then press T or choose Task Manager. In Windows 8, click on “More details” from the bottom of the task manager’s interface (if you need to do so).

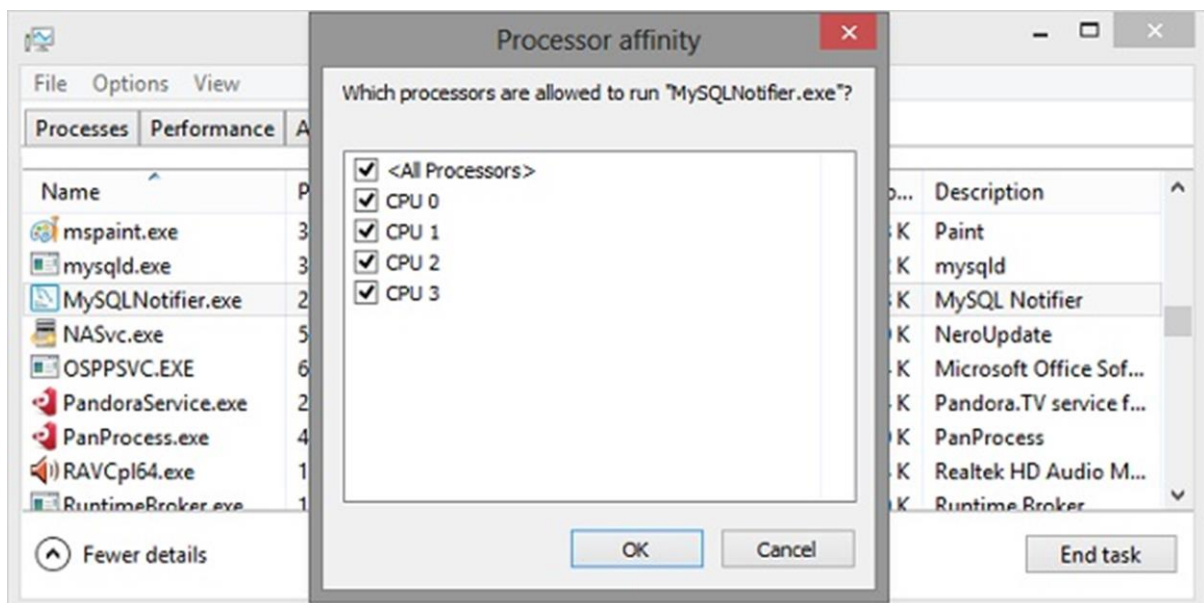
From the processes tab, click on the CPU column heading to display the running processes in order of CPU usage, in that case the app using the most CPU time would be on top of the list. You can stop the app by clicking on it and choosing “End Task” from the bottom of the screen or right click on it and choose End task.

If you want to allow the app to continue running, you can minimize the impact it has on your PC’s performance by lowering its priority. In Windows 7, right click on the process, click on Set Priority and choose Below Normal or Normal. In Windows 8, right click on the process from the Processes tab and click on “Go to details”. This opens the Details tab and selects the app’s service in that tab. Now, right-click the process’s service name (which might

be different from the name displayed on the Processes tab) selected by task manager itself, select Set Priority, and click “Below normal” or “Low” as shown below:



If you want to prevent an app from using 100% of your processor time, even if it's the only app running, you can adjust the process' affinity. Follow the series of steps described earlier to find the process in the Details tab, and then, right-click the process and select “Set Affinity” as shown below, use the Processor Affinity dialog box to specify which processors the app is allowed to use. To limit an app to half your total processing time, select only half of your processors.



Setting priority and affinity when starting an app

Windows doesn't remember your priority or affinity settings after you close an app. The next time you start it, Windows will run the app at normal priority on all processors. To manually set the priority or affinity for an app, use the **start** command-line tool. For example, to start Notepad with low priority and allow it to run on only one processor, you would run the following command at a command prompt:

start /low /affinity 0x1 %windir%\System32\Notepad.exe

Where %windir%\System32\Notepad.exe is the directory where notepad resides.

To have Notepad run with below normal priority on two processors, you would run the following command:

start /belownormal /affinity 0x3 %windir%\System32\Notepad.exe

Affinity codes for different processor quantities:

No	Affinity Code
1	0x1
2	0x3
3	0x7
4	0xF
5	0x1F
6	0x3F
7	0x7F
8	0xFF

To create a shortcut for starting the app with the desired configurations without opening command prompt, use this command from command prompt:

echo start /belownormal /affinity 0x3 %windir%\System32\Notepad.exe > slowapp.bat

This would create a file “**slowapp.bat**” in your C:\Windows\System32 folder or the directory the command prompt points to. To create it into another folder, change the **slowapp.bat** from the command above to the path you wish e.g. E:\slowapp.bat, the file would then be in your drive E. You can change the name of the file to whatsoever you wish but with the .bat extension.

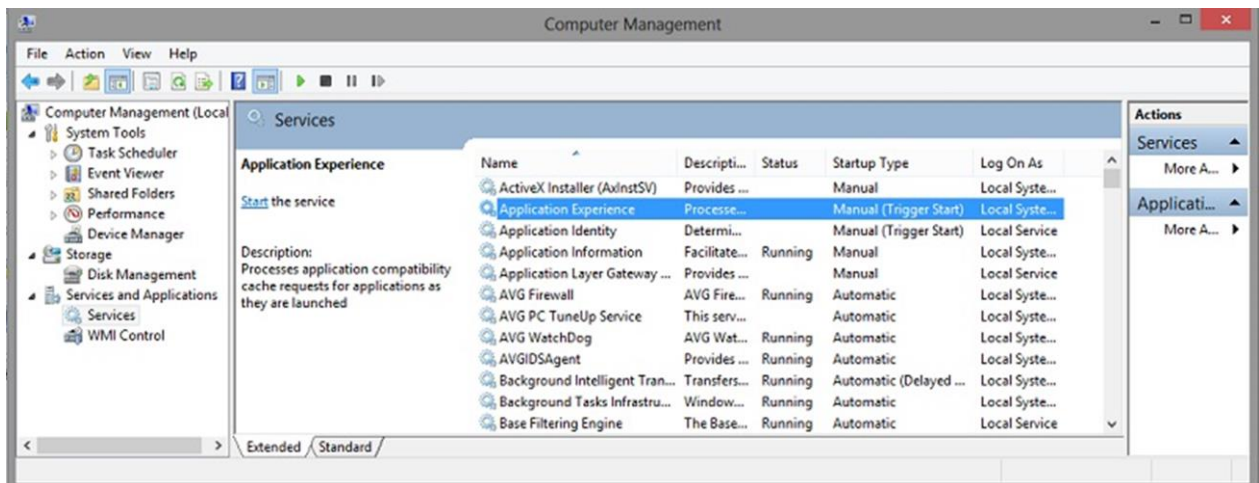
Speeding up start-up

To speed up the rate at which your PC starts, do the following:

- **Remove start-up apps:** Start-up apps reside in system configuration for Windows 7 and in task manager in Windows 8. For Windows 7, search for **msconfig** from the start

menu and then click on the start-up tab; and for Windows 8 launch the task manager and click on the start-up tab. Every app with “Enabled” in its status column would run immediately your PC start-up, Windows also have start-up impact column which tells you the impact that particular app have on your PC’s start-up. If you wish to disable the app, click on it and click on disable (be very careful when doing this, if you don’t know the function of the app, don’t stop it, especially antivirus softwares-don’t try it if you are not very sure-).

Some services (apps that run in background without a user interface) also start automatically when you start-up your PC and stopping some of them might help hasten up how your PC start. To see such services, open computer management (search for **compmgmt.msc** from start menu), expand “Services and Applications” and click on services to get the interface shown below.



If you scroll through the start-up type column, you’d notice four different options:

- **Manual/Manual (Trigger start):** This type would only start-up when they are needed i.e. when you run the app that uses that service. The service(s) would not start-up automatically.
- **Automatic (Trigger start):** This type would only start-up when they are needed i.e. when you run the app that uses that service. The service(s) would not start-up automatically.
- **Automatic (Delayed start):** would start-up once you have launched your first app successfully.
- **Automatic:** start-up automatically.

To change the start-up type, double click on the service and change it from the start-up type box. You can choose any type of start-up once you are sure of what the service does, just make sure you don’t change the start-up type of any service to manual or automatic (trigger start) if the service does not have an app that can launch it because

it would never start and might affect your PC. Similarly, make sure you don't change the start-up type of Windows services.

Choose Manual if you don't want the service to run automatically; you'll be able to start it manually whenever you need it by launching the app that owns the service. Choose Automatic (Delayed Start) to allow the service to start shortly after start-up so that it won't impact your start-up time but will still be available when you need it—as long as you don't need it immediately after logging on to your PC.

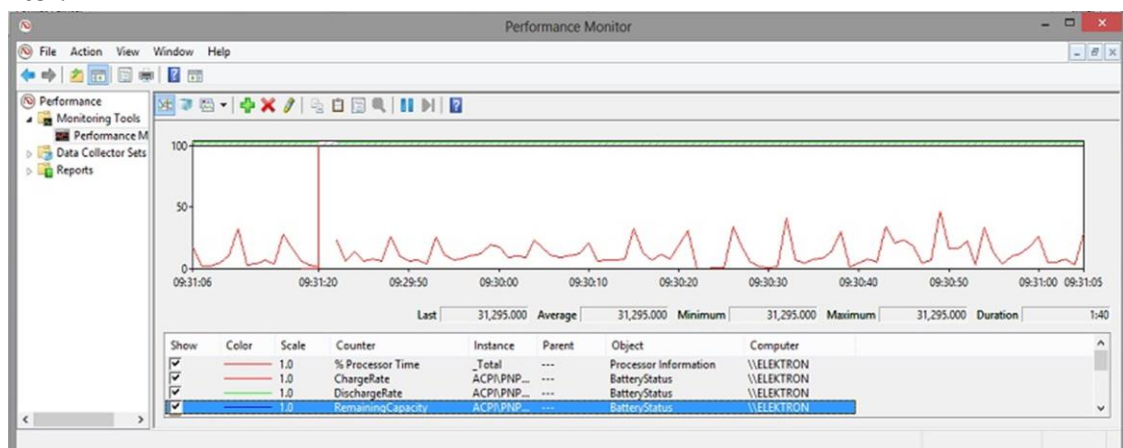
Windows 8 comes preconfigured to start the following services automatically. If you change their start-up type, something won't work right:

- Background Tasks Infrastructure Service
- Base Filtering Engine
- COM+ Event System
- Cryptographic Services
- DCOM Server Process Launcher
- DHCP Client
- Diagnostic Policy Service
- Distributed Link Tracking Client
- IP Helper
- Local Session Manager
- Multimedia Class Scheduler
- Network Location Awareness
- Network Store Interface Service
- Power
- Print Spooler
- Program Compatibility Assistant Service
- Remote Procedure Call (RPC)
- RPC Endpoint Mapper
- Security Accounts Manager
- Server
- Shell Hardware Detection
- SuperFetch
- System Event Notification Service
- Task Scheduler
- Themes
- User Profile Service
- Windows Audio

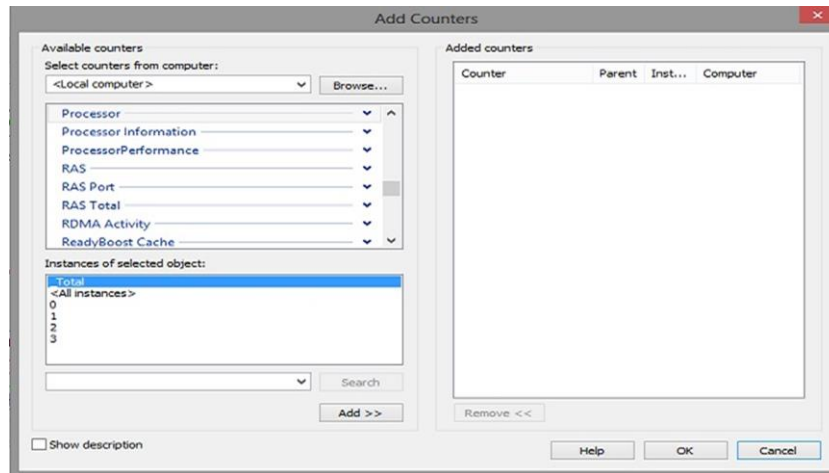
- Windows Audio Endpoint Builder
 - Windows Event Log
 - Windows Firewall
 - Windows Font Cache Service
 - Windows Image Acquisition (WIA)
 - Windows Management Instrumentation
 - Windows Modules Installer
 - WLAN AutoConfig
 - Workstation
 - And many more
- **Disable unused hardware:** When Windows starts, it loads software drivers that allow it to communicate with every hardware component in your PC. If your PC has hardware components you never use, disabling them in Device Manager can improve your start-up time, if only by a few milliseconds. To disable a hardware device, launch Device Manager by running **devmgmt.msc** at the Start screen. Then, right-click your device and click “Disable”. To use it again, right-click it and then click Enable.
- **Change your boot order:** When your PC starts, BIOS would search for a valid operating system in every device in your boot order until it finds one, so placing the hard disk at the top of that list would help speed up booting since BIOS would now check your hard disk first and should find an operating system therein (Refer to chapter one to change your boot order). The change might be fractions of a second though!

Using Performance Monitor

Performance monitor is a Windows utility for monitoring performance in real time. To start performance monitor, search for **perfmon.msc** from the start screen, in the performance monitor window, expand “Monitoring tools” from the left pane and select performance monitor.



Below the screen is the list of events/devices you are presently monitoring. To monitor more events/devices, click the plus (+) button on top of the screen.



Choose the device from the list and click “Add”, once you are through adding all devices, click “OK”.

You can change the graph type simply by double clicking on any of available device and choosing the graph type you want, selecting report would give an analysis of each device in form of report.

Virtual Memory

Sometimes your PC need more RAM to execute some task you wish to execute than your PC presently has, to make up for these types of events, Windows creates something known as page file on your hard disk which is used as a virtual memory. This hard disk space is used to complement for the PC’s RAM such that when you run out of memory Windows can make use of the specified space on the hard disk as a virtual memory.

Every PC running Windows has this configured automatically but you can decide to increase the size of your virtual memory.

To open virtual memory settings, open Windows explorer, right click on any white space and click properties to open system properties, from the left pane, select “Advanced System Settings” and open the “Advanced” tab. You should have the interface in Figure A below.

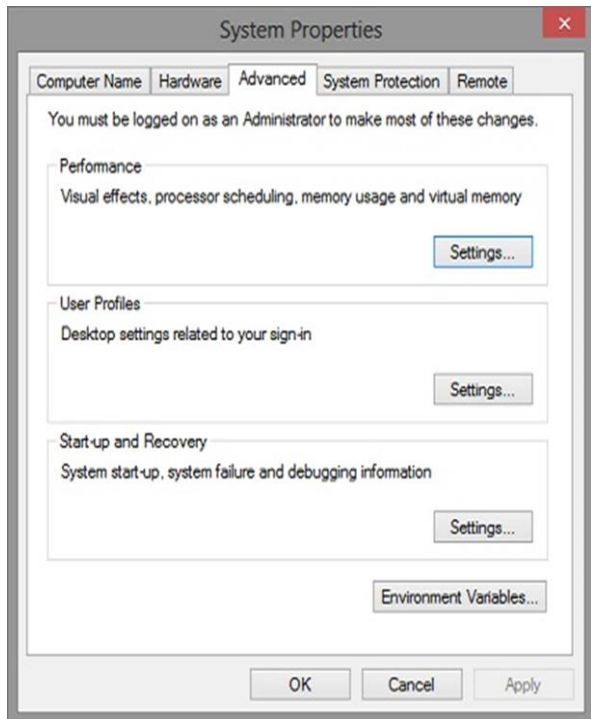


Figure A

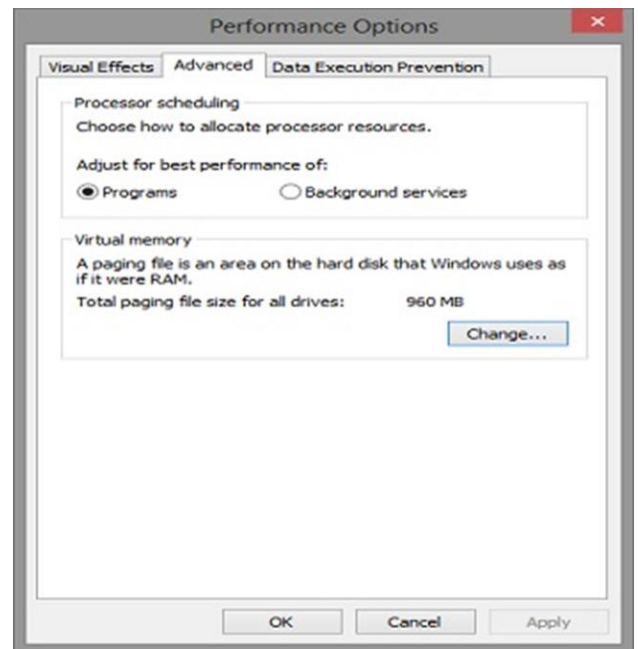
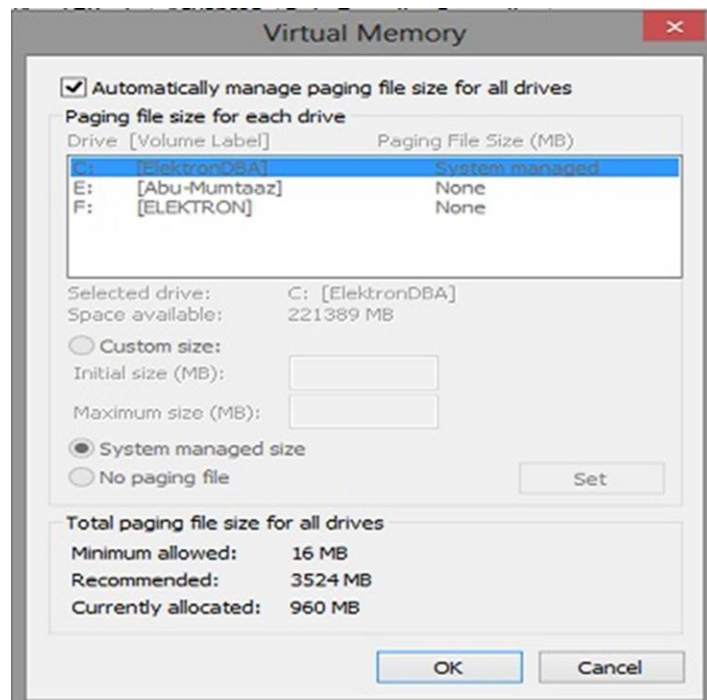


Figure B

From the “Performance” section in Figure A, click on settings to get the interface in Figure B which contains the “virtual memory” section, you can click “change” to open the virtual memory settings.



Notice that the virtual memory settings are managed automatically; to alter the settings uncheck the box “Automatically manage paging file size for all drives”. You can then select “Custom size” and change the initial size and maximum size to the value you want in MB.

Visual settings

The basic visual settings of a PC are always set to automatic which enables Windows to adjust visual settings as required by events.

Sometimes when the PC gets very slow, it would do a great favour to change the visual settings to be adjusted for best performance of the PC or adjusted for the best appearance of the PC or use custom setting.

To adjust visual settings, open control panel (view by large or small icons) and select “Performance information and tools”. From this interface, select “Advanced Tools” from the left pane and then select “Adjust the appearance and performance of Windows” from the list of options. You can then choose any of the available options and select “OK”. Once the visual settings change, the PC’s speed would also be altered.

Maintaining Your Battery

Rechargeable batteries lose capacity over time. If you can get nine hours of battery life when your PC is new, you might be down to five hours of battery life after two years. After five years, your battery might be completely unusable.

There are a few things you can do to improve the lifespan of your battery:

- ❖ Use software from your hardware manufacturer to intelligently charge your battery. Some hardware manufacturers offer settings that charge your battery more slowly above 50%, which can increase your battery’s lifespan. This feature might be called “desktop mode.”
- ❖ Store your batteries at 40% charge and keep them in a cool place. If you’re not going to be using your mobile PC for a few days or more, allow the battery to discharge to a little less than half before powering it down. Don’t store your PC in a hot car.
- ❖ Every two or three weeks, run the battery down to empty. Batteries work better when they go through a full cycle on a regular basis. If you often run your PC without being plugged in, this shouldn’t be a problem. If your PC is constantly plugged in, however, it’s a good idea to unplug it once in a while and wait until the battery is almost empty before plugging it back in.

BACKUP AND RECOVERY

This Chapter covers:

-  Why and how to backup files
-  Windows file recovery
-  Creating system image
-  Creating system repair disk
-  Restoring system images
-  Refreshing your PC
-  Resetting your PC
-  System restore
-  Last resort recovery

Losing files on a PC is always a tragedy and often times the user would not be aware of the possibility of such occurrence. The drive might sometimes fail completely such that recovery would no longer be possible or Windows crash and the user are left with few recovery options. To avoid these occurrences, it is pertinent to save a backup of files on a drive onto another drive known as the backup drive.

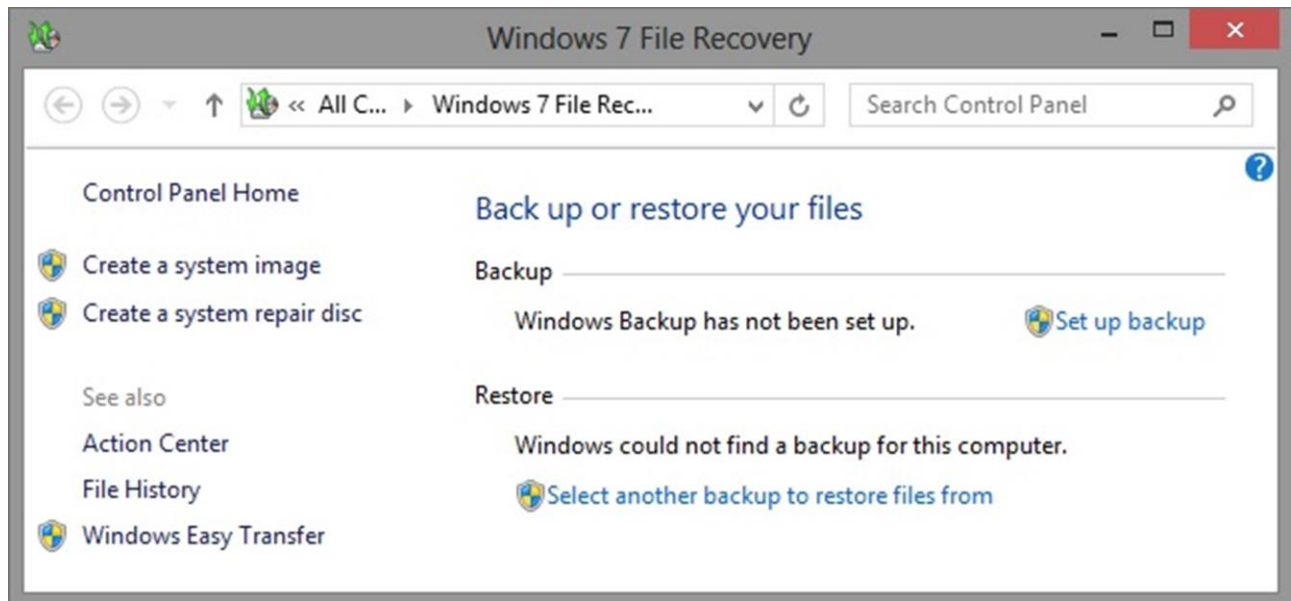
Windows has provided a means for continuous backup of the content of a drive occasionally to a removable drive, cd or another hard drive or creation of a system image from which Windows can be recovered in case of crash. Windows crash implies your copy of Windows would no longer be accessible and you can only recover it with an already created system image (i.e. your PC would be returned back to the state it was when you created the image) or reload Windows completely.

Now let's start backing up our files! What exactly do we need?

We would only simply need a backup drive. Your backup drive can be the same drive upon which Windows currently reside (provided you have a partition on it) but as a precaution, make sure you backup your files to a separate drive apart from the one your copy of Windows presently reside (i.e. A different hard disk or a removable media), this is because when you create a backup, you are preparing to recover your files from that backup in case the drive become inaccessible (as the worst case scenario). Now ask yourself; if your drive becomes inaccessible, and that same drive hold your backup files, where do you want to get your files from? Now you see the reason why you should keep your backups separate?

If you have gotten your backup media ready, then let's start learning.

Open your command prompt, type **sdclt** and press enter or open control panel and choose “Windows 7 file recovery” in Windows 8 (in Windows 7, open control panel, click on System and Security and click on Backup and Restore). Once that is done, you should have the interface below:

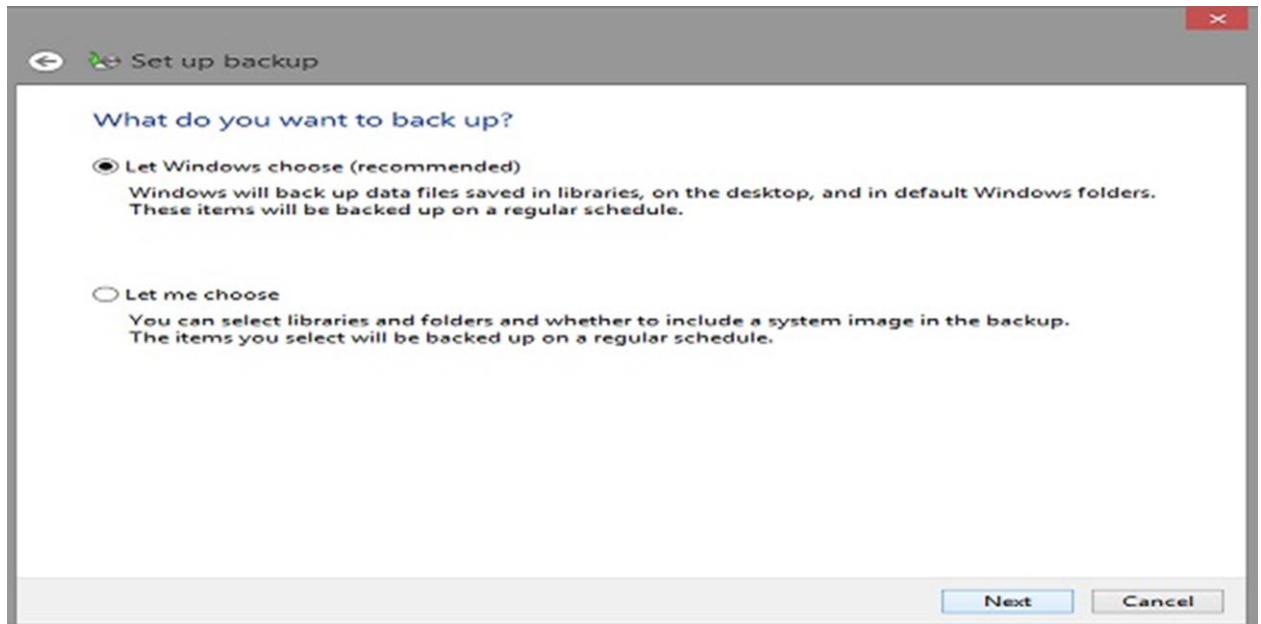


Notice from this interface that you’ve got an option to:

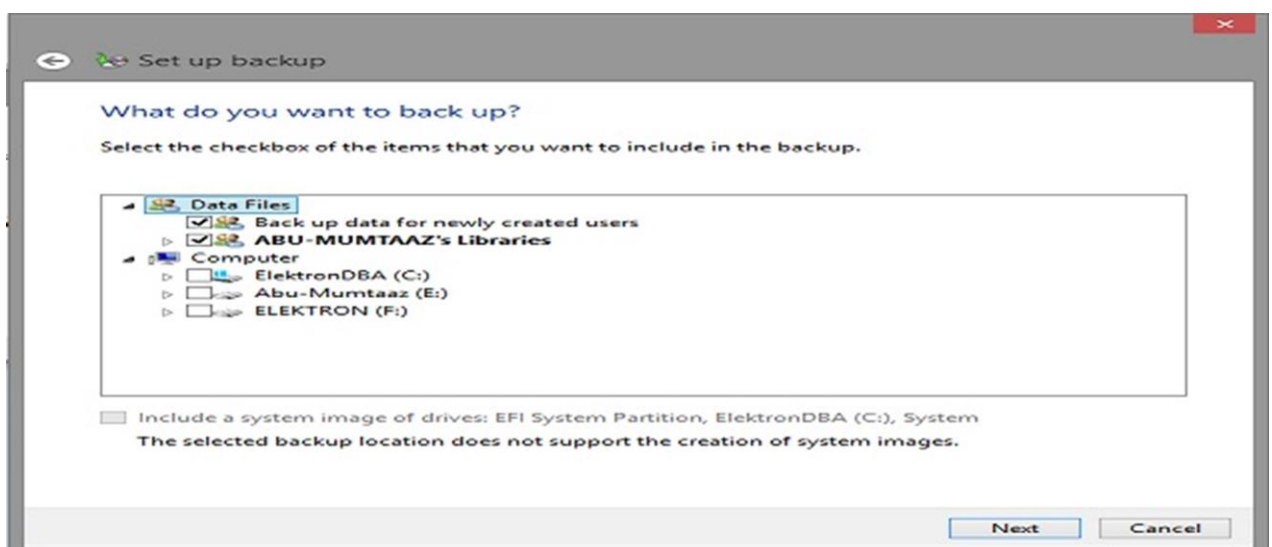
- **Set up backup:** This would enable you to choose a drive upon which backup would be made and then Windows would make backups on that drive subsequently i.e. once you start it, a backup would be made and occasionally (maybe weekly), backups would also be made to that drive.
- **Create a system image:** Just like I explained earlier, you can recover your copy of Windows from this image in case of a crash. It is done only once unlike a backup but you can do it whenever you wish i.e. you can decide to create an image every week.
- **Create a system repair disc:** A repair disc is used to repair Windows in case of failure. It’s always a good idea to create one as a precaution so that you can repair your copy of Windows with it especially when Windows fail to start. To do this (i.e. repair Windows after failure), you’d need to access your repair disc from Windows recovery environment (this is discussed later in this chapter).
- **Select another backup to restore files from:** Once you have a backup, you would use this option to recover your lost files or restore it back to an old copy in case of loss or any other occurrence.

Recovery simply means that you are restoring/returning your data or whatever you backed up to the state it was when you created that backup. So if you backup a document on the 10th of March, and your drive fails on the 25th of April, don't expect your documents to be in the state they were on the 24th of March after you restore it. This is an implication that you must make reasonable and frequent backup.

Now insert your backup media and click on "Set up backup", you should get the screen below:



From this screen, Windows expect you to choose what you want to backup; a decision which can either be made by Windows or you yourself. If you ask Windows to decide for you, then backup would begin but if you wish to decide on your own, click on "Let me choose" and click next. You should have a screen similar to the one below:



Check the checkboxes for all the items you wish to include in the backup and check the checkbox below the screen to add a system image file you created earlier (this would enable you to back up your system image so that you can recover it in case of disaster. The box would be enabled only if the drive you insert can accommodate image files).

Maybe I need to talk about this. How can data loss occur on a PC?

The common disasters that can lead to data loss on a PC include:

- **Deleting a file accidentally:** Deleted files are sent to the recycle bin by default (if you don't choose the permanent delete option), but you can also configure the recycle bin such that files would not stay there, once you delete it, it is deleted permanently.
To configure your recycle bin, open recycle bin, from the menu bar in recycle bin, choose recycle bin properties, click on "Don't move files to the recycle bin. Remove files immediately when deleted" and click on OK. Once you do this, pressing delete key on your keyboard would automatically delete the selected item permanently.
- **Accidentally messing up with a file:** Let me explain this with this scenario: when I started learning MS Word, my boss gave me a question to start typing (a whole 60 leaves for that matter) but to continue from where someone stopped without introducing me to "how to use a mouse", somehow I deleted the whole document (about 50 pages. I think I pressed ctrl+A and then delete) and continued typing only to find out after about 2 hours that I had messed up with the document. There was no way to make any recovery since there was no back up and I had saved the document several times and we also had power outage (meaning the document closed and reopened) without a backup. It was a great mess, but they couldn't beat me!
- **File corruption:** Files can also get corrupted due to failing hard drive, fluctuating power etc. which would require recovery from a backup.
- **Failed disk:** Hard disk can fail either partially or totally leading to a great loss to some or all of its content.
- **Theft or loss:** Your PC can be stolen or your hard drive stolen, if you have no backup it's going to be a great loss.
- **Natural disaster:** Fires, earthquakes, flooding, hurricanes, and many other events can cause catastrophic damage. Your PC might get lost, but your data doesn't need to be if you back it up properly.

The backup method discussed earlier is just sufficient to save you from most disaster. When you create a backup as explained earlier, you are backing up the data on your drive but what if Windows crash? Windows also reside in the same drive you backed up data from but

you didn't back Windows up. To back up your copy of Windows, you have to create something known as system image (I mentioned this earlier).

To create a system image, attach a hard drive, and open "Windows 7 file recovery" as described above, choose "system image recovery" from the left pane and follow the prompt to create the image.

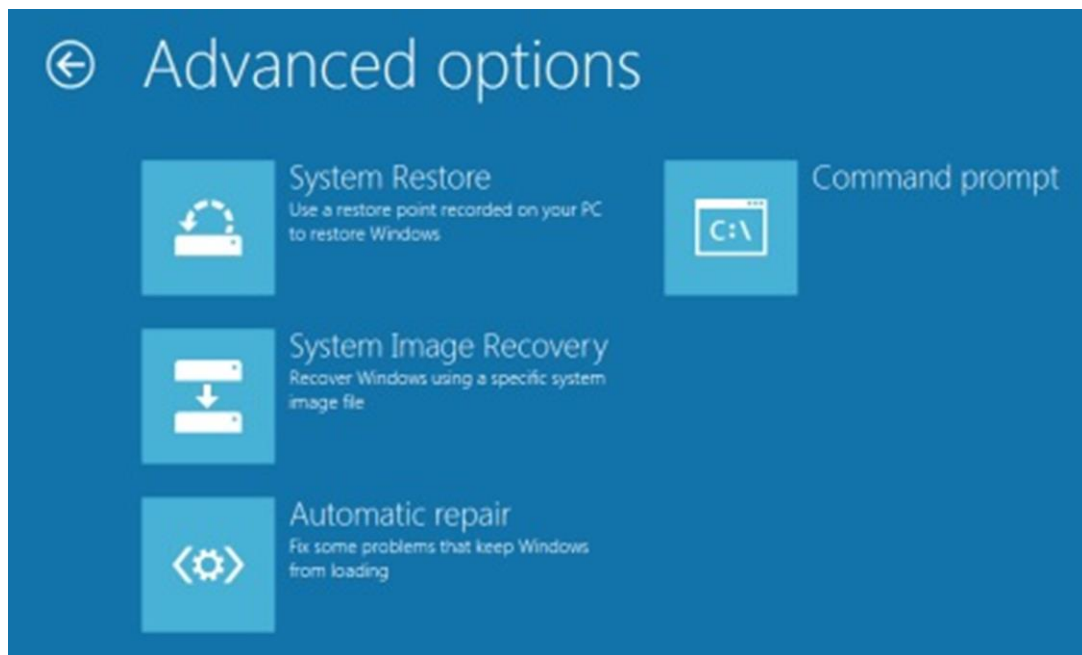
Another option that is available in Windows 8 is "File History" which is also available in control panel. Open it in control panel and follow the prompt just as described for backup.

You should test your backup once in a while by restoring it to confirm that you made a valid backup. It's not going to be funny if you fall into any of the aforementioned disaster just to find out that your backup isn't working; it's going to be a catastrophic tragedy!

Restoring System Image

If Windows still start, then boot to advanced boot menu. How?

If your copy of Windows is Windows 8, then press a combination of Windows logo key and "I" key and then click "Change PC Settings" to open PC settings and then click on "general" from the left pane, scroll down the right pane and click on "restart now" under "Advanced start-up" to launch the advanced boot menu. Once the advanced boot menu option is displayed, click on "Troubleshoot" > click on "Advanced Options" > click on "System Image Recovery" to restore with a system image (The path might differ in your own copy of Windows but would be very similar). The screen is shown below:



For Windows 7, restart and hold down F8 or open command prompt and type **shutdown /r /o /t 0** and hit enter (if both does not work, check with your PC's manufacturer to see how to boot to advanced boot menu). Once the advanced boot menu is up, choose to recover from a system image.

If Windows don't start, then do these:

Get a Windows installation disc or a flash containing a copy of Windows that is the same as the copy of Windows on the PC you are trying to recover and then boot from that media until you see a screen similar to the one shown below (Read chapter 1 –clean installation- for instructions on how to do this):



Once this screen is up, click on “Repair your computer” and then choose to recover from a system image. If that option is not included in the next screen, then choose next to view other options.

If you would love to see the start menu shortly before Windows load every time you boot your PC, then run command prompt as an administrator and type the command below (do you still remember how to do that? Search start menu for cmd, right click on it and choose run as administrator or press ctrl + shift + Enter):

Bcdedit /set {bootmgr} displaybootmenu yes

This would display the start menu for 30s every time you boot your PC.

Windows 8 also come with two other recovery options for your consumption, these are “Refresh” and “Reset”.

Refreshing your PC

When you refresh your PC, you are restoring it back to the state it was when a particular refresh image was created (something similar to using system image) if you had created one. By default, Windows 8 include a refresh image of your operating system with your apps (but not desktop apps) on your hard drive; this image would be used if you have not created one yourself. To create a system refresh image yourself, run command prompt as an administrator and type the command below:

recimg /CreateImage C:\Folder

Where C:\Folder is the location where the image would be saved (Windows would create it if it doesn't exist yet), you can change this to whatsoever you wish.

To Refresh your PC:

- Press a combination of Windows logo key and “I” and then choose “Change PC settings”
- From the left pane, select “General”
- From the right pane, scroll down and select “Get started” under “Refresh your PC without affecting you files”.

Note that the last refresh image you create (when you have created multiple images) or the default (created by Windows) is what Windows would use for refreshing. You can decide which image to use when you have created multiple images. To do that, run command prompt as an administrator and type:

recimg /setcurrent <directory>

Where <directory> is the location of the image you want to set as the current image (since your images exists in different directories e.g. C:\folder1 for image 1, C:\folder2 for image 2 and so on).

To delete an image you created, use the command **recimg /deregister** or simply locate the image in the directory you specified (it should have a .wim extension) and delete it.

When you choose to refresh your PC, Windows 8 would do the following:

- Starts Windows RE (Recovery Environment)
- Saves user settings and files to a location on the same disc.
- Reinstalls a fresh copy of Windows 8
- Restores user files and settings
- Restarts the new installation of Windows 8

NB: You are not going to see all these happening, Windows would only tell you that refresh is in progress and once it is done, you'd see your PC start up.

The items that would be restored back are:

- Windows 8 applications
- Desktop wallpaper and other personalization settings
- Wireless network connections
- Mobile broadband connections
- Drive letter assignments (i.e. the letter you assign to your partitions)
- BitLocker and BitLocker To Go settings

You would therefore be required to reinstall your desktop apps after the refresh is complete.

Resetting your PC

If refreshing your PC does not solve your problems, you can reset your PC. This process removes all your applications, files, and settings. This allows Windows to fix any software-related problems, but you need to have a backup of your data available to restore it later and be ready to reinstall any applications and drivers. Be sure to have any media and product keys required to reinstall applications.

Resetting is the option to go for when you need to format your Windows 8 PC, you don't need to format with any installation media and start looking for drivers to download; resetting would turn your PC to a completely new one.

When you choose to reset your PC, Windows 8:

1. Starts Windows RE.
2. Formats the system drive.
3. Reinstalls a fresh copy of Windows 8.
4. Restarts the new installation of Windows 8.

Resetting would only work if the installation media you used for installing your copy of Windows 8 already saved settings required for resetting (those copies that come pre-installed on the PC usually have it) else you would be prompted to insert the installation media. This might be the best option if you wish to perform a clean installation of Windows 8 on your PC probably because of a deadly virus.

To reset your PC, click General in PC Settings as described under “refresh” above (make sure your system is plugged into a power supply). Under “Remove everything and reinstall Windows”, click Get Started.

Once you launch the reset, you would be presented with two options as shown below:



If you choose “Just remove my files”, Windows would remove your files and re-install Windows such that your data can still be recovered technically but when you choose “Fully clean the drive”, Windows format your drive and write random pattern over the data to make recovering impossible.

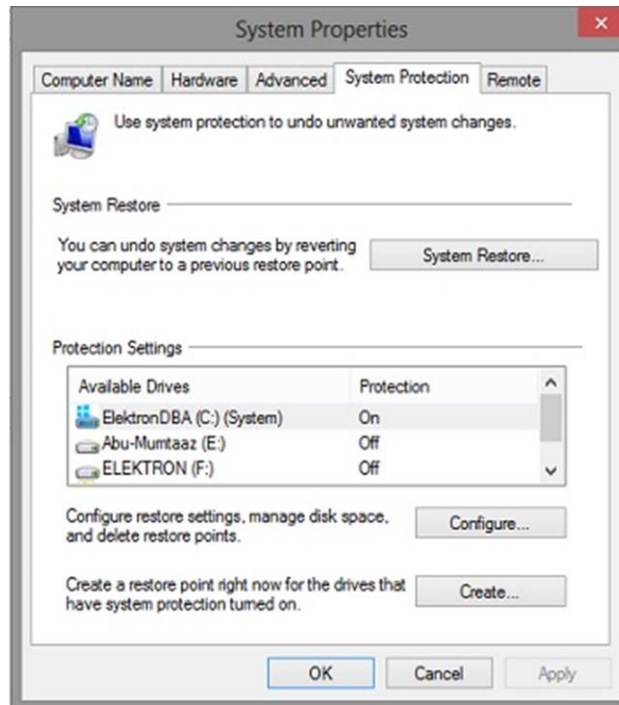
Resetting your PC turns it to a new PC so don’t expect to have anything that has been on your PC before back, except those items that came with your copy of Windows.

Restoring your PC

If you’ve got some funny problems with your PC and you’ve not created a custom refresh image, then you can restore your PC with something known as restore point. Restore points are similar to system images; they are created either by the user or during some events like installing software, installing a driver and so on. Your PC should contain multiple restore points you can always choose from so that if your PC worked excellently on the 25th of May but started behaving funny on the 15th of June, then just get ready to restore your PC back to the state it was on or before the 25th of May; that is just how restoring works.

System restore would not restore personal files; it would only restore system files, registry settings and some other settings; so if you lose a file without backup, system restore is not going to be your saviour.

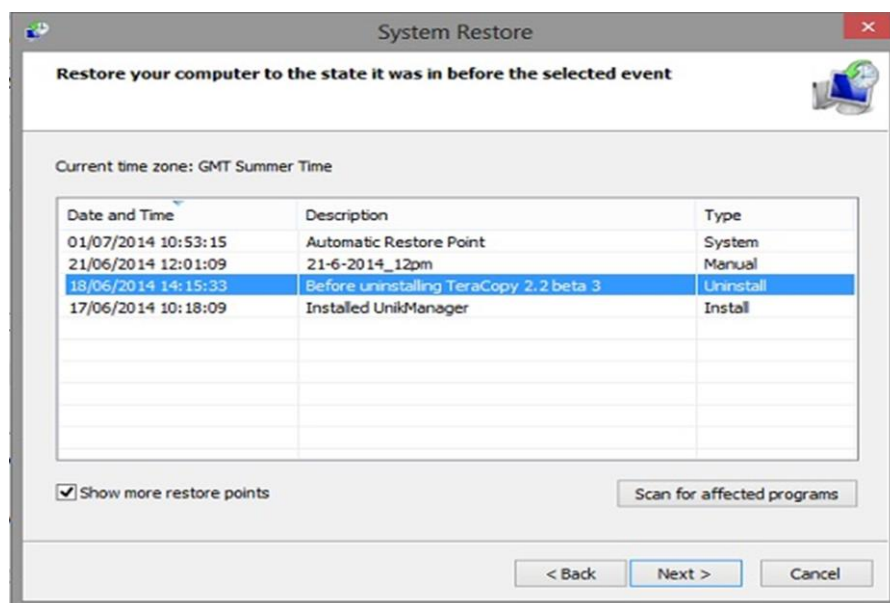
To launch system restore, open system settings (Go to my computer, right click on any white space and choose properties) and from the left pane click on “System protection”. You should be presented with a screen similar to the one shown below:



From this interface, you can choose to create a new restore point (by clicking on Create), this would take the image of your present system condition and save it as a restore point so that you can restore back to it in case of any disaster. Creating restore points often is always a good idea.

You can also change settings related to system protection (by clicking Configure) such as deleting restore points, turning off system protection on drives and changing the size on your drive (10G default) used to save restore points.

If you would wish to restore your PC, then click on “System Restore” and watch Windows start system restore. Once the restore interface is up, click “Next” to get an interface similar to the one below:



Be sure to choose the restore point corresponding to the date your PC worked fine last or a date before that (if the restore point you wish is not listed, check the box “Show more restore points”), you can click on “Scan for affected programs” after selecting a restore point to see the list of programs and drivers that would be installed/uninstalled. Click “Next” once you are ready and follow the prompt until you see “Finish”, Click “Finish” to get started with the restore.

Note that if you just concluded a restore, you would be presented with a screen before the screen above, to either undo system restore or choose another restore point; if you would wish to choose another restore point, click on that option and click next to get the interface shown above.

Note also that antivirus programs can affect system restore; it is therefore a good idea to disable your antivirus program permanently before restoring. Once system restore begins, you can no longer work with your PC; make sure you get it plugged in as it can take between few minutes to few hours.

Last Resort Recovery

What if your PC doesn't start-up any longer probably because the drive is failing and can't start your copy of Windows anymore and you've got a lot of important files on your drive without any backup? Chai! That could be deadly but there is still one solution I'm going to mention here.

You can still recover your files from that drive before attempting to revive the drive or reinstall Windows. For this operation, you would need a backup drive (a new drive) and a Windows installation media (we are going to boot from that media). If you don't know how to boot with an installation media or you've forgotten it, refer back to the chapter on clean installation. Once the tools are ready, let's start our training!

We are going to be making use of commands, so be prepared not to type nonsense so that you would not cause a greater havoc than has already occurred.

Boot with that media till you see the “Install Now” screen, from that screen click on “Repair your computer” below the screen. Windows would start the Recovery Environment (Windows RE) just click next to see recovery tools which should include a command prompt as shown below:



Click on command prompt and let's start the business.

Let me inform you of what we are planning to do; we are going to be copying all your files from your old drive to the new drive you have with you, so you should attach that new drive to your PC before booting at all. By default everything you save in Windows should be in your C:\Users folder but you might also have other files in other folders or even other partitions apart from C. I'd talk about how to copy the content of C:\Users and you should be able to use the same procedure to copy files from other locations.

Note that once you start Windows RE, it is possible that drive letter assignment has changed, i.e. your drive C might now be drive E, you therefore need to be sure which drive you wish to work on. Let's start like this, from the command prompt window; type the command below and press enter:

wmic logicaldisk get name

This would give you the list of drive letters currently on your PC (maybe C, D, E and so on). You can now check the content of each drive by typing the command **dir <drive-letter>**: e.g. **dir E**:

Any drive with the "Windows" directory is the old drive C (let's say your drive C is now drive D in Windows RE, you would recognize each drive with their content).

Now, to copy the content of D:\Users (since D is now your old drive C), type this command:

xcopy D:\users\ <backup-drive>:\backup /E /C /Y

where <backup-drive> is the drive letter of your backup drive, use the technique described above to get the drive letter of your backup drive, assuming the backup drive letter is G, the command would be:

xcopy D:\users\ G:\backup /E /C /Y

You can now take that backup drive to another PC and you'd find your files located in a folder named backup. You can rename that folder from the command if you wish.

You can now use this same procedure to copy other files in other location.

If you now wish to revive your hard drive (maybe because it's failing), type this command:

chkdsk D: /f /r

Where D is the drive letter of your old drive C. Note that if your drive is close to total failure, that command might be enough to cause it to completely fail, so beware!

If you think this option is too technical, you can take your hard disk off your PC and attach it as an external hard drive to another PC to access its content.

What if you can still start Windows, but you suddenly lose a file from your PC without any backup? Then the only option you are left with is to use recovery tools (command-line tools, recovery softwares etc.). You might need to consult an expert to do that for you or just search online for one of such tools if you think you can do it yourself.

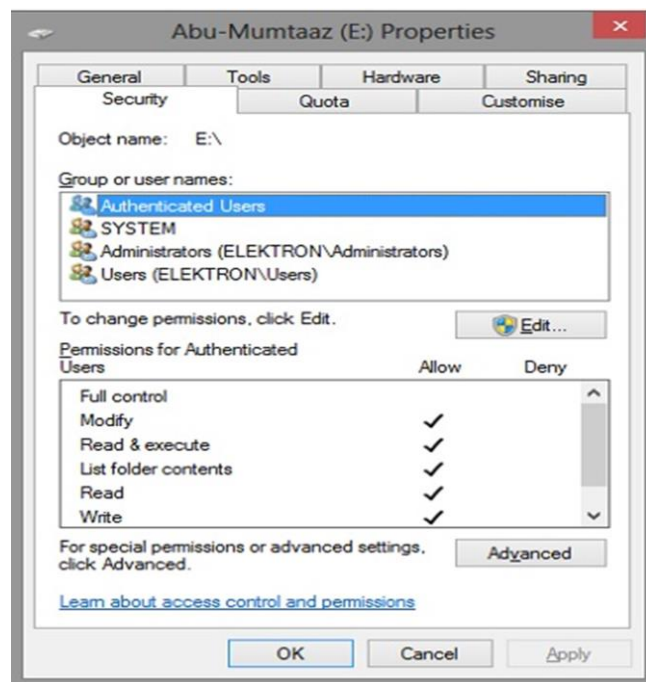
TECHNICAL FIXES

This Chapter covers:

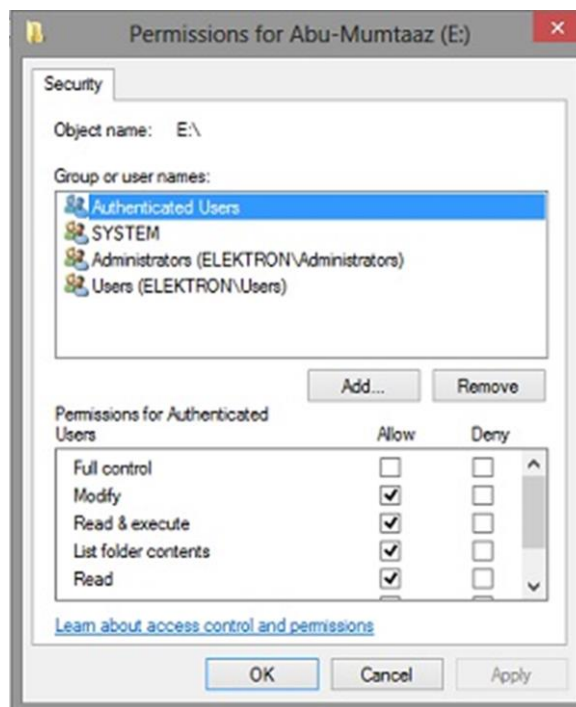
- ✚ File security issues
- ✚ Speeding up Windows shutdown
- ✚ Managing Windows search
- ✚ Resetting Internet Explorer
- ✚ Repairing corrupt boot menu in Windows 8
- ✚ Setting file associations
- ✚ Adding features to Windows
- ✚ Event Viewer
- ✚ Multi-booting problems
- ✚ Setting password age
- ✚ Changing administrator's username
- ✚ Resolving "BOOTMGR is missing" problem

File Security Issues

Sometimes, Windows inform you that you do not have enough privilege to write to a particular disk even when you are logged on as an administrator. To solve issues like this, open Windows explorer, right click on that drive and choose properties. From the properties tab, click on security to get an interface similar to the one below (the number of tabs on the system drive – drive C – would be less than what we have below):

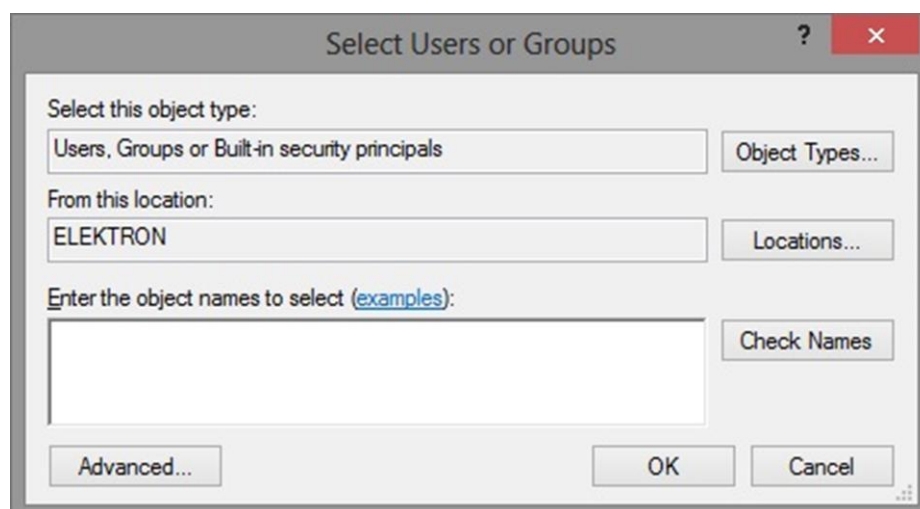


From this interface, choose the user name of the user you wish to edit permission settings for (e.g. Users (ELEKTRON\Users) from the example above) and click on “Edit”. You should get a screen similar to the one shown below:

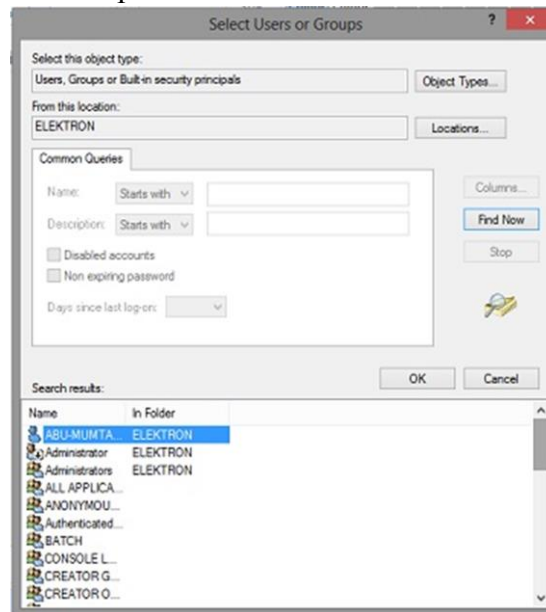


Notice the check boxes in one of the sections in that interface; notice that they belong to two columns Allow and Deny. The implication of this is that the user only has access to do the permission whose checkbox under column “Allow” is checked. To change this setting, check the appropriate boxes and click Apply.

If the user you wish to edit settings for is not listed, click on Add.



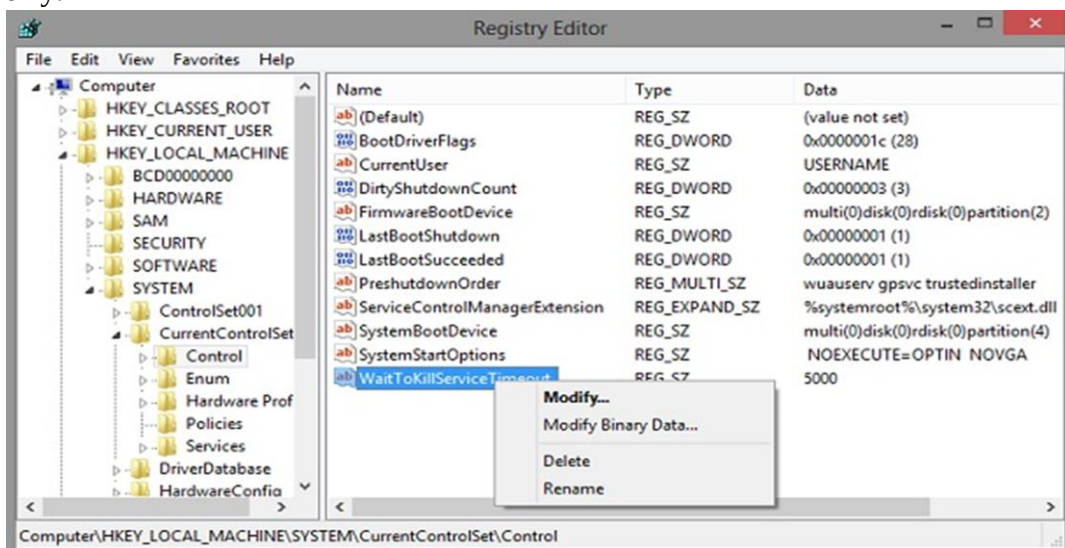
Click on Advanced to proceed...



From this screen click on “Find Now” to populate the search results, choose the user from that list and click OK.

My PC shutdown very slowly

This is because Windows wait for some seconds to kill background task before shutting down. The time Windows take is saved in the registry as “WaitToKillServiceTimeout” and it is given in milliseconds. You can edit the value of this timer to speed up shutdown or even delay it more. To edit it, search start menu for “**regedit**” or type **regedit** in command prompt and press enter, this opens the registry editor interface. From the left pane of registry editor, expand HKEY_LOCAL_MACHINE, expand SYSTEM, expand CurrentControlSet and click on Control. From the right pane, right click on WaitToKillServiceTimeout and choose modify.

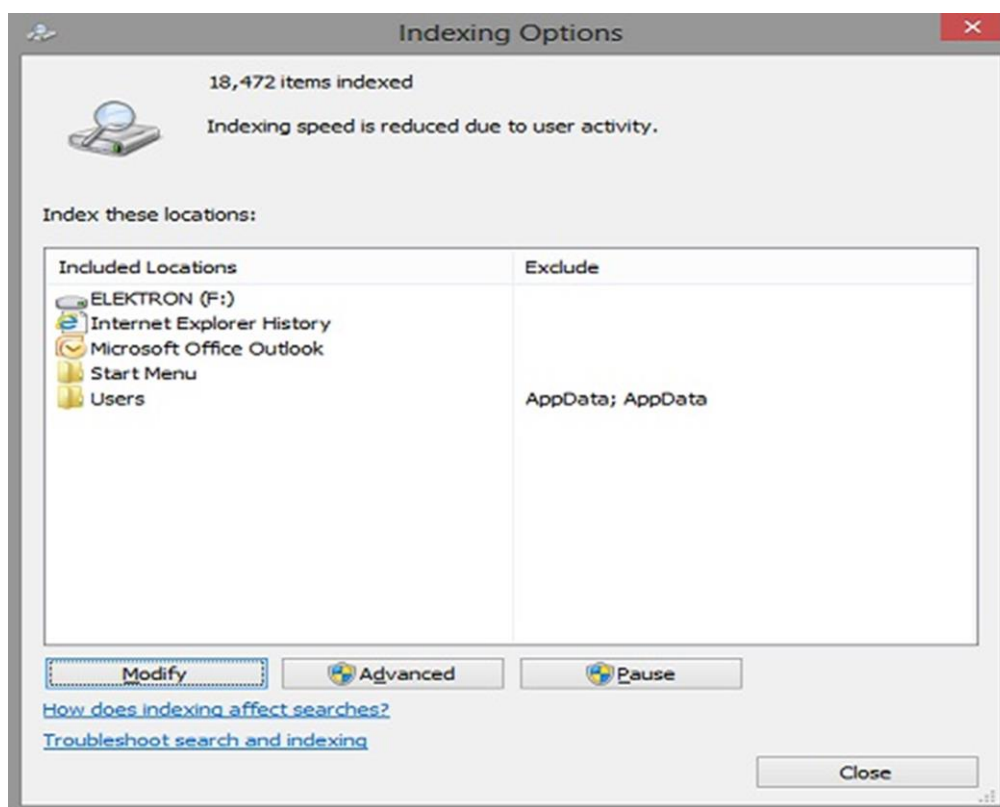


This would edit the value of this registry key whose default value is 12000 for Windows 7 and 5000 for Windows 8 (if this differs, it's because of your edition of Windows), change this value to a smaller value in milliseconds (note that milliseconds is 1/1000 of a second), the recommended value is 3000.

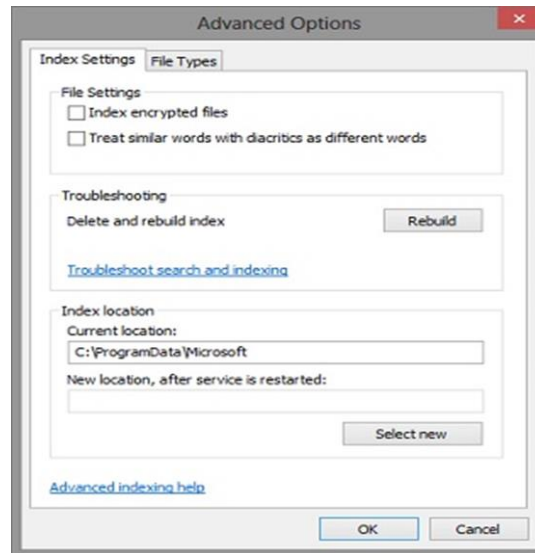
Managing Windows Search: Search Isn't Finding My Files!

Sometimes you search for something that actually exist on your PC at the start screen but Windows doesn't show it, this is because Windows limit the directories that are indexed to make searching faster. Indexed locations on Windows are locations Windows would check when you try to find something, so if a file is located in your drive E and E is not indexed then Windows can't find that file.

To change Windows indexing settings, search start menu for "Indexing options", you should have a screen similar to the one below:



From this screen, you can index more locations by clicking modify and checking the box beside that location in the modify interface, you can also change the settings for searching by clicking "Advanced".



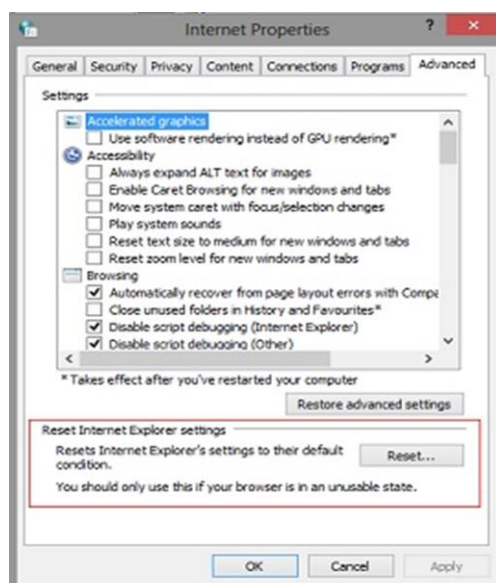
To index encrypted files, check the box that says “Index encrypted files”. To treat diacritic words as different words, check that box also. When diacritic words are treated together as a single word (which is the default), fiancée and fiancée are treated as the same word but when you check that box they are treated as different words i.e. when you search for fiancée, fiancée would not be returned.

When you notice any problem with your index probably as a result of manipulating its settings, click **rebuild** to delete and rebuild it with default settings.

To change File association settings, manipulate the content under “File Types” tab.

Resetting Internet Explorer (IE)

Sometimes accessing IE becomes a problem; you might even launch it and be unable to use it. When this occurs, the last option you might be left with is to reset your internet explorer; to do this open control panel and click on “Internet Options”. From the internet options interface, click on the advanced tab to get the interface below and then click on reset.



Repairing corrupt boot menu in Windows 8

When your boot menu get corrupt, one of those things you'd notice is you can't boot into Windows anymore, Windows might inform you that your boot menu is corrupt. Another problem that can occur is only one OS is displayed in a multi-boot environment.

To do this, you'd need to boot your PC with an installation media (disc or USB) and then click on "Repair my Computer" once the "Install now" interface is displayed to open the recovery environment. Choose command prompt from the recovery tools and then follow the following instructions OR boot your PC to advanced boot menu, choose troubleshoot, choose advanced options and then choose command prompt.

To repair boot files for BIOS based systems, do the following:

- Type **BcdEdit /export C:\backup** (where backup is the location where you want your backup saved) and press Enter.

This creates a backup of the current boot options menu so that it can be restored if needed.

- Type **C:** and press enter

If you are using the command prompt from the recovery tools environment, then your boot files might not be in drive C. You can use the command **wmic logicaldisk get name** to see the list of drives you have and then type **dir C:** to check the content of drive C and then do the same for other drives.

- Type **cd Boot** and press enter. This navigates to the Windows 8 boot options
- Type **attrib bcd -s -h -r** and press Enter. This allows you to modify and overwrite the boot options
- Type **ren C:\Boot\bcd C:\Boot\bcd.old** and press Enter. This renames the current boot file
- Type **Bootrec /RebuildBCD** and press Enter to force Windows 8 to rebuild the boot menu from scratch.

If you need to re-import the boot file then use the command **BcdEdit /import C:\backup**

To repair boot files for UEFI based systems, do the following:

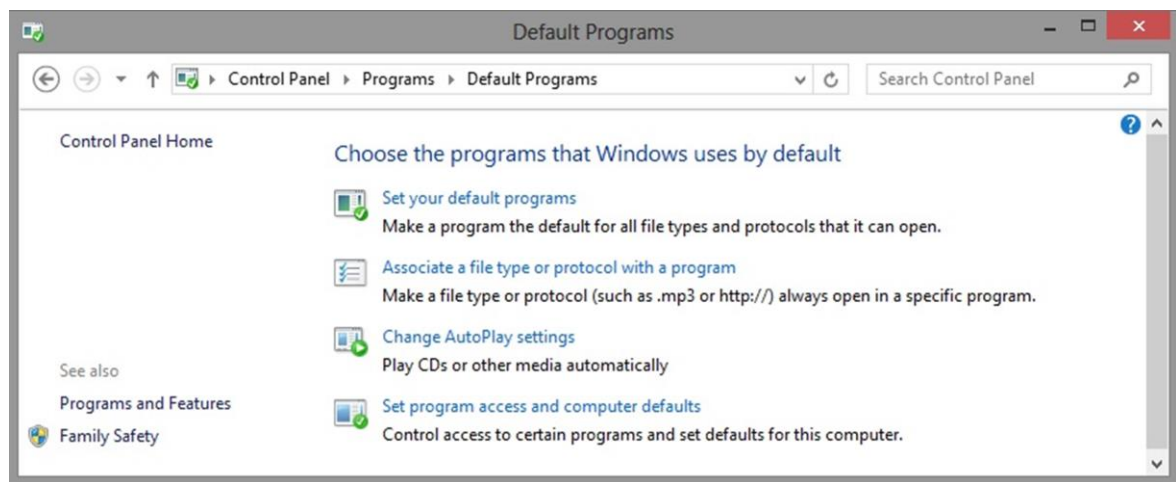
- Type **BcdEdit /export C:\backup** (where backup is the location where you want your backup saved) and press Enter.
- Type **Bootrec /RebuildBCD** and press Enter to force Windows 8 to rebuild the boot menu from scratch.

If you wish to check your type of PC, maybe UEFI or BIOS based, then open BIOS settings on your PC (as described in chapter one) and then check boot device options. If UEFI is listed there, then your PC is UEFI based.

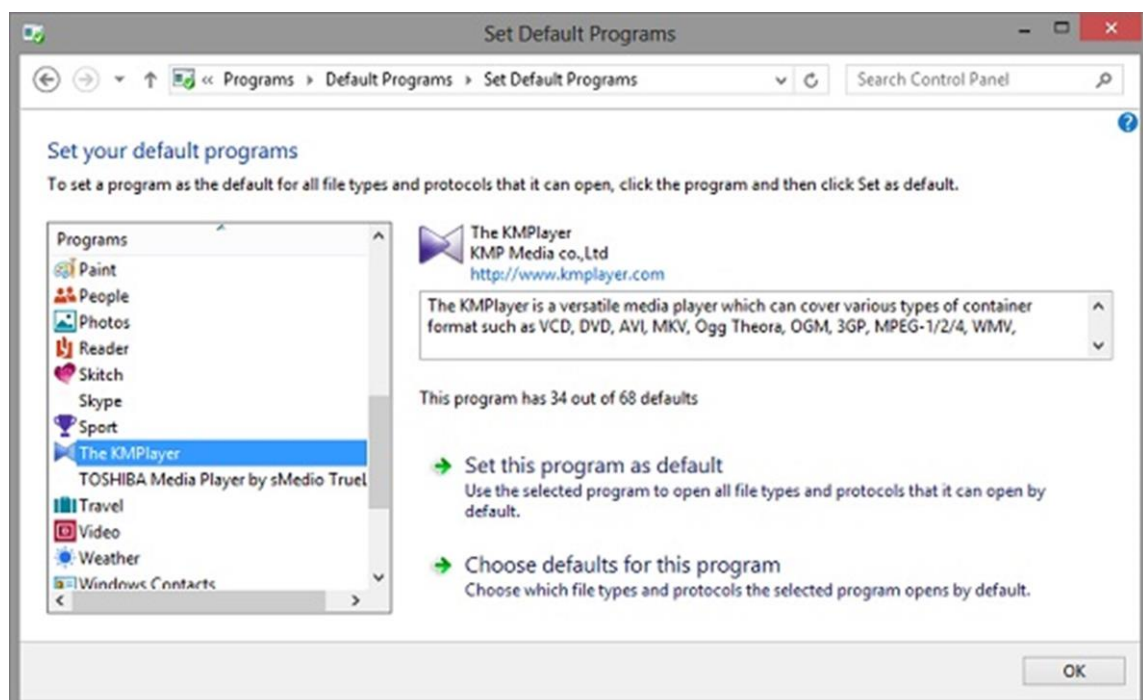
Setting File Associations

Sometimes running a file on your PC becomes a problem when a word document file opens with VLC media player; in fact I've witnessed a PC which launches every program with VLC and nothing can be viewed on that PC except music and videos.

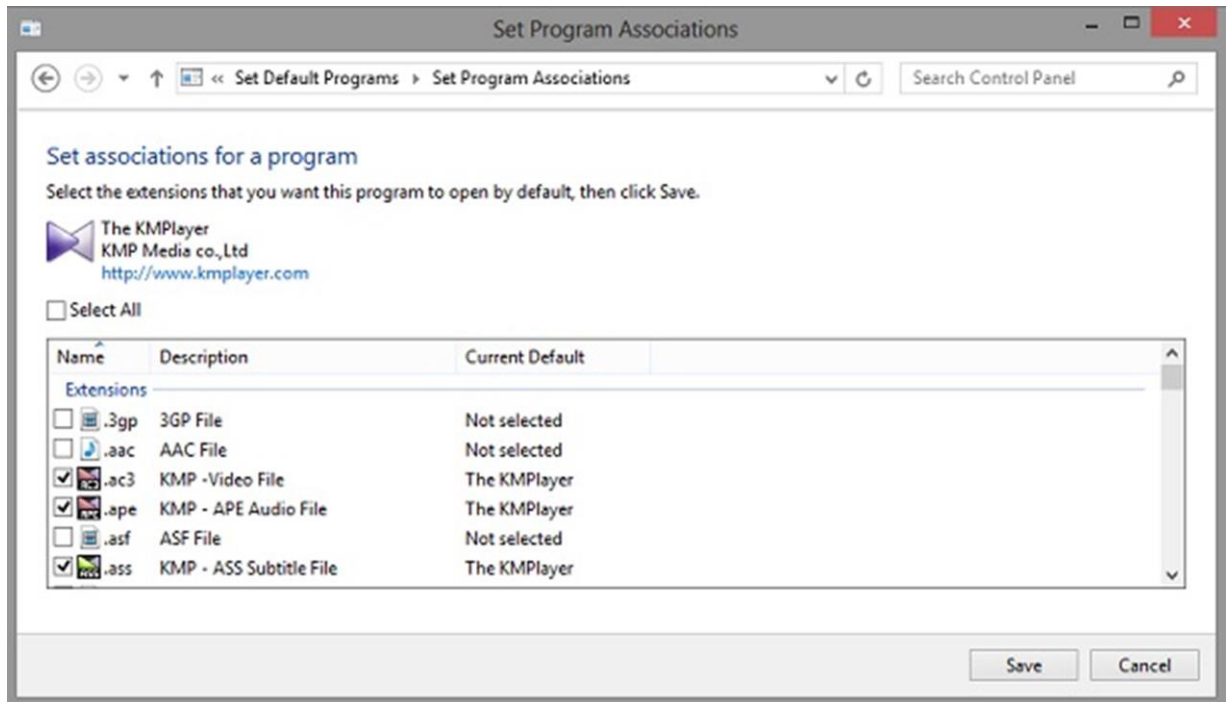
To solve problems like this, all you need do is to change file association of file extension types on your PC. Open control panel (view by category), choose programs and from Programs choose "Default programs".



To decide what extension types you want to open with a particular application, click on "Set your default programs" and select a program you wish to edit its setting.

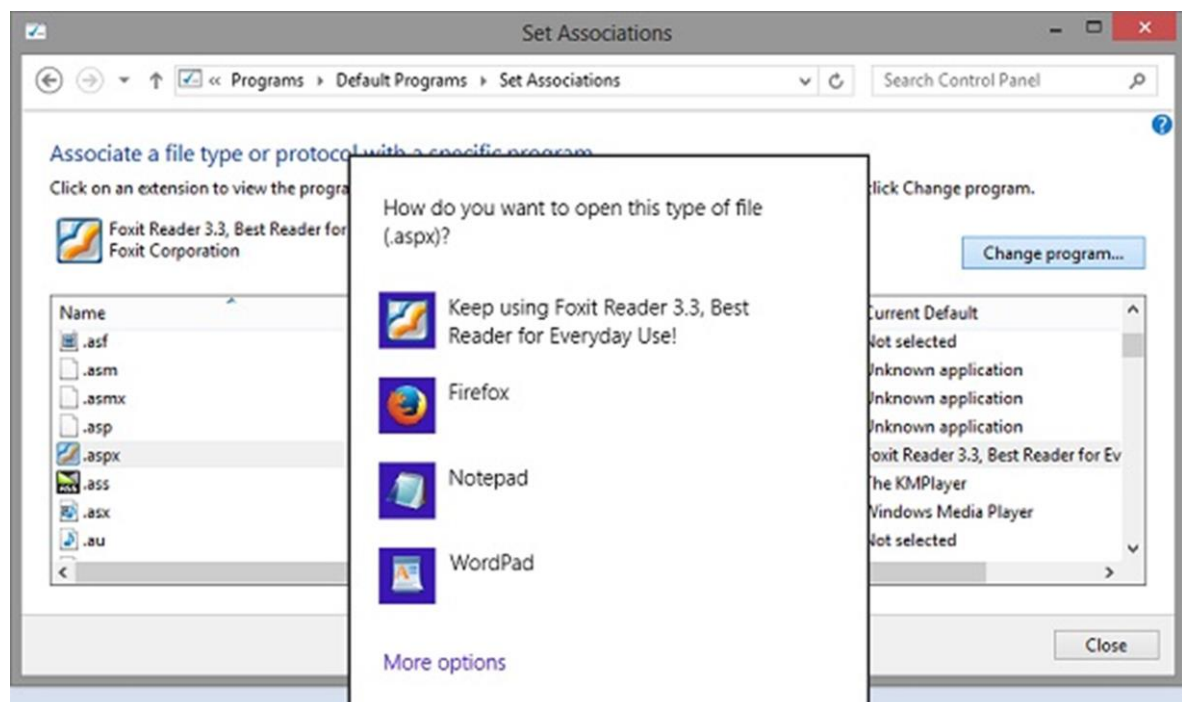


You now have two options to change the setting for your desired program. You can either set the program as default or choose file extensions you wish to open with that program (which is what we want to edit now). Click on “Choose defaults for this program”.

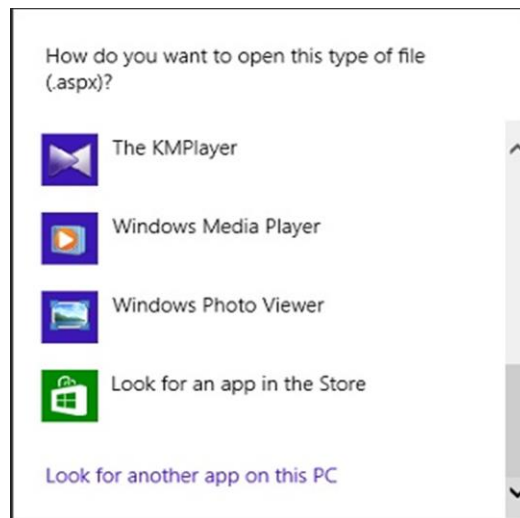


Select the extensions and then click Save.

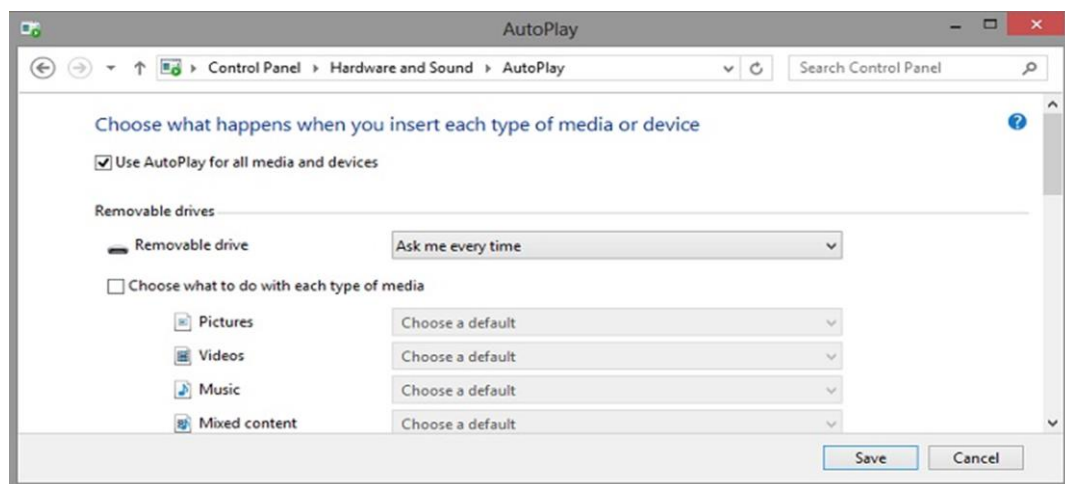
The second option in the “Default program” interface is “Associate a file type or protocol with a program”. This would give you the option to decide the program that would run each of the file extensions on your PC once you click on the extension and choose “Change program”.



If the program you wish to use is not listed, click on “More options” to see more programs. If the program is still not listed, click on “Look for another app on this PC” to open file dialog and select the program from your program folder in drive C.



Another option from the “Default Programs” interface is “Change AutoPlay settings” with which you can change what happens when a removable media is attached to your PC.



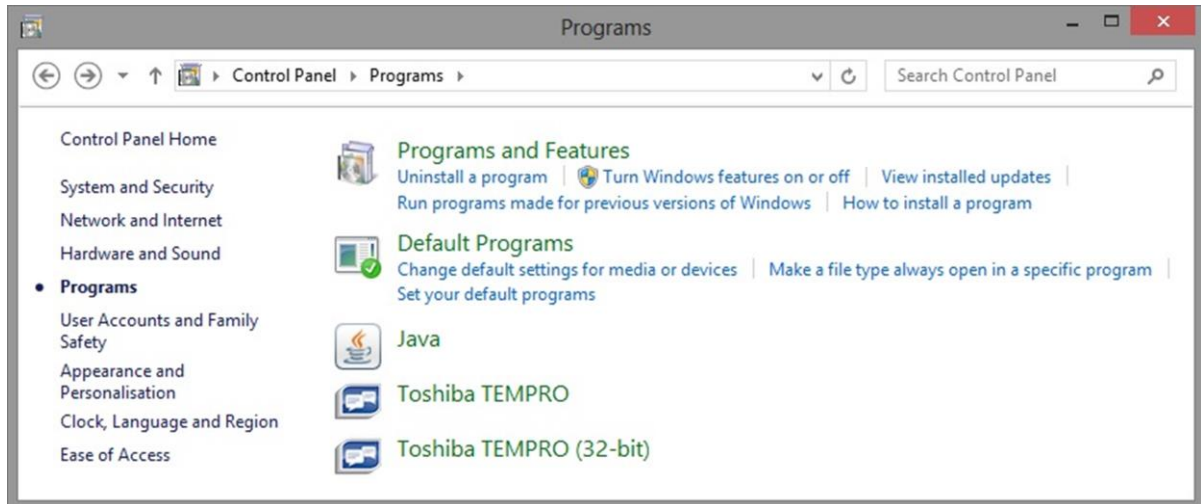
It is very important that you change this setting for a new Windows 8 PC because the default setting for removable drives is to open it in Windows explorer which would make your PC prone to getting infected with a virus as viruses on removable drives often come with auto-runs which would run once you open that drive but if you don't open it you are safe. You should therefore change the setting to “Ask me every time”.

Another setting worth changing is that of video and music, if you wish to open music or videos you insert into your PC with a particular program, you can also set it from AutoPlay.

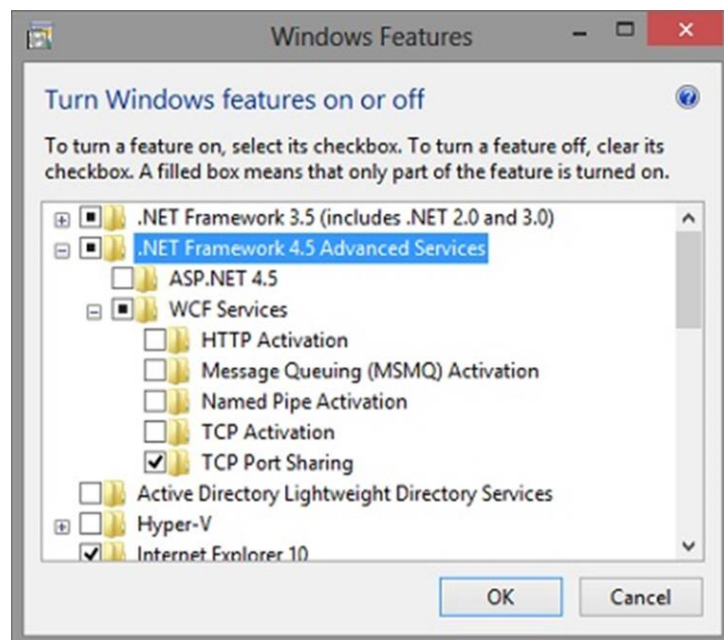
Adding Features to Windows

Some Windows features are disabled by default since not every user would want to use it and turning it on might make their services run automatically every time you start Windows reducing your booting speed e.g. the .NET framework on some PC might be off by default and should be turned on to make use of it.

To turn on features in Windows, open control panel and click on “Programs”.



Click on “Turn Windows features on or off” under Programs and Features.

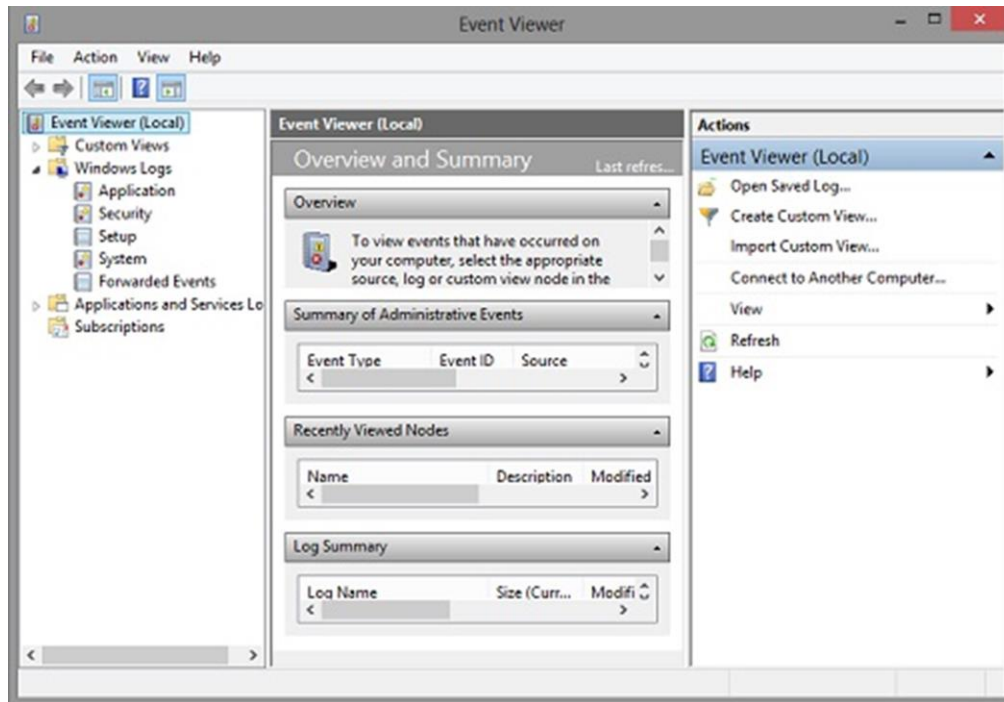


Select the programs and/or features you wish to turn on and then click OK. Windows would check for the required files to install the app/feature on your PC and if it's not found, Windows would attempt to download it from the internet if you have internet access.

You might need to allow Windows restart your PC to fully configure such app/feature.

Event Viewer

With the event viewer, you can monitor every event that takes place on your PC. To start event viewer, search the start screen for **eventvwr.msc** or simply event viewer



As a glance at the console tree confirms, events are recorded in one of several logs:

- **Application:** Application events are generated by applications, including programs you install, programs that come with Windows, and operating system services. Program developers decide which events to record in the Application log and which to record in a program-specific log under Applications and Services Logs.
- **Security:** Security events include logon attempts (successful and failed) and attempts to use secured resources, such as an attempt to create, modify, or delete a file. You can use the event section to check when somebody attempts to logon to your PC or attempts to make changes to your files.
- **Setup:** Setup events are generated by application installations.
- **System:** System events are generated by Windows itself and by installed features, such as device drivers. If a driver fails to load when you start a Windows session, for example, that event is recorded in the System log.
- **Forwarded Events:** The Forwarded Events log contains events that have been gathered from other computers.
- **Applications and Services Logs:** The Applications and Services Logs folder contains logs for individual applications and services. The other logs generally

record events that are system-wide in nature, but each log in Applications and Services Logs records the events related only to a particular program or feature. Within the Applications and Services Logs folder resides a Microsoft\Windows folder, which contains a folder for each of many features that are part of Windows. Each of these folders contains one or more logs.

Even more logs appear in Applications and Services Logs when you open the View menu and click Show Analytic and Debug Logs. These additional logs are generally needed only for a hard-core troubleshooting session, so they're hidden by default. Also note that these logs do not record events by default. To use either the Analytic or Debug log, right-click it and choose Enable Logging

Multi-booting Problems

I can't install another OS!

As explained under installation, you'd need to create partitions on your hard disk before you can run multiple operating system; so that you can install different operating systems on the various partitions, but sometimes after partitioning your hard drive, you still can't install an OS in the other partitions, when you attempt to do this Windows inform you that the partition is dynamic and Windows can't be installed in a dynamic volume. What do you do?

You see, basically there are two types of disk as mentioned earlier in disk management:

- Basic disk and
- Dynamic disk

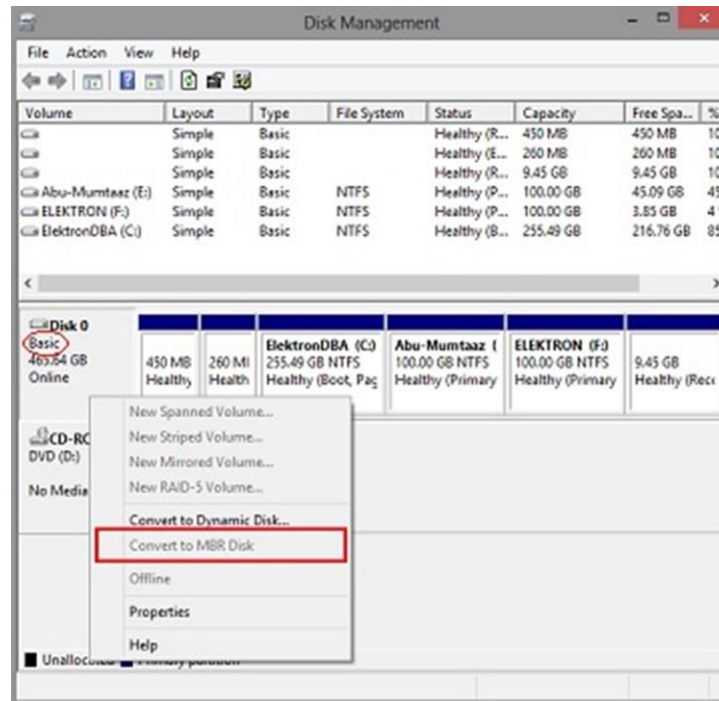
The foundation knowledge you should have is that Windows can't be installed on a dynamic disk. Now, each of these types of disk has a way of allocating disk spaces to files saved on them; this allocation method is managed in two different ways in disks, these are the MBR (Master Boot Record) and GPT (GUID Partition Table).

By default, MBR can only have up to a maximum of four (4) primary partitions while GPT can have more than four primary partitions. That implies that the partitions on a disk can either be primary or secondary, the secondary partition on an MBR disk is called an extended partition which can take up to 2million logical partitions.

The problem is this: Windows can only be installed in primary partitions on basic disk. Basic disks uses MBR (though some use GPT), when you attempt to partition your hard disk, if it's an MBR disk you can't use more than four primary partitions and remember some partitions come with PCs by default (system reserved partitions), if adding that partition to your hard drive would make the number of partition greater than four, then Windows would inform you that the operation you are trying to perform would convert your hard drive to a dynamic volume/disk which you might unknowingly agree to. Once you do this, it implies

that you can no longer install Windows to other partitions on that drive (since they are now dynamic), but if your hard drive is basic but uses GPT, then you can have as many as possible primary partitions on it and installing Windows would just be easy.

To check your type of disk, start **diskmgmt.msc**, right click on your hard disk (where you have it listed as disk 0 (the number might differ on your hard disk)).



Notice from the interface above that the disk is a basic disk (not dynamic) therefore it can still install Windows. Notice also the squared area, where we have convert to MBR Disk, this implies that the disk is presently GPT meaning it can take as many as possible primary partitions, with this type of hard disk you are save since creating more partitions would affect nothing.

Now your disk has been converted to a dynamic disk or it's still basic but an MBR type and Windows inform you that creating more partition would turn it to a dynamic disk and you need to install more OS, what do you do?

The simple answer is you have to convert it back to a basic disk if it's now dynamic. If it's not but you can't create more partition, then you would have to find a way of removing some of the partitions on it, this might often times be impossible except you convert the whole disk to a single partition.

To convert the disk to a basic disk, right click on it in disk management as shown above and choose "Convert to basic disk". If this doesn't work, you'd have to use the method described below.

To convert the whole partition to a single one, you'd need to take the following step which would format everything on your drive and recovering might not be possible anymore (because disk allocations would be altered) so be sure you have backup(s) of your data, it's also going to remove your Windows installation so be prepared to re-install Windows and every app on your PC.

1. Boot your PC with an installation media
2. Click on repair your computer and start the Windows recovery environment
3. Choose command prompt from the recovery tools
4. Type the commands we used to make a USB bootable (check the chapter on installation if you've forgotten the commands).
5. You can skip the format command (i.e. `format fs = fat32 quick`) because the volume on the drive might be too large to be formatted like that but if you'd use it make sure the file system you use is ntfs and not fat32 (i.e. **`format fs = ntfs quick`**). You may and you may not add the quick option but adding it would speed up the process.
6. To convert the disk to GPT, type the command **`convert GPT`** and press Enter.

Your disk is now a basic disk with a single partition; you can now re-install Windows and then use disk management to create more partitions for multi-booting.

I have more than one OS on my PC but I need only one.

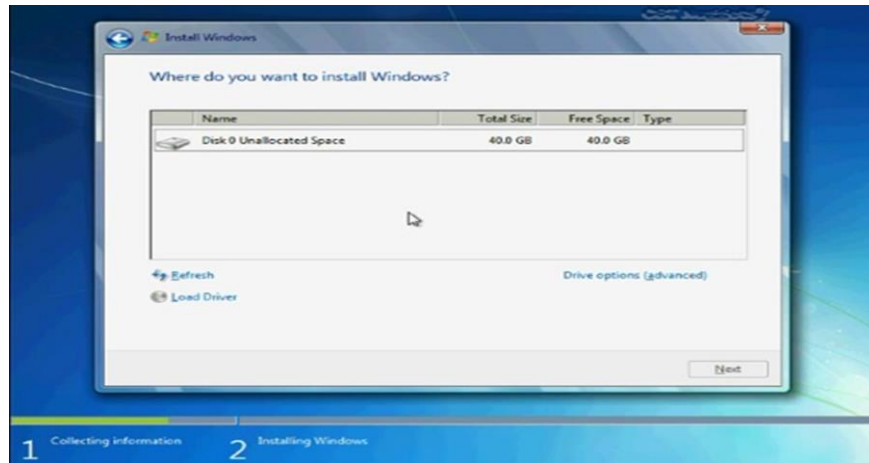
If you have multiple OS on your hard drive and you need only one of them, then follow the steps below to remove all other ones. Note that this method would clear every data on those partitions where the Operating Systems reside.

1. Log in with the operating system you want to continue to use.
2. Start disk management console
3. Format the partitions housing the other operating systems.

You should now have a single OS the next time you login.

If this doesn't solve the problem, then

1. Boot your PC with an installation media
2. Click on install now and proceed with the installation till the interface where you are prompted to select disk for installation appears as shown below.



3. Click on the partitions you wish to format (i.e. remove OS from), click on Drive options and then click on Format.

You can also use command prompt (in Windows or from recovery tools) if you are sure of what you want. Use your knowledge so far to do that.

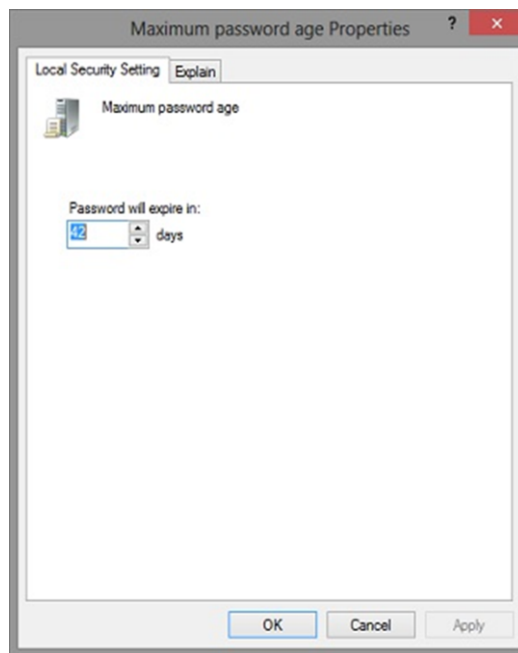
Setting password age

Changing passwords regularly is one security measure security experts have mentioned, they argued that stealing passwords would become useless if you change your password often – and I quite agree with them --.

One thing Microsoft included in Windows 8 is password age which specifies how long you can use a specific password before it becomes obsolete. Once the password expires, you would not be able to use it any longer and Windows would inform you when next you attempt to login to your PC that your password has expired and would then prompt you to change it; but not before you have supplied the old one correctly.

To change how long your passwords can live before they expire, take the following steps (this option is not available in Windows 8 standard edition):

1. At the Start screen, type **secpol.msc** and press Enter.
Windows opens the Local Security Policy desktop app.
2. Select Security Settings\Account Policies\Password Policy.
3. Double-click Maximum Password Age, and choose a time span (between 1 and 999 days) after which the user will be required to change their password, as shown below. Windows will notify the user several days in advance that they need to change their password.
Click OK.



4. Double-click Enforce Password History. Set the value to 10, and then click OK. This prevents users from choosing a password that they have recently used, which is a concern because many people simply alternate between passwords when required to change their password.

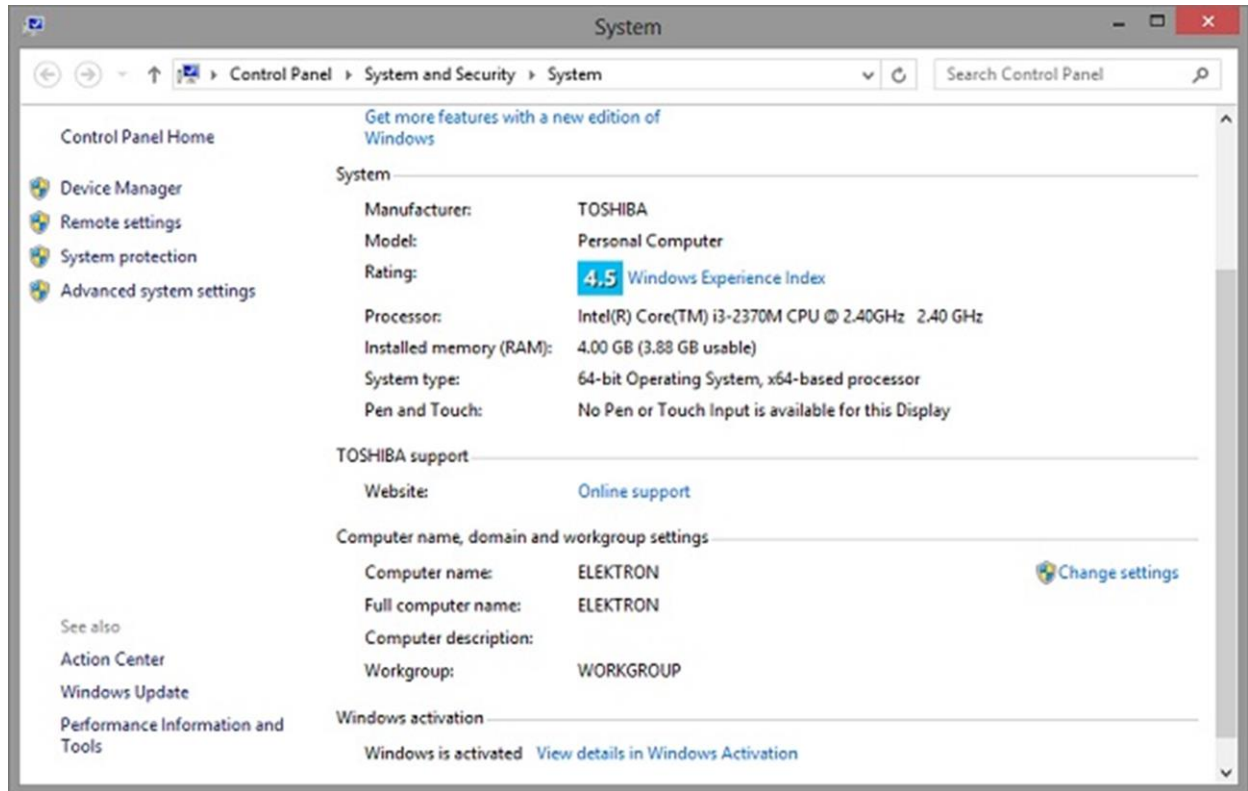
The implication of the value supplied (i.e. 10) is that the user can't re-use the last 10 passwords he/she used, but can use the last 11th password.

5. Double-click Minimum Password Age, and set it to 1 day and then click OK. Working with Enforce Password History, this setting prevents the user from changing their password to something new and then immediately changing it back; at least not on that same day anymore.
6. Double click "Password must meet complexity requirements" and choose enable
7. Close the Local Security Policy app.

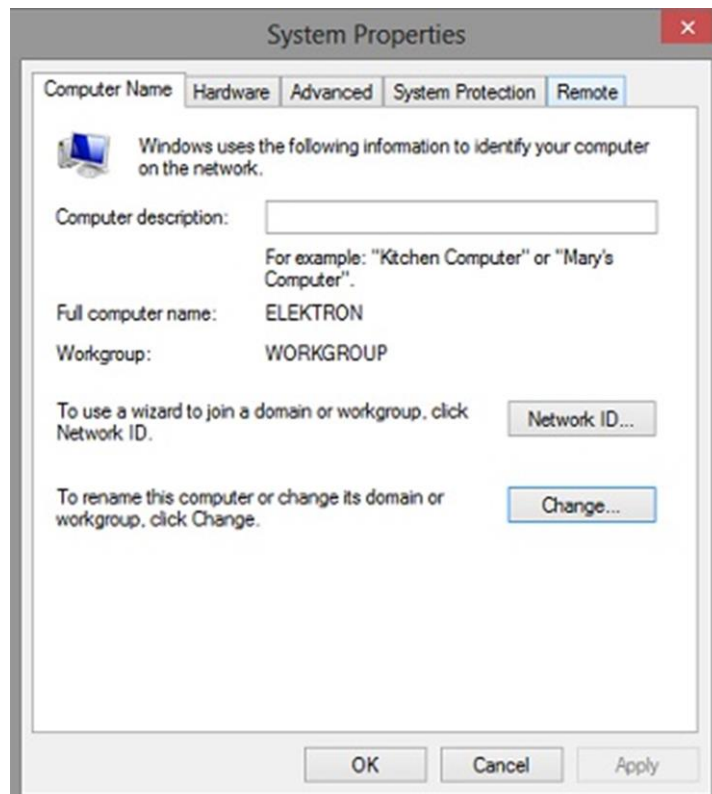
Changing administrator's username in Windows

One of the most tedious or technical thing to do in Windows is to change the username of the PC's administrator. This is the name you would see when you navigate to C:\users folder on your PC.

You can easily change your computer name when you open system properties (open Windows explorer, right click on any white space and choose properties) as shown below and then click change settings in front of the computer name.

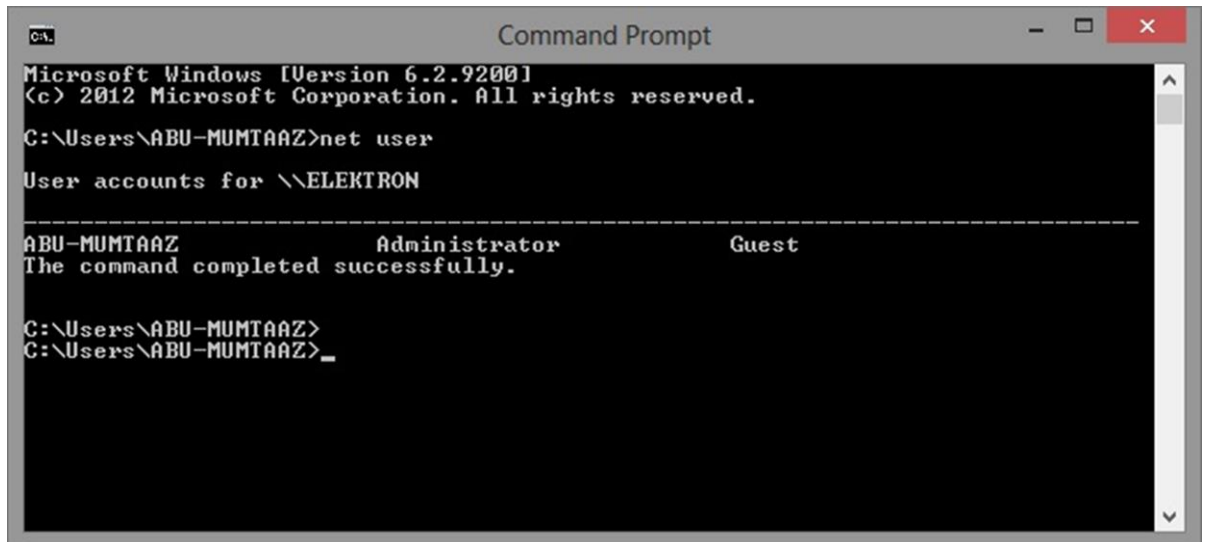


When you click change settings, Windows would present you with a screen similar to the one below:



And you'd simply click change to change the name.

When you follow the steps above, you have only changed your PC's name and not the administrator's username. You can check your C:\Users again to see if the name has changed or open command prompt window, type **net user** and press enter; this would show you the username of every user on your PC and would tell you what type of user it is. The possible output of the command is shown below:



```
C:\Users\ABU-MUMTAZ>net user

User accounts for \\ELEKTRON

-----
ABU-MUMTAZ           Administrator           Guest
The command completed successfully.

C:\Users\ABU-MUMTAZ>
C:\Users\ABU-MUMTAZ>
```

Now how would you change the admin username?

Follow the steps below:

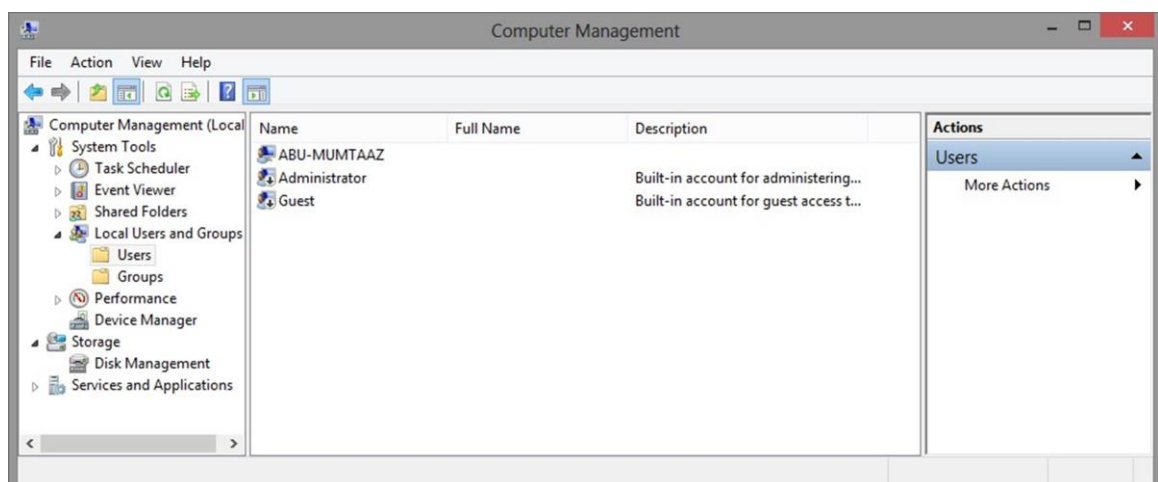
1. At the start screen, type **compmgmt.msc** and press Enter.

This would open the computer management console.

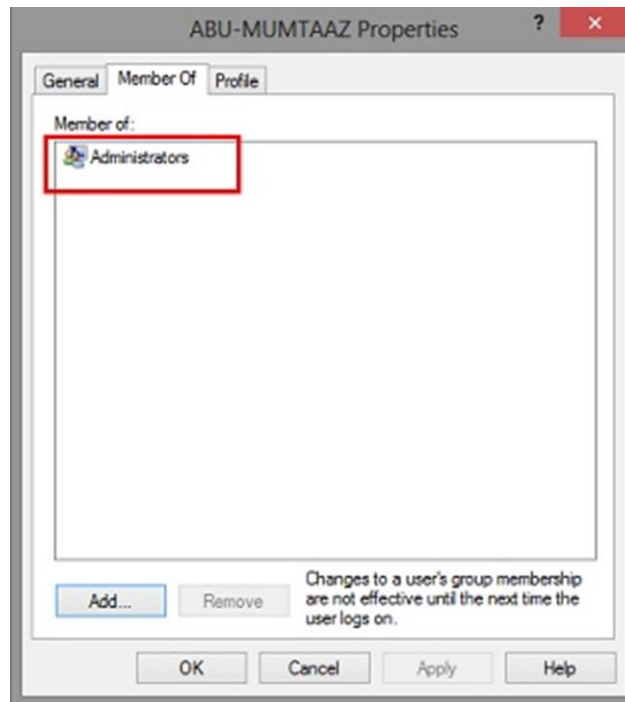
2. From the left pane, navigate to:

Computer Management\System Tools\Local users and groups\Users

You should get a screen similar to the one below:



3. The first name in the list should be the name you are planning to change. You can confirm if the user is an administrator by right clicking and choosing properties, from the properties tab, click on Member of as shown below:



To rename this user, Right click on the name again and choose Rename.

4. To change the name for other users too, right click on the user and choose Rename.
5. Close the computer management console.

To confirm the success of the above procedure, open command prompt window and type **net user** as shown above.

BOOTMGR is missing

If you suddenly power on your PC and instead of seeing it load Windows you found a message on the screen saying “BOOTMGR is missing” or something similar, then don’t get scared; something just went wrong, but what?

Let me explain like this:

Your PC was searching for a valid operating system but could not find one. I know you are going to say that you’ve got a valid operating system in your hard disk, but this is what you have to understand:

Remember I talked about the boot order in chapter one, your PC's BIOS would check the devices in that list (boot order) for a valid operating system in ascending order but the way it checks is this; it's going to search that device if it's available, if it is, then it would check if it has got a primary partition, if it does it would then check for boot files which should be supplied by the boot manager on that device.

Often times, problems like this occur when you attach a bootable USB drive or a similar drive to your PC but the drive does not have boot files on it presently. Why is it checking it and not my hard drive? Because you have placed that device on top of hard disk in your boot order and BIOS is configured to stop searching once it has found any, I mean any device with a bootable primary partition.

The solution to this problem is this: remove everything you attach to your PC presently and reboot your PC; everything should then work fine.

If that doesn't solve your problem, you might need to reload your boot menu using the commands discussed above under "repairing corrupt boot menu".

If that doesn't work either, then consider repairing your Windows by using the Windows Recovery environment (discussed under backup and recovery, and technical fixes), let Windows scan your PC for errors.

If that doesn't solve it, then you can consider reloading Windows but I would advise you see an expert before doing so..... We just might still be able to solve that problem somehow.

CUSTOMIZING WINDOWS

This Chapter covers:

-  Windows shortcuts
-  Customizing Windows 8
-  Uninstalling Windows 8 apps

Windows 8 shortcuts (Some are applicable to other versions of Windows)

Shortcut	What it does
Windows logo key  +start typing	Search your PC
Ctrl+plus (+) or Ctrl+minus (-)	Zoom in or out of a large number of items, like apps pinned to the Start screen
Ctrl+scroll wheel	Zoom in or out of a large number of items, like apps pinned to the Start screen
Windows logo key  +C	Open the charms In an app, open the commands for the app
Windows logo key  +F	Open the Search charm to search files
Windows logo key  +H	Open the Share charm
Windows logo key  +I	Open the Settings charm
Windows logo key  +J	Switch the main app and snapped app
Windows logo key  +K	Open the Devices charm
Windows logo key  +O	Lock the screen orientation (portrait or landscape)
Windows logo key  +Q	Open the Search charm to search apps
Windows logo key  +W	Open the Search charm to search settings
Windows logo key  +Z	Show the commands available in the app
Windows logo key  +spacebar	Switch input language and keyboard layout
Windows logo key  +Ctrl+spacebar	Change to a previously selected input
Windows logo key  +Tab	Cycle through open apps (except desktop apps)
Windows logo key  +Ctrl+Tab	Cycle through open apps (except desktop apps) and snaps them as they are cycled
Windows logo key  +Shift+Tab	Cycle through open apps (except desktop apps) in reverse order
Windows logo key  +PgUp	Move the Start screen and apps to the monitor on the left (Apps in the desktop won't change monitors)
Windows logo key  +PgDown	Move the Start screen and apps to the monitor on the right (Apps in the desktop won't change monitors)
Windows logo key  +Shift+period (.)	Snaps an app to the left
Windows logo key  +period (.)	Snaps an app to the right
Esc	Stop or exit the current task
Alt+F4	Close the active item, or exit the active app
Alt+Esc	Cycle through items in the order in which they were opened
Alt+underlined letter	Perform the command for that letter
Alt+Enter	Display properties for the selected item

Alt+Spacebar	Open the shortcut menu for the active window
Alt+Left arrow	Back
Alt+Right arrow	Forward
Alt+Page Up	Move up one screen
Alt+Page Down	Move down one screen
Alt+Tab	Switch between open apps (except desktop apps)
Ctrl+F4	Close the active document (in apps that allow you to have multiple documents open simultaneously)
Ctrl+A	Select all items in a document or window
Ctrl+C (or Ctrl+Insert)	Copy the selected item
Ctrl+D (or Delete)	Delete the selected item and move it to the Recycle Bin
Ctrl+R (or F5)	Refresh the active window
Ctrl+V (or Shift+Insert)	Paste the selected item
Ctrl+X	Cut the selected item
Ctrl+Y	Redo an action
Ctrl+Z	Undo an action
Ctrl+plus (+) or Ctrl+minus (-)	Zoom in or out of a large number of items, like apps pinned to the Start screen
Ctrl+ mouse scroll wheel	Change the size of desktop icons or zoom in or out of a large number of items, like apps pinned to the Start screen
Ctrl+Right arrow	Move the cursor to the beginning of the next word
Ctrl+Left arrow	Move the cursor to the beginning of the previous word
Ctrl+Down arrow	Move the cursor to the beginning of the next paragraph
Ctrl+Up arrow	Move the cursor to the beginning of the previous paragraph
Ctrl+Alt+Tab	Use the arrow keys to switch between all open apps
Ctrl+arrow key (to move to an item)+Spacebar	Select multiple individual items in a window or on the desktop
Ctrl+Shift with an arrow key	Select a block of text
Ctrl+Esc	Open the Start screen
Ctrl+Shift+Esc	Open Task Manager
Ctrl+Shift	Switch the keyboard layout when multiple keyboard layouts are available
Ctrl+Spacebar	Turn the Chinese input method editor (IME) on or off
Shift+F10	Display the shortcut menu for the selected item
Shift with any arrow key	Select more than one item in a window or on the desktop, or select text within a document
Shift+Delete	Delete the selected item without moving it to the Recycle Bin first
Right arrow	Open the next menu to the right, or open a submenu
Left arrow	Open the next menu to the left, or close a submenu
Windows logo key  +B	Set focus in the notification area
Windows logo key  +C	Open the Charms
Windows logo key  +D	Display and hide the desktop
Windows logo key  +E	Open Computer
Windows logo key  +F	Open the Search charm and search for files
Windows logo key  +G	Cycle through Gadgets on the desktop
Windows logo key  +H	Open the Share charm
Windows logo key  +I	Open the Settings charm
Windows logo key  +J	Switch the main app and snapped app
Windows logo key  +K	Open the Devices charm
Windows logo key  +L	Lock your PC or switch users

Windows logo key  +M	Minimize all Windows
Windows logo key  +O	Lock device orientation
Windows logo key  +P	Choose a presentation display mode
Windows logo key  +Q	Open the Search charm and search for apps
Windows logo key  +R	Open the Run dialog box
Windows logo key  +T	Cycle through apps on the taskbar
Windows logo key  +U	Open Ease of Access Center
Windows logo key  +V	Cycle through notifications
Windows logo key  +Shift+V	Cycle through notifications in reverse order
Windows logo key  +W	Open the Search charm and search for settings
Windows logo key  +X	Open the Quick Link menu
Windows logo key  +Z	Show the commands available in the app
Windows logo key  +,	Temporarily peek at the desktop
Windows logo key  +Pause	Display the System Properties dialog box
Windows logo key  +Ctrl+F	Search for PCs (if you're on a network)
Windows logo key  +Shift+M	Restore minimized Windows on the desktop
Windows logo key  +number	Start the desktop app pinned to the taskbar in the position indicated by the number. If the app is already running, switch to that app.
Windows logo key  +Shift+number	Start a new instance of the desktop app pinned to the taskbar in the position indicated by the number
Windows logo key  +Ctrl+number	Switch to the last active window of the desktop app pinned to the taskbar in the position indicated by the number
Windows logo key  +Alt+number	Open the Jump List for the desktop app pinned to the taskbar in the position indicated by the number
Windows logo key  +Ctrl+Shift+number	Open a new instance of the desktop app located at the given position on the taskbar as an administrator
Windows logo key  +Tab	Cycle through open apps (except desktop apps)
Windows logo key  +Ctrl+Tab	Cycle through open apps (except desktop apps) and snap them as they are cycled
Windows logo key  +Shift+Tab	Cycle through open apps (except desktop apps) in reverse order
Windows logo key  +Ctrl+B	Switch to the app that displayed a message in the notification area
Windows logo key  +Up arrow	Maximize the desktop window
Windows logo key  +Down arrow	Minimize the desktop window
Windows logo key  +Left arrow	Maximize the desktop window to the left side of the screen
Windows logo key  +Right arrow	Maximize the desktop window to the right side of the screen
Windows logo key  +Home	Minimize all but the active desktop window (restores all Windows on second stroke)
Windows logo key  +Shift+Up arrow	Stretch the desktop window to the top and bottom of the screen
Windows logo key  +Shift+Down arrow	Restore/minimize active desktop Windows vertically, maintaining width
Windows logo key  +Shift+Left arrow or Right arrow	Move a window in the desktop from one monitor to another
Windows logo key  +Spacebar	Switch input language and keyboard layout
Windows logo key  +Ctrl+Spacebar	Change to a previously selected input
Windows logo key  +Enter	Open Narrator

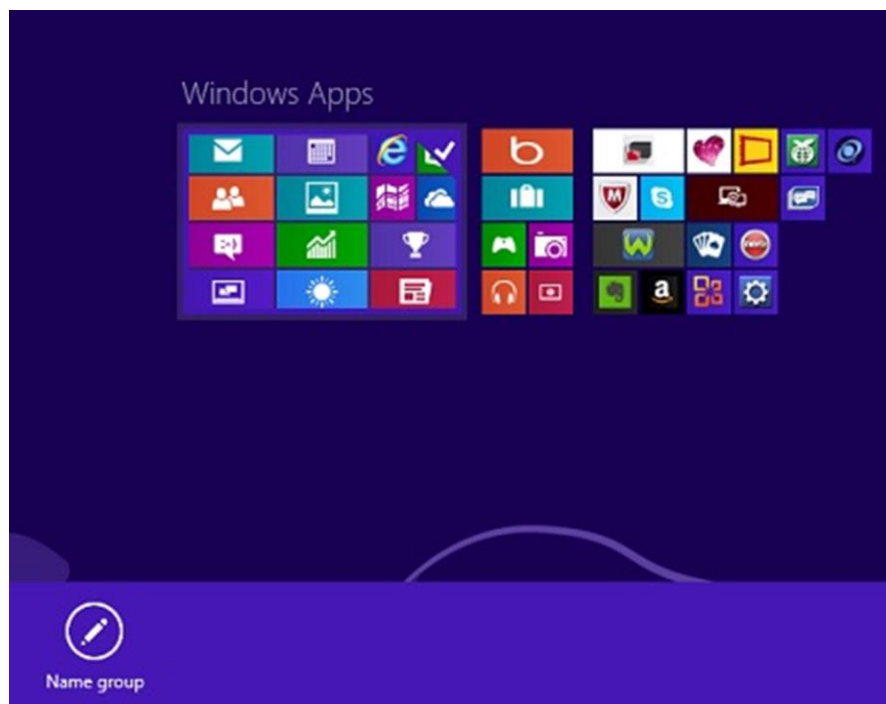
Windows logo key +Page Up	Move the Start screen and apps to the monitor on the left (apps in the desktop won't change monitors)
Windows logo key +Page Down	Move the Start screen and apps to the monitor on the right (apps in the desktop won't change monitors)
Windows logo key +Shift+period (.)	Snap an app to the left
Windows logo key +period (.)	Snap an app to the right
Windows logo key +/	Initiate IME reversion
Windows logo key +Alt+Enter	Open Windows Media Center
Windows logo key +plus (+) or minus (-)	Zoom in or out using Magnifier
Windows logo key +Esc	Exit Magnifier

CUSTOMIZING WINDOWS 8

Grouping apps

You can decide to group apps that show up on your start screen and give each group a name. To do this,

1. Open start menu (press the window key)
2. Click on the minus sign at the bottom right side of your screen (Windows now display your apps in groups)
3. Right click on each group and click on "Name group" to give the group a name.



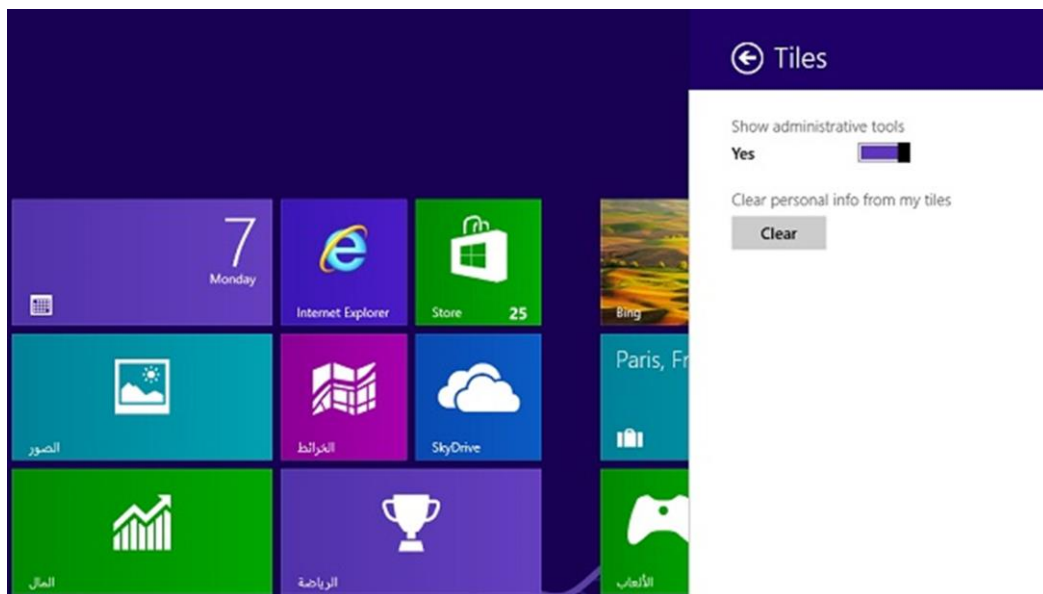
To reorganize the apps in each group, click and drag the app to the group you want it to belong.

Showing Admin Tools on Start Screen

Windows comes with administrative tools you can use to perform administrative functions, these tools reside in Control Panel\System and Security\Administrative Tools. They are displayed by default in the start screen in Windows 7 (Start\All Programs\Accessories\System Tools) but they are hidden from the start screen in Windows 8.

To turn it on:

1. Open the start screen
2. Press a combination of window key and I (Windows logo key +I)
3. Click on Tiles
4. Click to turn on admin tools



The tools are then displayed in your start screen.

Adding a Shortcut to the Start Screen

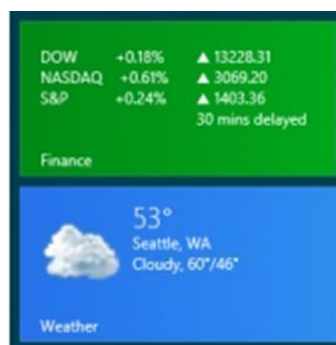
The Windows 8 start screen, unlike Windows 7 is made up of tiles that can be clicked to launch the real applications. You can also create a shortcut and add it as a tile to the start screen. For example, Windows does not have an icon you can click to shut down or hibernate your PC; you can create this yourself and add it to the start screen. To do this:

1. Right click on any white area of your desktop and click New. From the listed options, choose "Shortcut".

2. In the box where Windows asks you to type the location of the item, type
 - i. **shutdown /s /t 0** to shutdown
 - ii. **shutdown /r /t 0** to restart
 - iii. **shutdown /g /t 0** to restart Windows & registered appsand then click Next.
3. Give the shortcut your desired name (like shutdown or restart) and then click finish.
4. Right click on the created shortcut and choose properties. From the properties tab, click on “shortcut” (if it’s not the selected tab) and then click “Change icon”. Choose your desired icon from the list of icon and then click “Apply” and “OK”.
5. Right click on the shortcut and choose “Cut”.
6. Now open Windows explorer and type this in the address bar:
%UserProfile%\AppData\Local\Microsoft\Windows\Application Shortcuts and then press enter.
7. Right click on any white space in this location (Application Shortcuts) and then choose paste.
The shortcut should now be in your start screen.
8. Scroll to the last end of your start screen to locate the new shortcut.
9. Drag the new shortcut to where you want it placed in the start screen.

Live Tile

Windows 8 tiles (icons in the start screen) have been designed in such a way that it can show notifications and updates by default without having to launch the app that has the tile (this is restricted to Windows 8 apps only except you add your app as a tile/live tile). These tiles are known as live tiles.



You can completely turn off these live tiles if you don’t want updates again by right clicking on the tile and then choosing “Turn live tile off”.

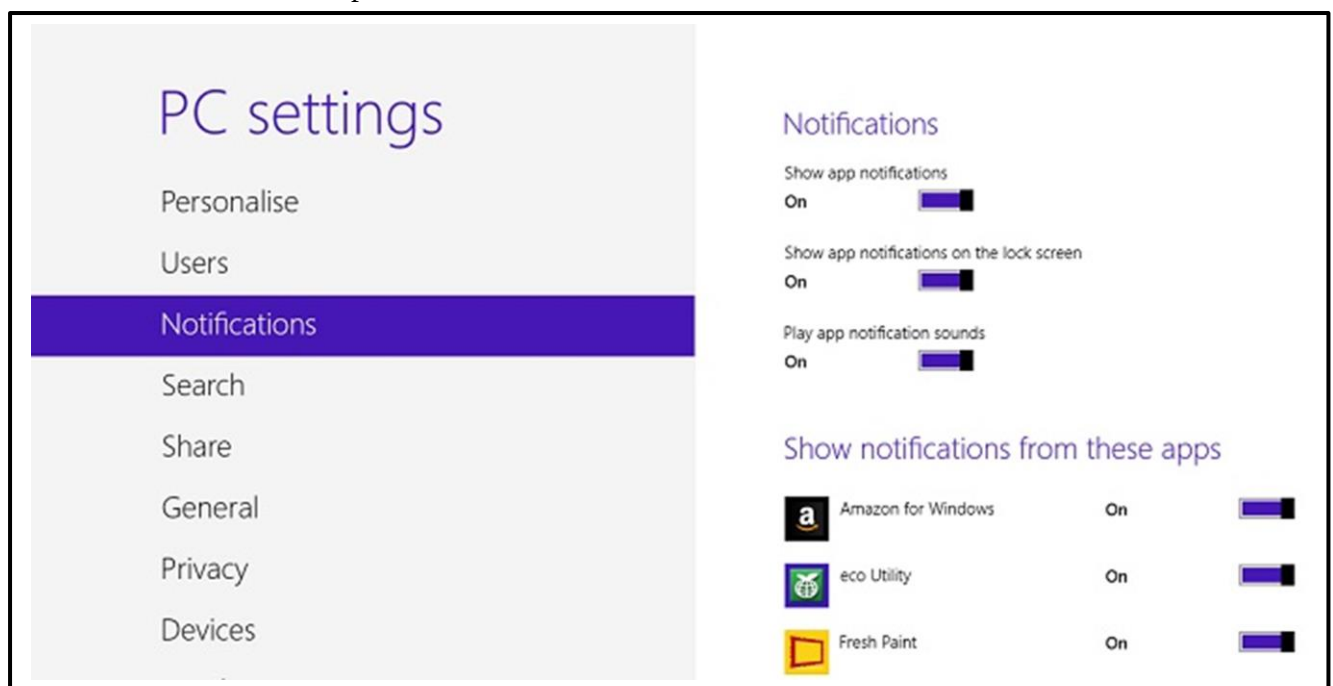
You can also decide to change the setting of your live tile so that every time you log off your PC, the data stored by the live tile would be cleared off and the tile shows its default icon, the tile would only show notifications again when it gets updated. To do this:

1. Type **gpedit.msc** in the start screen and press enter. This would launch the group policy editor which is not available in Windows 8 standard.
2. From the left pane, expand User Configuration
3. Expand Administrative Templates
4. Click on Start Menu and Taskbar
5. From the right pane, double click on “Clear history of tile notifications on exit”
6. Click on “Enabled” from the interface Windows display and then click “OK”.

Notifications

Windows 8 apps can automatically display notifications on top of any other app when they have something to inform you about; for example, the instant messaging app can notify you of a new message a friend sent to you. Just like this could be helping, it could also be annoying sometimes and you can decide you don't want such notification anymore. To disable an app's notification

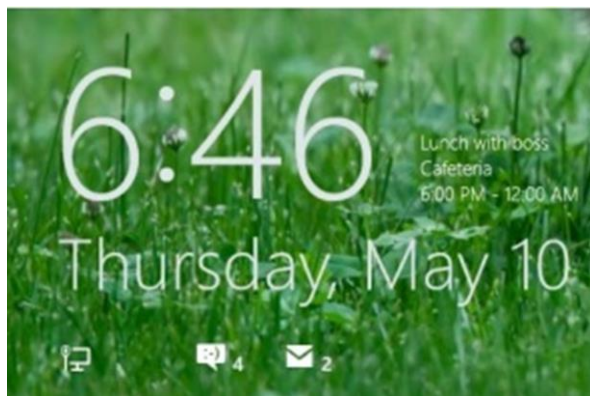
1. Open PC settings (press Windows logo key + I and then click on Change PC settings).
2. From the left pane, choose Notifications.



3. From this interface, you can decide to turn off notifications for all apps, or scroll down to see the annoying app and then turn off notification for it.

When the notifications icon shows a clock over it, notifications are disabled. Click the icon to re-enable it.

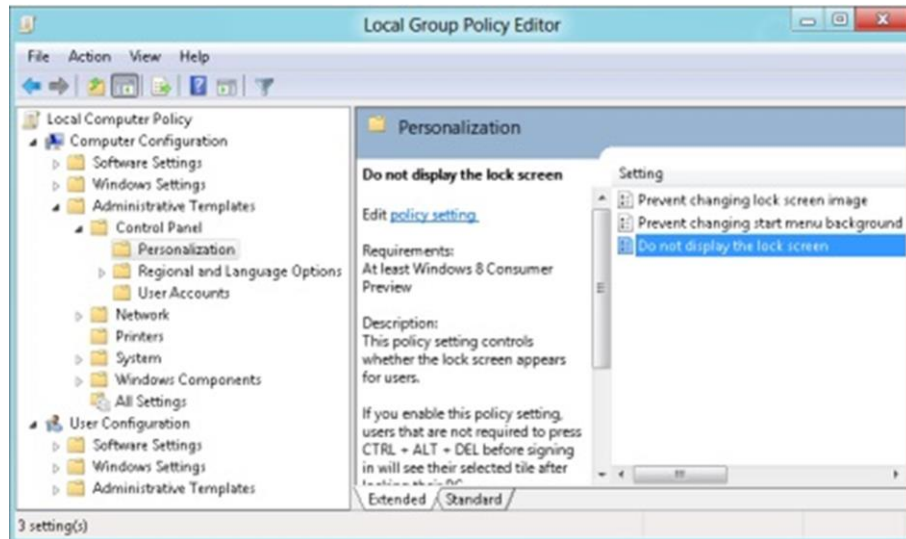
Lock Screen



The lock screen is the first screen you see when you start Windows. By default it shows notification from some apps. You can modify the lock screen settings (change the display image, change which app shows notification on it and so on) from the lock screen interface. Press window logo key and I and choose Change PC Settings, choose Personalize and then click the lock screen tab (if it's not the selected tab).

If you don't want to see the lock screen the next time you start your PC, then:

1. Run **gpedit.msc** (not available in Windows 8 standard)
2. Expand Computer Configuration
3. Expand Administrative Templates
4. Expand Control Panel
5. Click on Personalization
6. From the right pane, double click on "Do not display the lock screen"
7. Click "Enabled" and then OK.



The lock screen would be skipped next time Windows start up.

Start Screen Apps

You can create a folder on your desktop that contains all apps in your start screen so that whenever you want to launch apps, you just open that folder from desktop and then double click its shortcut.

To do this

1. Right click on your desktop and create New Shortcut
2. In the box “Type the location of the item”, type:
%windir%\explorer.exe shell:::{4234d49b-0245-4df3-b780-3893943456e1}
3. Click next and then name the shortcut and click finish
4. Windows 8 apps might not be visible since there background is white, select all (ctrl+A) to see them.
5. To make it easily accessible, right click the shortcut and click on “Pin to Taskbar”.

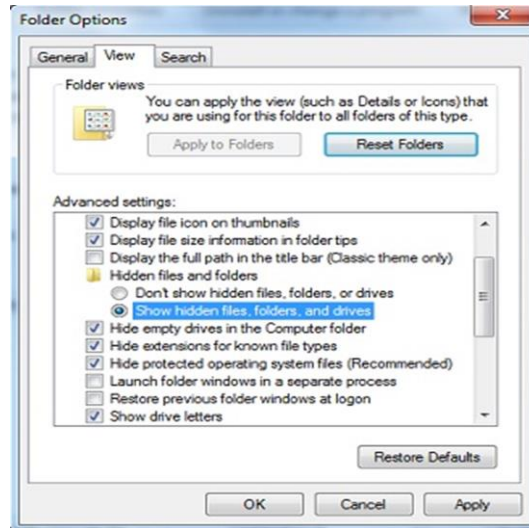
Recycle Bin (applicable to other versions of Windows)

When you delete files from your PC, Windows move them to the recycle bin (except you decide otherwise). To send files to the recycle bin very fast, you can simply drag the file to the recycle bin icon but this might not be very easy if your recycle bin is on the desktop, therefore adding recycle bin to the taskbar is a great fun.



To add recycle bin to the task bar, do the following:

1. In Windows 8, open Windows explorer, from the tabs click “View” and then check the box for “Hidden items”. In Windows 7, open Windows explorer, click on Organize from the menu bar, click on Folder and search options, click on the view tab and choose “Show hidden files, folders, and drives” under Advanced settings as shown below.



2. Right click on the taskbar and click on “Lock the taskbar” if it’s marked, if not skip this step.
3. Right click on the taskbar and choose Toolbars > New toolbar
4. From the screen that appears, open your drive C, open users, open your Username, open AppData, open Roaming, open Microsoft, open Internet explorer, click on Quick Launch and then click OK. i.e.
C:\Users\<username>\AppData\Roaming\Microsoft\Internet Explorer\Quick Launch
5. From the task bar, right click on Quick Launch and unmark Show Text, unmark Show Title and click on View > Large icons.
6. Drag the Quick Launch toolbar to resize it by grabbing the dotted lines on the taskbar and dragging it towards the left.
7. Drag the recycle bin to the quick launch toolbar.
8. Resize the Quick Launch toolbar by grabbing the dotted lines and dragging it to your desired location and size, you can right click on other shortcuts in it and delete them.
9. Right click taskbar and click Lock the taskbar.

You can now delete items by dragging them to the recycle bin. You can empty recycle bin by right clicking it in the taskbar and choosing Empty recycle bin.

Uninstalling Windows 8 apps

Windows 8 apps are those apps that came pre-installed in Windows 8 such as the Mail app, People, Calendar etc.

To uninstall any of these apps, right click on the app in the start screen and choose uninstall.

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❖ Trainings on

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- CCNA
- and more

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